EXACT COUNTEREXAMPLES TO THE BARKER–LARMAN PROBLEM IN EVERY DIMENSION AT LEAST FOUR

KNOWLEDGE SENSE

Abstract. Let $K \subset \mathbb{R}^d$ be a convex body containing the unit ball B^d in its interior such that every hyperplane tangent to B^d cuts K in a section of the same $(d-1)$-dimensional volume. The Barker–Larman problem asks whether K must be a ball. We show that the answer is negative in every dimension $d \geq 4$. For each fixed $d \geq 4$ we construct origin-symmetric, strictly convex bodies of revolution with real-analytic boundary which strictly contain B^d, are not balls, and have the same volume on all sections at distance one from the origin. In dimension four they form a sequence tending to the ball of radius two, and the set of attained volumes has positive measure near $4\sqrt{3}\,\pi$. In dimension $d \geq 5$ they are parametrized by a compact set of frequencies of length $h - o_d(h)$ inside $[\omega_{0,d} + h, \omega_{0,d} + 2h]$, as the restriction of a C^1 family of analytic profiles, and the common volumes fill a set of positive Lebesgue measure. The proofs combine finite-order formal solutions of the section equation with a Newton scheme in analytic Wiener algebras. Small divisors are controlled by circle arithmetic based on linear forms in logarithms, by a smooth diagonal elimination for real nonsymmetric matrices, and by the exclusion of a set of frequencies of measure $o_d(h)$.

1. Introduction

1.1. **The problem.** Let $d \geq 2$ and let B^d be the closed Euclidean unit ball. Barker and Larman [2] asked the following question. Let K and L be convex bodies in $\mathbb{R}^d$ containing B^d in their interiors and assume that

$$\mathrm{vol}_{d-1}(K \cap H) = \mathrm{vol}_{d-1}(L \cap H) \qquad \text{for every hyperplane } H \text{ tangent to } B^d.$$

Does it follow that $K = L$? The problem and several modifications are recorded in [12]. In the plane the answer is affirmative when L is a centered disk, a result of Santaló from 1951 (see [7]). For central sections the analogous statement for origin-symmetric bodies is a classical consequence of the injectivity of the spherical Radon transform; no such tool is available for sections at distance one. Taking for L a ball centered at the origin one

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obtains the special case in which K is required to have constant section volume over all tangent hyperplanes of B^d, and one asks whether K is a ball.

The case $d = 4$ of bodies of revolution was studied by Haddad and Ryabogin [7], who proved rigidity under arithmetic conditions on the continued fraction expansion of a rotation number computed from the volume. Haddad [6] then constructed, for every N, approximate bodies $K_{\lambda,N}$ in $\mathbb{R}^4$ whose tangent sections have volume within $c(|\lambda-\lambda_{\mathrm{b}}|/r_N)^{N+1}$ of λ while the difference between outradius and inradius is at least $c|\lambda-\lambda_{\mathrm{b}}|$, where $\lambda_{\mathrm{b}} = 4\sqrt{3}\,\pi$ [6, Theorem 1]. Convergence of the underlying formal series as $N \to \infty$ would give a negative answer to the problem in dimension four; this question is posed in [6, Section 6.1].

1.2. **Main results.** Throughout, κ_k is the volume of the unit ball of $\mathbb{R}^k$ and $|S^k|$ the area of the unit sphere $S^k \subset \mathbb{R}^{k+1}$. A body of revolution is a body invariant under all rotations fixing the first coordinate axis. For a body K with $B^d \subset \operatorname{int} K$ and a number $V > 0$ we say that K has *constant distance-one sections of volume* V if $\mathrm{vol}_{d-1}(K \cap H) = V$ for every hyperplane H tangent to B^d.

Theorem 1.1 (Exact sequence in dimension four)**.** *Put $\lambda_{\mathrm{b}} = 4\sqrt{3}\,\pi$. There are an integer m_0 and, for every integer $m \geq m_0$, an origin-symmetric strictly convex body $K_m \subset \mathbb{R}^4$ of revolution with real-analytic boundary and a number $\lambda_m > 0$ such that*

(i) *B^4 lies in the interior of K_m;*
(ii) *K_m has constant distance-one sections of volume λ_m;*
(iii) *K_m is not a Euclidean ball;*
(iv) *K_m tends to the ball of radius 2 in every fixed C^j norm of the radial function, and $\lambda_m \to \lambda_{\mathrm{b}}$ as $m \to \infty$.*

Theorem 1.2 (Positive measure of volumes in dimension four)**.** *There is $\varepsilon_0 > 0$ such that for each sign $\pm$ and each $0 < \varepsilon < \varepsilon_0$ there is a compact set $V_\varepsilon^{\pm} \subset \mathbb{R}$ with*

$$\mathrm{Leb}(V_\varepsilon^{\pm}) \;\geq\; 18\pi^2\varepsilon, \qquad \pm(\lambda-\lambda_{\mathrm{b}}) \in [\,40\pi^2\varepsilon,\, 100\pi^2\varepsilon\,] \quad (\lambda \in V_\varepsilon^{\pm}),$$

such that every $\lambda \in V_\varepsilon^{\pm}$ is the common distance-one section volume of a body K with properties (i)–(iii) *of Theorem 1.1. The bodies tend to the ball of radius two in every fixed C^j norm of the radial function as $\varepsilon \to 0$, uniformly in $\lambda \in V_\varepsilon^{\pm}$.*

For $d \geq 5$ put $\omega_{0,d} = 2\arccos(1/\sqrt{d})$, $I_{d,h} = [\omega_{0,d}+h, \omega_{0,d}+2h]$ and $q_d = 32(d+1)$. For $\rho > 0$ let $\mathcal{A}_\rho^q$ be the Banach space of real, odd, π-periodic functions $v(u) = \sum_k v_k e^{iku}$ with $\|v\|_{\rho,q} = \sum_k (1+|k|)^q e^{\rho|k|}|v_k| < \infty$. Every v with $\|v\|_{\rho,1}$ small defines, for every $\omega = 2\theta$ near $\omega_{0,d}$, an origin-symmetric strictly convex body of revolution $K(v,\omega) \subset \mathbb{R}^d$ by the centered

parametrization of Definition 2.3 below; its radial function is real analytic and close to $\sec\theta$.

Theorem 1.3 (Positive measure of volumes in dimension $d \geq 5$)**.** *Fix an integer $d \geq 5$. There is $\rho_\infty = \rho_\infty(d) > 0$ such that for every sufficiently small $h > 0$ there are a compact set $E_{d,h} \subset I_{d,h}$ with $\mathrm{Leb}(E_{d,h}) \geq h - o_d(h)$ and a C^1 map $\omega \mapsto v_\omega$ from $I_{d,h}$ to $\mathcal{A}^{q_d}_{\rho_\infty}$ with the following properties, where $K_\omega = K(v_\omega, \omega)$.*

(i) *For every $\omega \in I_{d,h}$, K_ω is an origin-symmetric, strictly convex body of revolution with real-analytic boundary, strictly contains B^d and is not a ball.*

(ii) *For every $\omega \in E_{d,h}$, K_ω has constant distance-one sections, of volume $V(\omega)$.*

(iii) *The radial profiles of K_ω converge to the constant $\sqrt{d}$, uniformly in $\omega \in I_{d,h}$, on a fixed complex neighborhood of the real axis as $h \to 0$.*

(iv) *Let $V(\omega)$ denote, for every $\omega \in I_{d,h}$, the mean of the distance-one section volume of K_ω over all unit normals. Then $V \in C^1(I_{d,h})$ and*

$$V'(\omega) = -2\,|S^{d-2}|\,(d-1)^{(d-2)/2} + O_d(h).$$

Consequently $V(E_{d,h})$ has positive Lebesgue measure.

In particular $\omega \mapsto v_\omega$ restricted to $E_{d,h}$ is a Whitney C^1 family in $\mathcal{A}^{q_d}_{\rho_\infty}$. The section equation is asserted only on $E_{d,h}$; the extension to $I_{d,h}$ is a family of bodies with a C^1 mean volume, not a family of solutions. A set of exact volumes of positive measure is obtained; an interval of exact volumes is not asserted. The constants and smallness thresholds depend on the dimension.

Corollary 1.4 (Negative answer to the Barker–Larman problem for $d \geq 4$)**.** *Let $d \geq 4$ and let K be one of the bodies of Theorem 1.1, Theorem 1.2 or Theorem 1.3*(ii)*, with common volume V. Let $R_V = \big(1 + (V/\kappa_{d-1})^{2/(d-1)}\big)^{1/2}$. Then K and the ball $R_V B^d$ are distinct convex bodies containing B^d in their interiors which have equal volume on every hyperplane tangent to B^d.*

Proof. A hyperplane at distance one from the center cuts $R_V B^d$ in a $(d-1)$-ball of radius $\sqrt{R_V^2 - 1}$, whose volume is $\kappa_{d-1}(R_V^2 - 1)^{(d-1)/2} = V$. Since K is not a ball it differs from $R_V B^d$. □

Dimension three is not covered by these results. The bodies depend on the parameter and the distance is fixed; no statement is made about two fixed bodies and several distances.

The exact solutions constructed here do not arise by summing Haddad's formal series. In a companion paper [8] we show that this series, in the normalization recalled in Section 3.2, has radius of convergence zero:

Theorem 1.5 ([8])**.** *Let $\lambda_{\mathrm{b}} = 4\sqrt{3}\,\pi$, $z = (\lambda - \lambda_{\mathrm{b}})/(\sqrt{3}\,\pi)$ and let $h(z,t) = t - \pi/3 + \sqrt{3}\sum_{n\geq 1} R_n(t) z^n$ be the normalized formal solution*

of the dimension-four section equation with $R_1 = \frac{1}{48} + \frac{1}{12}\cos 2t$. *Then* $\limsup_{n\to\infty} \|R_n\|_{C^0}^{1/n} = \infty$. *The same holds in every fixed* C^j *and Sobolev* H^s *norm, in the Fourier* ℓ^1 *norm and in every positive-width analytic Wiener norm.*

1.3. **Relation to earlier work.** We list what is used and what is new.

- From [6] we use the section equation in dimension four and the recursion producing finite trigonometric solutions to arbitrary finite order; we fix the order $N = 12$ in Section 3.2.
- The rigidity theorem of [7] (Theorem 1, Definition 8 and Theorem 9 there) holds on a particular arithmetic set of volumes, with $A > 8\kappa_3$ in their Definition 8. Our base volume $\lambda_{\mathrm{b}} = 4\sqrt{3}\,\pi$ lies below $8\kappa_3 = 32\pi/3$. Moreover the rotation attached to the volume in [7] differs from the rotation ω of the endpoint map used here.
- In [5] one fixed pair of bodies is considered with infinitely many distances having an interior accumulation point (Theorem 6.2 there), and Theorems 4.2–4.4 there need information over a distance interval. Here the body varies and the distance is fixed.
- Lazutkin [9] proved that a sufficiently smooth strictly convex planar billiard has a family of caustics near the boundary whose rotation numbers form a set of positive measure; his KAM-type method is a general ancestor of the small-divisor arguments below. The reduction of a variational equation to two successive differences and a scalar compatibility condition, the quasi-Newton terms proportional to the residual, and the use of finite-order initializers after fixing the analytic losses have direct antecedents in [10, 11]. The hypotheses of [10, 11] concern a full caustic collar of a single planar billiard, which is not supplied by the single-distance section equation for varying bodies. The geometry-specific forced-zero division, the scalar calculations and the retained-family inverse estimates are proved here.
- The conjugacy of analytic circle maps with the quantitative a posteriori theorems of [4] is used in dimension four.
- Lower bounds for $|\sin k\theta_0|$ and for the even-dimensional round symbols come from the coefficient-height consequence of the theorem of Baker and Wüstholz [1], recalled in Lemma 9.6.

1.4. **Outline of the proofs.** A body of revolution is described by its meridian profile. Parametrizing the profile by the angles of the tangent lines to B^d through its points (the *centered parametrization*, Section 2) turns the constancy of the section volume into a functional equation for an odd lift p of the circle, in which the unknown enters through two translates by $\pm\omega/2$, where ω is the rotation number of the endpoint map. This holds in every dimension.

In dimension four the base ball of radius two has rational rotation $\omega = 2\pi/3$, and the linearization at the ball has infinitely many resonances. We use a finite-order formal solution, adjust the rotation to a nearby Diophantine number with continued fraction $[0; 3, m, 1, 1, \dots]$ by an a posteriori conjugacy theorem, and solve the linearized equation by two successive cohomological equations with a forced zero. The scalar compatibility condition is solved by the volume and is nonzero because of a contribution from the third resonance. A Newton scheme with fixed width loss then gives exact solutions.

In dimension $d \geq 5$ the ball of radius $\sqrt{d}$ has irrational rotation and a one-dimensional bifurcation in the degree-two zonal harmonic. A finite-order bifurcation expansion in the amplitude $\varepsilon \sim h$ gives initializers to every finite order. The linearization, suitably normalized, has the form $D + A$ with a diagonal D built from Gegenbauer values (odd d) or from cotangents (even d), and an off-diagonal part $A = O(h)$. A smooth diagonal elimination followed by a three-class analysis of the effective diagonals gives inverses on a set of frequencies of measure $h - o(h)$, with costs controlled in two regimes; this makes a Newton iteration with infinitely many steps converge. The surviving set of frequencies is nested and has a positive limiting threshold.

1.5. **Results used without proof.** Besides standard facts about Gegenbauer polynomials [3], the proofs use the following results which are not reproved: the finite-order solutions of [6]; Theorems 3.1 and 4.1 of [4] together with their proofs, with the parameter budget discussed in Remark 4.2; and the coefficient-height consequence of [1] recalled in Lemma 9.6. The companion result Theorem 1.5 is not used in the proofs of the existence theorems.

1.6. **Organization.** Section 2 collects the geometry and function spaces common to all dimensions. Part I treats $d = 4$: Sections 3–6 prove Theorem 1.1 and Section 7 proves Theorem 1.2. Part II treats $d \geq 5$: Section 8 sets up the section operators, Section 9 treats resonances, arithmetic and finite initializers, Section 10 constructs the normalized operators for odd and even d, Section 11 proves the high-mode inverse theorem, Section 12 treats the finite center and Section 13 carries out the Newton iteration and proves Theorem 1.3.

2. Conventions and common geometry

2.1. **Function spaces.** For a 2π-periodic function $f = \sum_n \hat{f}_n e^{inu}$ and $\rho \geq 0$ put

$$|f|_\rho = \sum_{n\in\mathbb{Z}} |\hat{f}_n| e^{\rho|n|}, \qquad |f|_{\rho,1} = |f|_\rho + |f'|_\rho.$$

This is a Banach algebra norm which is invariant under real translations, and $|f'|_{\rho-\eta} \leq \eta^{-1}|f|_\rho$ for $0 < \eta \leq \rho$. More generally $\|f\|_{\rho,q} = \sum_n (1 +$

$|n|)^q e^{\rho|n|}|\hat{f}_n|$; since $1+|n+m| \le (1+|n|)(1+|m|)$, these norms are submultiplicative as well. Spaces of functions holomorphic in the interior of a strip (or a Bernstein ellipse) and continuous on its closure are equipped with the supremum norm. When we speak of the parity of a function on the circle (even, odd, π-periodic, π-antiperiodic) we mean its parity as a function of the angular variable in which it is written.

2.2. **Bodies of revolution.** Let $K \subset \mathbb{R}^d$ be a star body of revolution about the x_1-axis with radial function r. We write r both for the function on S^{d-1} and for the function $t \mapsto r(\cos t, \sin t, 0, \dots, 0)$ of the angle t from e_1; K is origin symmetric exactly when the latter is even and π-periodic.

Lemma 2.1 (Analyticity at the poles)**.** *Let r be even, π-periodic and holomorphic on a strip $|\operatorname{Im} t| < \rho$. Then $r(u) = F(2u_1^2 - 1)$ for a function F holomorphic on a Bernstein ellipse around $[-1,1]$; in particular the radial function is real analytic on all of S^{d-1}, including $u_1 = \pm 1$.*

Proof. Write $r(t) = \sum_{n\ge 0} a_n \cos 2nt$. Holomorphy on the strip gives $|a_n| \le C_{\rho'} e^{-2n\rho'}$ for every $\rho' < \rho$. Since $\cos 2nt = T_n(\cos 2t)$ with the Chebyshev polynomials T_n, and $|T_n| \le e^{2n\rho''}$ on the Bernstein ellipse with parameter $e^{2\rho''}$, the series $F(x) = \sum a_n T_n(x)$ converges uniformly on that ellipse for $\rho'' < \rho'$. With $\cos 2t = 2u_1^2 - 1$ the claim follows. □

Lemma 2.2 (Global strict convexity)**.** *For every $d \ge 2$ and $R > 1$ there is $\delta_1(d,R) > 0$ with the following property. Let r be a C^2 radial function on S^{d-1} with $\|r - R\|_{C^2(S^{d-1})} \le \delta_1$. Then $K = \{su : u \in S^{d-1},\ 0 \le s \le r(u)\}$ is strictly convex, its boundary is a C^2 hypersurface (real analytic if r is), and $B^d \subset \operatorname{int} K$.*

Proof. Put $\Phi(x) = |x|^2 / r(x/|x|)^2$ for $x \ne 0$ and $\Phi(0) = 0$. On the unit sphere the Hessian of Φ depends continuously on the 2-jet of r and equals $2R^{-2}I$ for $r \equiv R$; hence for small δ_1 it is at least $R^{-2}I$ there, and by homogeneity of degree two (so that the Hessian is homogeneous of degree zero) at every $x \ne 0$. Also $|\Phi(x)| \le C|x|^2$, so Φ is C^1 at 0 with zero gradient. On a line $s \mapsto a + sb$ the function $\phi(s) = \Phi(a+sb)$ is C^1, and C^2 with $\phi'' \ge R^{-2}|b|^2$ except at most at one point; hence ϕ' is strictly increasing and ϕ is strictly convex. Thus Φ is strictly convex, $K = \{\Phi \le 1\}$ is strictly convex, and since $x \cdot \nabla\Phi = 2$ on $\{\Phi = 1\}$ the boundary is a C^2 (resp. analytic) hypersurface. Finally $\min r \ge R - \delta_1 > 1$. □

2.3. **The centered parametrization.** Suppose $r > 1$ and write $r(t) = \sec\psi(t)$ with $0 < \psi < \pi/2$. The tangent lines from the point $r(t)(\cos t, \sin t)$ of the meridian curve to the unit circle touch it at the points with normal angles $t \mp \psi(t)$.

Definition 2.3 (Centered parametrization)**.** Fix $\theta \in (0, \pi/2)$ and put $\omega = 2\theta$. Let $p(u) = u + v(u)$ with v real analytic, odd and π-periodic, and $p' > 0$.

Define

$$\chi(y) = \tfrac{1}{2}\big[p(y-\theta) + p(y+\theta)\big], \qquad \alpha(y) = \tfrac{1}{2}\big[p(y+\theta) - p(y-\theta)\big],$$
$$\hat{r}(y) = \sec\alpha(y),$$

and let r be the function with $r(\chi(y)) = \hat{r}(y)$. We write $K(v,\omega)$ for the body of revolution with radial function r.

If $v = 0$ then $\chi(y) = y$, $\alpha = \theta$ and $r \equiv \sec\theta =: R$ is the ball of radius R centered at the origin. For general p the pair (χ, α) describes the profile point of polar angle $\chi(y)$ whose two tangent lines to the unit circle have normal angles $\chi - \alpha = p(y-\theta)$ and $\chi + \alpha = p(y+\theta)$. Consequently the chord cut from the meridian curve by the tangent line with normal angle $p(u)$ has end points with parameters $y = u - \theta$ and $y = u + \theta$: in the parameter y the endpoint map is the rotation $y \mapsto y + \omega$. This holds at every θ with no arithmetic assumption.

Lemma 2.4. *Let $\mathcal{J} \subset (0, \pi/2)$ be a compact interval, $d \geq 2$ and $0 < \rho \leq 1$. There is $\delta_0 = \delta_0(d, \mathcal{J}, \rho) > 0$ such that the following holds for $\theta \in \mathcal{J}$ and $|v|_{\rho,1} \leq \delta_0$. The map χ is an analytic diffeomorphism of $\mathbb{R}/\pi\mathbb{Z}$ commuting with $y \mapsto -y$, it maps the strip $|\operatorname{Im} y| < \rho/2$ biholomorphically onto a domain containing $|\operatorname{Im} t| < \rho/4$, and α takes values in a compact subset of $(0, \pi/2)$ on $|\operatorname{Im} y| \leq \rho/2$, so that $\sec\alpha$ is unambiguous. The radial function r is even, π-periodic, holomorphic on $|\operatorname{Im} t| < \rho/4$ and satisfies $\|r - \sec\theta\|_{C^2} \leq C(\mathcal{J}, \rho)|v|_{\rho,1}$; the body $K(v, 2\theta)$ is origin symmetric, strictly convex, has analytic boundary and strictly contains B^d.*

Proof. On $|\operatorname{Im} y| \leq \rho$ we have $|\chi(y) - y| + |\chi'(y) - 1| \leq 2|v|_{\rho,1}$ and $|\alpha - \theta| \leq |v|_\rho$. Injectivity of χ on the convex strip of width $\rho/2$ follows from $|\chi' - 1| < \frac{1}{2}$ by integration along segments, and the inverse on the strip of width $\rho/4$ is obtained from the contraction $y = t - (\chi(y) - y)$ on disks of radius $\rho/4$ contained in the larger strip; the Cauchy estimates on the remaining buffer bound its derivatives. Oddness and π-equivariance of χ, and evenness and π-periodicity of α, follow from those of v. The bound on $r - \sec\theta = \sec\alpha \circ \chi^{-1} - \sec\theta$ follows from these estimates and Cauchy's estimate on the buffer. The geometric statements follow from Lemma 2.1 and Lemma 2.2 with $R = \sec\theta$, after reducing δ_0. □

Part I. Dimension four

3. The section equation in $\mathbb{R}^4$ and its finite-order solutions

Throughout Part I put $\alpha = \pi/3$, $\lambda_{\mathrm{b}} = 4\sqrt{3}\,\pi$ and $w = 1/16$.

3.1. **The equation and its converse.** A real increasing lift h of the circle with $h(t+\pi) = h(t) + \pi$ and $h(t) - h(-t) = 2t$ can be written $h(t) = t - \psi(t)$ with ψ even and π-periodic. Assume that ψ is close to α in an analytic norm. The planar curve with polar equation $t \mapsto \sec\psi(t)$ encloses the unit

disk and is close to the circle of radius 2; revolving it about the first axis gives a body $K \subset \mathbb{R}^4$. Put $a(x) = \tan\bigl(h^{-1}(x) - x\bigr)$ and

$$W(h)(x) = \pi \sin x \left[a(x)^2 - a(-x)^2\right] + \tfrac{2\pi}{3} \cos x \left[a(x)^3 + a(-x)^3\right]. \tag{3.1}$$

Let $A_K(x)$ be the three-dimensional volume of the section of K by the hyperplane at distance one with unit normal $(\cos x, \sin x, 0, 0)$. In [6, Section 2.2], $A_K(x)$ is written as an explicit integral over the chord cut from the meridian curve by the tangent line, using the inverse-angle parametrization of [6, Lemma 1]; our ψ satisfies the assumptions of that lemma. Differentiating this integral gives [6, equation (6)]: for $0 < x < \pi$,

$$\frac{d}{dx}\bigl[\sin x\, A_K(x)\bigr] = W(h)(x). \tag{3.2}$$

Proposition 3.1 (Equation and converse)**.** *Let K be as above, with ψ analytic and so close to α that K is strictly convex, has analytic boundary and contains B^4 in its interior. Then $A_K \equiv \lambda$ on all hyperplanes tangent to B^4 if and only if $W(h) = \lambda \cos$ on $\mathbb{R}$.*

Proof. If $A_K \equiv \lambda$ then (3.2) gives $W(h) = \lambda \cos$ on $(0, \pi)$, and on $\mathbb{R}$ by analyticity. Conversely, if $W(h) = \lambda \cos$ then $\sin x\,[A_K(x) - \lambda]$ has zero derivative on $(0, \pi)$, so it is constant. Section volumes are bounded and continuous near $x = 0$: for a smooth strictly convex body containing B^4 in its interior, nearby affine sections are uniformly transverse compact sections, so their radial functions in these hyperplanes depend continuously on the normal. Hence the constant is zero (let $x \downarrow 0$), so $A_K = \lambda$ on $(0, \pi)$ and, by continuity, at both endpoints. Revolution symmetry covers all normals $u \in S^3$. □

3.2. **The formal branch and a finite initializer.** Put $\lambda_z = \lambda_{\mathrm{b}} + \sqrt{3}\,\pi z$. The section equation $W(h) = \lambda_z \cos$ has a formal solution

$$h_z(t) = t - \alpha + \sqrt{3} \sum_{j \geq 1} R_j(t) z^j,$$

where each R_j is a real even π-periodic trigonometric polynomial, normalized by $R_1 = \frac{1}{48} + \frac{1}{12}\cos 2t$; this is the formal branch of [6, Sections 3–5]. The first coefficients are

$$R_1 = \tfrac{1}{48} + \tfrac{\cos 2t}{12}, \qquad R_2 = \tfrac{3}{256} - \tfrac{\cos 2t}{144} + \tfrac{\cos 4t}{72}. \tag{3.3}$$

We use only a fixed finite order. Let

$$h_z(t) = t - \alpha + f_z(t), \qquad f_z(t) = \sqrt{3} \sum_{j=1}^{12} R_j(t) z^j. \tag{3.4}$$

By the construction of the R_j, the coefficients of $z^1, \ldots, z^{12}$ in the expansion of $W(h_z)(x) - \lambda_z \cos x$ vanish; this is an identity between finite trigonometric polynomials for real x and extends by analytic continuation to complex x.

Lemma 3.2 (Common complex domain)**.** *There is $r_* > 0$ such that for $|z| < r_*$ and $|\operatorname{Im} t| < 4w$ one has $|f_z(t)| \le w/2$ and $|\partial_t f_z(t)| \le 1/4$. For $|\operatorname{Im} x| < 3w$ the equation $t = x + \alpha - f_z(t)$ is a contraction on the closed disk $|t - (x+\alpha)| \le w$, whose unique fixed point $t = h_z^{-1}(x)$ is holomorphic in (x, z). After reducing r_* the function $W(h_z)$ is jointly holomorphic on $|\operatorname{Im} x| < 2w$, and*

$$W(h_z) - \lambda_z \cos = O(z^{13}) \tag{3.5}$$

uniformly on $|\operatorname{Im} x| \le w$; every fixed finite number of derivatives satisfies the same bound on smaller fixed strips.

Proof. Since f_z is a polynomial in z with trigonometric polynomial coefficients vanishing at $z = 0$, the bounds on f_z hold for small z. The contraction maps the disk into the subdisk of radius $w/2$ and has Lipschitz constant at most $1/4$; the holomorphic implicit function theorem gives holomorphy in (x, z), and periodicity identifies the local inverses as one inverse lift. This holds for complex z. Holomorphy of $W(h_z)$ requires the two tangent arguments to stay away from their poles, which holds after reducing r_* since $h_z^{-1}(x) - x$ is close to α. The first twelve Taylor coefficients in z of $W(h_z) - \lambda_z \cos$ vanish, so Cauchy's formula on a smaller z-disk gives (3.5). The radius r_* depends on the fixed order 12. □

4. Circle arithmetic and the approximate conjugacy

4.1. **Explicit arithmetic.** For every integer $m \ge 3$ put

$$\theta_m = [0; 3, m, 1, 1, 1, \dots], \qquad \nu_m = -2\pi\theta_m, \qquad \gamma_m = \frac{1}{m+2}. \tag{4.1}$$

These numbers are irrational. If $t_m = m + 1/\phi$ with $\phi = (1+\sqrt{5})/2$ then $\theta_m = t_m/(3t_m+1)$, so $\nu_m + \frac{2\pi}{3} = 2\pi/[3(3t_m+1)] > 0$ is comparable to $1/m$.

Lemma 4.1 (Diophantine bound)**.** *For every nonzero integer q, $\|q\theta_m\| \ge 1/((m+2)|q|)$, and consequently*

$$|1 - e^{iq\nu_m}| \;\ge\; \frac{4\gamma_m}{|q|}. \tag{4.2}$$

Proof. All partial quotients of θ_m are at most m, so the convergent denominators satisfy $q_{k+1} \le (m+1)q_k$. By best approximation and the convergent-error inequality, for $q_k \le |q| < q_{k+1}$ one has $\|q\theta_m\| \ge \|q_k\theta_m\| > 1/(q_k + q_{k+1}) \ge 1/[(m+2)q_k] \ge 1/[(m+2)|q|]$; the case $q = 1$ follows directly. Then $|1 - e^{iq\nu_m}| = 2\sin(\pi\|q\theta_m\|) \ge 4\|q\theta_m\|$ since $2\sin\pi x \ge 4x$ for $0 \le x \le \frac{1}{2}$. □

In particular θ_m and $-\theta_m$ satisfy the Diophantine condition $D(\gamma_m, 1)$ of [4]. For the centered difference used below a Fourier mode n has divisor $2i\sin(n\nu_m/2)$, whose modulus equals that in (4.2); on π-periodic functions only even n occur.

4.2. **The circle lift of the approximate body.** Put

$$\mathcal{S}_z(t) = -h_z^{-1}\big(-h_z(t)\big). \tag{4.3}$$

It is analytic on $|\operatorname{Im} t| < 2w$, because h_z maps this strip into $|\operatorname{Im} x| < 5w/2 < 3w$; it is real and increasing for real z, π-equivariant, and reversible under $t \mapsto -t$. Uniform bounds for its derivatives up to total order two on the closed strip of width $2w$ follow from the extra buffer in Lemma 3.2 after reducing r_*. A direct substitution using only (3.3) gives, with

$$\begin{aligned} k_z(t) &= t + \tfrac{z}{12}\sin 2t - \tfrac{z^2}{48}(\sin 2t + \sin 4t), \\ \bar\nu(z) &= -\tfrac{2\pi}{3} + \tfrac{\sqrt3}{24}z + \tfrac{11\sqrt3}{1152}z^2, \\ \mathcal{S}_z(k_z(t)) - k_z(t+\bar\nu(z)) &= O(z^3). \end{aligned} \tag{4.4}$$

Explicitly $\mathcal{S}_z(t) = t - \frac{2\pi}{3} + z s_1(t) + z^2 s_2(t) + O(z^3)$ with

$$\begin{aligned} s_1 &= \tfrac{\sqrt3}{24} + \tfrac{\sqrt3}{24}\cos 2t - \tfrac18 \sin 2t, \\ s_2 &= \tfrac{5\sqrt3}{384} - \tfrac{\sqrt3}{72}\cos 2t + \tfrac{1}{48}\sin 2t + \tfrac{\sqrt3}{144}\cos 4t + \tfrac{1}{24}\sin 4t, \end{aligned}$$

and (4.4) is a finite algebraic identity after expanding every sine and cosine.

Since $\bar\nu'(0) = \sqrt3/24 > 0$, for all large m there is a unique small $z_m > 0$ with $\bar\nu(z_m) = \nu_m$. Then z_m, $\nu_m + 2\pi/3$ and γ_m are pairwise comparable with constants independent of m, and all later $O(z_m^j)$ bounds have constants independent of m.

4.3. **Preparation for the a posteriori conjugacy theorem.** We use the unit-circle conventions of [4]:

$$F_z(x) = \frac{\mathcal{S}_z(2\pi x)}{2\pi}, \qquad H_z(x) = \frac{k_z(2\pi x)}{2\pi}, \qquad \vartheta_m = \frac{\nu_m}{2\pi}.$$

Take the fixed parameter interval $\mathcal{A} = (-r_*/2, r_*/2)$ and widths

$$\hat\rho = w/\pi, \quad \rho = w/(2\pi), \quad \delta = \rho/8, \quad \rho_\infty = \rho/2. \tag{4.5}$$

Condition G1 of [4, Theorem 3.1] holds by the mixed derivative bounds above and G2 holds by (4.2). For small r_* the map H_z has zero mean displacement, derivative within $1/8$ of 1 and displacement at most $w/(8\pi)$; hence its image of the strip of width ρ is at distance at least $3w/(8\pi)$ from the boundary of the strip of width $\hat\rho$, and the derivative and reciprocal-derivative bounds of H1 hold with $\sigma_1 = \sigma_2 = 2$. The function of H2, which we denote by β, is

$$\beta(x) = \frac{(\partial_z F_z)(H_z(x))}{H_z'(x+\vartheta_m)}, \qquad \text{mean } \beta \text{ at } (z,\vartheta) = (0, -\tfrac13) \text{ equals } \beta_0 = \frac{\sqrt3}{48\pi},$$

because the angular mean of $\partial_z \mathcal{S}_z$ at $z = 0$ is $\sqrt3/24$. By continuity $|\text{mean}\,\beta - \beta_0| \le \beta_0/4$ after a fixed restriction, and we take $\sigma_b = 2/\beta_0$. Also $z_m < r_*/4$ eventually, so z_m has distance at least $r_*/4$ from $\partial\mathcal{A}$.

Remark 4.2 (Uniformity as $\gamma_m \to 0$ and the parameter budget). Equation numbers in this remark refer to the 2019 author manuscript of [4]. With the widths (4.5) the constants $a_1 = 2$, $a_2 = 2$, $a_3 = 8$ of [4, (3.37)–(3.38)] are fixed, and in [4, (3.21), (3.22), (3.26)], with $\tau = 1$, all terms containing γ other than the displayed inverse powers carry nonnegative powers of γ. The iteration ratio κ of [4, (3.40)] is $O(z_m^3/\gamma_m^2) = O(z_m)$, hence eventually at most $\frac{1}{2}$, and the sums $\Sigma_{\kappa,p}$ of [4, (3.41)], $p = 0, 1, 2$, are then bounded by 2. The positive denominators in [4, (3.48)] have the fixed margins above. In the parameter budget [4, (3.43)] and in the corresponding entry of [4, (3.48)] the numerator is written $\sigma_2\sigma_1$, while the single-step bound [4, (3.23)] is $|\Delta_\alpha| < \sigma_b\sigma_2\|e\|_\rho$. We therefore use the numerator $\sigma_2 \max(\sigma_1, \sigma_b)$ in these two places; equivalently we sum the step bounds of [4, Lemma 3.2]:

$$\sum_j |\Delta z_j| \le \sigma_b\sigma_2\, \Sigma_{\kappa,2}\, \|e_0\|_\rho \le 2\sigma_b\sigma_2\|e_0\|_\rho = O(z_m^3),$$

which is eventually below $r_*/4$, so the parameter stays in $\mathcal{A}$ at every step.

Proposition 4.3 (Approximate conjugacy)**.** *For all sufficiently large m there are a real number z_m^* and an analytic, odd, π-equivariant diffeomorphism k_m of the circle, holomorphic on the angular strip of width $w/2$, such that*

$$\begin{aligned} z_m^* &= z_m + O(z_m^3), \qquad & k_m &= k_{z_m} + O(z_m^2), \\ \mathcal{S}_{z_m^*} \circ k_m &= k_m \circ T_{\nu_m}, \qquad & \operatorname{mean}(k_m - \mathrm{id}) &= 0, \end{aligned} \tag{4.6}$$

where $T_\nu(s) = s + \nu$. The parameter estimate is of the order of the initial error, and the conjugacy estimate of the order of the initial error divided by γ_m.

Proof. The initial error $e = \mathcal{S}_{z_m} \circ k_{z_m} - k_{z_m} \circ T_{\nu_m}$ is $O(z_m^3)$ by (4.4). The hypotheses of [4, Theorem 3.1] and of the iteration in its proof (with the parameter budget of Remark 4.2) hold by the preceding paragraphs; part T1 of that theorem gives existence and part T2 gives $|z_m^* - z_m| \le C_3\|e\| = O(z_m^3)$ and $\|k_m - k_{z_m}\| \le C_2\|e\|/(\gamma\rho^\tau) = O(z_m^2)$.

A normalized conjugacy to an irrational rotation is unique for its fixed circle map: the quotient of two such conjugacies commutes with the irrational rotation and is therefore a translation by density of orbits, and a phase shift c adds c to the mean displacement, so the normalization fixes it. Applying this to $-k_m(-s)$, using the reversibility of $\mathcal{S}_z$, proves oddness; applying it to $k_m(s + \pi) - \pi$ proves π-equivariance. □

Define

$$G_m(y) = h_{z_m^*}\big(k_m(y - \nu_m/2)\big). \tag{4.7}$$

The conjugacy equation and $h_z(-t) = h_z(t) - 2t$ give

$$\begin{aligned} G_m(-y) &= -G_m(y), \qquad & G_m(y + \pi) &= G_m(y) + \pi, \\ k_m(s) &= \tfrac{1}{2}\big[G_m(s - \tfrac{\nu_m}{2}) + G_m(s + \tfrac{\nu_m}{2})\big]. \end{aligned} \tag{4.8}$$

For instance $h_z(-k_m(s+\nu_m)) = -h_z(k_m(s))$ gives oddness after setting $y = s + \nu_m/2$. Expanding (4.7) with (3.3), (4.4) and (4.6),

$$G_m(y) = y - \tfrac{z_m}{6}\sin 2y + O(z_m^2). \tag{4.9}$$

All these maps are holomorphic with uniform estimates on the fixed strip of width $w/4$, because k_m is $O(z_m)$-close to the identity on the strip of width $w/2$.

4.4. **Transfer to Wiener norms.** Set

$$\rho_0 = w/8 = \tfrac{1}{128}. \tag{4.10}$$

If a 2π-periodic function has supremum norm M_0 on the strip of width $w/4$, its Fourier coefficients obey $|\hat{f}_n| \le M_0 e^{-|n|w/4}$, so its Wiener norm at ρ_0 is at most $M_0 \sum_n e^{-|n|w/8}$; multiplying by $|n|^j$ gives the same transfer for every fixed j. Consequently (4.9), including any fixed number of derivatives, holds in the norm $|\cdot|_{\rho_0}$. The residual (3.5), composed with G_m, is $O(z_m^{13})$ on the strip of width $w/4$ after a fixed reduction of r_*; the parameter z_m^* is comparable to z_m, and the volume parameter is $\lambda_{z_m^*}$. The prepared residual is therefore $O(z_m^{13})$ in the norm $|\cdot|_{\rho_0}$.

5. The fixed-shift Newton scheme

In this section $\varsigma = 10^{-5}$, $\rho_0 = 1/128$ as in (4.10), and $b_* = 1/(96\pi)$.

5.1. **The equation and a bounded class.** Let $D_\nu Y = Y - Y(\cdot + \nu)$ and $\Delta_\nu v = v(\cdot + \nu/2) - v(\cdot - \nu/2)$. We consider real parameters with

$$\begin{aligned} &0 < \gamma \le 1, \qquad |\nu + \tfrac{2\pi}{3}| \le \varsigma, \qquad |\lambda - \lambda_{\mathrm{b}}| \le 1, \qquad \tfrac{\rho_0}{2} \le \rho \le \rho_0, \\ &G(y) = y + g(y), \qquad g \text{ real, odd, } \pi\text{-periodic}, \qquad |g|_{\rho,1} \le \varsigma, \end{aligned} \tag{5.1}$$

and assume (4.2), i.e. $|1 - e^{iq\nu}| \ge 4\gamma/|q|$ for $q \ne 0$. Put

$$\begin{aligned} A_-(y) &= \tfrac{1}{2}[G(y-\nu) - G(y)], \qquad A_+(y) = \tfrac{1}{2}[G(y) - G(y+\nu)], \\ a_\pm &= \tan A_\pm, \\ E(G,\lambda) &= \pi \sin G\,(a_-^2 - a_+^2) + \tfrac{2\pi}{3}\cos G\,(a_-^3 + a_+^3) - \lambda \cos G. \end{aligned} \tag{5.2}$$

The differences of lifts in $A_\pm$ are periodic with constant term $-\nu/2$, within $\varsigma/2$ of α, and remaining Wiener norm at most ς, so their distance in the algebra from α is at most $3\varsigma/2$. Hence $\cos A_\pm$ differs from $1/2$ by at most 3ς and is invertible by a Neumann series with inverse of norm < 3; no tangent pole occurs on the complex strip. The map E is analytic in (g, λ) with uniformly bounded second derivative as a map from the Wiener algebra to itself throughout (5.1), and on any straight segment inside the class; its value is even and π-antiperiodic. The constants are independent of γ, ν, ρ.

For G in the class put

$$k(s) = \tfrac{1}{2}\big[G(s - \tfrac{\nu}{2}) + G(s + \tfrac{\nu}{2})\big], \qquad h(k(s)) = G(s + \tfrac{\nu}{2}). \tag{5.3}$$

Real derivatives of G and k are positive, k is odd and π-equivariant, $h(t) - h(-t) = 2t$ and $-h^{-1}(-h(k(s))) = k(s+\nu)$. At $x = G(y)$ one has $h^{-1}(x) -$

$x = A_-(y)$ and $h^{-1}(-x) - (-x) = A_+(y)$, so the two tangent arguments in (3.1) are exactly $A_-(y)$ and $A_+(y)$. This proves:

Lemma 5.1. *For G in the class* (5.1), $E(G,\lambda)(y) = \big[W(h) - \lambda\cos\big](G(y))$.

The substitution is exact, and every shift in (5.2) is fixed rather than depending on the unknown.

5.2. **The forced-zero multiplier and the derivative identity.** Put

$$\begin{aligned} q(s) &= \tfrac{1}{2}[G(s-\tfrac{\nu}{2}) - G(s+\tfrac{\nu}{2})],\\ P(s) &= \tfrac{1}{2}G'(s-\tfrac{\nu}{2})G'(s+\tfrac{\nu}{2})\,\tan q(s)\sec^3 q(s)\,\sin k(s). \end{aligned} \tag{5.4}$$

This fixes the holomorphic expression without ambiguity of a fractional power. P is odd and π-antiperiodic. If $k = s + \mathfrak{k}$, then $\mathfrak{k}$ is odd and π-periodic with $|\mathfrak{k}|_{\rho,1} \le \varsigma$.

Lemma 5.2 (Division by sin)**.** *For an odd sine series f one has $|f/\sin|_\rho \le |f'|_\rho$. Moreover $P(s) = \sin s\, B(s)$ with $|B - 4\sqrt{3}|_\rho < \frac{1}{10}$, B^{-1} exists with norm < 1, and for odd π-antiperiodic Y*

$$|Y/P|_{\rho-\eta} \le \eta^{-1}|Y|_\rho \qquad (0 < \eta \le 1). \tag{5.5}$$

Proof. For $n \ge 1$ the Laurent polynomial $\sin(ns)/\sin s$ has n coefficients of modulus one with exponents between $-(n-1)$ and $n-1$; this gives the first claim. By $\sin(s+\mathfrak{k}) = \sin s\cos\mathfrak{k} + \cos s\sin\mathfrak{k}$,

$$\frac{\sin k(s)}{\sin s} = \cos\mathfrak{k} + \cos s\,\frac{\mathfrak{k}}{\sin s}\,\operatorname{sinc}\mathfrak{k},$$

which is close to 1 in the same algebra without loss of width: $|\cos s|_\rho = \cosh\rho < 2$, $|\cos\mathfrak{k} - 1| \le \varsigma^2$, $|\operatorname{sinc}\mathfrak{k}| \le 2$, so its difference from 1 is at most 5ς. Next $|\tan q\sec^3 q - 8\sqrt{3}| < 1/20$: the quantities $\sin q - \sin\alpha$ and $\cos q - \cos\alpha$ are at most 3ς in norm, the inverse of $\cos q$ has norm < 3, and along the joining segment the derivative of $w \mapsto w^{-4}$ is bounded by $4\cdot 3^5$, whence the difference is at most $3\varsigma\cdot 3^4 + 4\cdot 3^5\cdot 3\varsigma < 1/20$. Expanding $G'_-G'_+$ as 1 plus a term of norm $\le 2\varsigma+\varsigma^2$ and multiplying these bounds gives $|B - 4\sqrt{3}| < 1/10$. The Neumann series gives B^{-1}. Thus P has exactly the zeros of $\sin s$ on the working strip, all simple, and (5.5) follows from the first claim and the Cauchy estimate. □

Lemma 5.3 (Derivative identity)**.** *Let v be odd and π-periodic and $Y = P\,\Delta_\nu v$. Then*

$$\begin{aligned} &DE(G,\lambda)[G'v, \delta\lambda](y)\\ &\qquad = -\frac{2\pi}{G'(y)}\big[Y(y-\tfrac{\nu}{2}) - Y(y+\tfrac{\nu}{2})\big] + E'(y)v(y) - \delta\lambda\cos G(y). \end{aligned} \tag{5.6}$$

Proof. Direct differentiation of (5.2). With $X = G(y)$, $A = A_-(y)$, $D = A_+(y)$, the partial derivatives of E with respect to the two neighboring values $G(y\mp\nu)$ are $\pi\tan A\sec^3 A\sin(X+A)$ and $\pi\tan D\sec^3 D\sin(X-D)$.

Subtracting $E'(y)v(y)$ the central derivative cancels, and the neighboring coefficients times $v(y \pm \nu) - v(y)$ are exactly the two terms built from (5.4). □

5.3. **Two differences, one scalar.** For the current E define

$$Q_0(s) = \frac{G'(s+\frac{\nu}{2})E(s+\frac{\nu}{2})}{2\pi}, \qquad Q_\lambda(s) = -\frac{G'(s+\frac{\nu}{2})\cos G(s+\frac{\nu}{2})}{2\pi}. \tag{5.7}$$

Both are π-antiperiodic, hence of zero mean, and satisfy $Q(-s-\nu) = Q(s)$. Their unique zero-mean solutions Y_0, Y_λ of $D_\nu Y = Q$ are odd and π-antiperiodic (apply $s \mapsto -s$ and change sign; the reflection identity preserves the equation and uniqueness gives oddness). Define the scalar

$$\Lambda(G,\nu) = \operatorname{mean}(Y_\lambda / P), \tag{5.8}$$

and assume $|\Lambda| \geq b_*$. Put

$$\begin{gathered} \delta\lambda = -\frac{\operatorname{mean}(Y_0/P)}{\Lambda}, \qquad H = \frac{Y_0 + \delta\lambda\, Y_\lambda}{P}, \\ \Delta_\nu v = H, \qquad \operatorname{mean} v = 0, \qquad \delta G = G'v. \end{gathered} \tag{5.9}$$

The numerator of H is odd and π-antiperiodic and vanishes at the zeros of sin, so H is even and π-periodic and by construction of zero mean; its centered-difference inverse is odd and π-periodic. Everything preserves reality. Substituting (5.7)–(5.9) in (5.6) gives

$$DE[\delta G, \delta\lambda] = -E + E'v, \tag{5.10}$$

and there is no other compatibility condition in these symmetry spaces.

5.4. **Estimates.** Put $\rho_* = \rho_0/4$, let ρ range over $[\rho_0/2, \rho_0]$, and put $e = |E|_\rho$. The zero-mean inverses of D_ν and Δ_ν cost $C\gamma^{-1}\eta^{-1}$ in passing from ρ to $\rho - \eta$: divide each Fourier coefficient by $1 - e^{in\nu}$, resp. $2i\sin(n\nu/2)$, and use (4.2) and $\sup_{n\geq 1} n e^{-\eta n} \leq \eta^{-1}$. Using in succession the cohomological inverse from ρ to $\rho - \eta$, (5.5), and the second cohomological inverse, there are constants $C_p, C_v, C_D, C_B, C_L \geq 1$ independent of m, γ and the iterate such that for $0 < \eta \leq \min(1, \rho/4)$:

$$|\delta\lambda| \leq C_p \gamma^{-1} e, \tag{5.11}$$

$$|v|_{\rho-3\eta} + |\delta G|_{\rho-3\eta} \leq C_v \gamma^{-3}\eta^{-3} e, \tag{5.12}$$

$$|\delta G|_{\rho-4\eta,1} \leq C_D \gamma^{-3}\eta^{-4} e, \tag{5.13}$$

$$|E(G+\delta G, \lambda+\delta\lambda)|_{\rho-4\eta} \leq C_B \gamma^{-6}\eta^{-6} e^2, \tag{5.14}$$

whenever the correction segment stays in (5.1). In (5.13) the whole product $G'v$ is differentiated on one further strip, so no bound on G'' on the original boundary is needed. For (5.14), Taylor's formula for the analytic map (5.2), (5.10) and the Cauchy estimate $|E'|_{\rho-\eta} \leq \eta^{-1} e$ show that the Taylor term is bounded by a constant times $(|\delta G| + |\delta\lambda|)^2$, while the retained term $E'v$ is at most $C\gamma^{-3}\eta^{-4}e^2$; both are bounded by the right side of (5.14).

Scalar stability. For two maps G in the class on any strip of width $\geq \rho_0/2$, compare Q_λ on ρ_* in the Wiener norm; the difference is $\leq C|G_{\mathrm{new}} - G|_{\rho_*,1}$. Apply the cohomological inverse from ρ_* to $\rho_*/2$ and division by sin from $\rho_*/2$ to $\rho_*/4$. The multiplier B of Lemma 5.2 changes by at most $C|G_{\mathrm{new}} - G|_{\rho_*,1}$, since the quotient of the variation of $\mathfrak{k}$ by sin is controlled by the first part of Lemma 5.2 and all other factors are analytic functions of g, g'. Expanding the difference of $(Y_\lambda/\sin)B^{-1}$ gives

$$|\Lambda(G_{\mathrm{new}}, \nu) - \Lambda(G, \nu)| \leq C_L \gamma^{-1} |G_{\mathrm{new}} - G|_{\rho_*,1}, \tag{5.15}$$

since $Y_\lambda/\sin$ is itself $O(\gamma^{-1})$ on fixed smaller strips. All widths here are fixed independently of the iteration and strictly below $\rho_0/2$.

5.5. **The Newton lemma.** Put $A_D = C_D(16/\rho_0)^4/(1 - 2^{-6})$ and $A_p = C_p/(1 - 2^{-10})$, and let $c_* > 0$ satisfy

$$c_* \leq \frac{2^{-10}}{C_B(16/\rho_0)^6}, \qquad c_* \leq \frac{\varsigma}{2A_D}, \qquad c_* \leq \frac{b_*}{C_L A_D}, \qquad c_* \leq \frac{1}{2A_p}. \tag{5.16}$$

Lemma 5.4 (Quantified Newton lemma)**.** *Assume* (4.2) *with* $0 < \gamma \leq 1$ *and* $|\nu + 2\pi/3| \leq \varsigma$. *Let* g_0 *be real, odd,* π*-periodic and* λ_0 *real with*

$$\begin{gathered} |g_0|_{\rho_0,1} \leq \varsigma/2, \qquad |\lambda_0 - \lambda_{\mathrm{b}}| \leq \tfrac{1}{2}, \\ |\Lambda(G_0, \nu)| \geq 2b_*, \qquad e_0 = |E(G_0, \lambda_0)|_{\rho_0} \leq c_* \gamma^6. \end{gathered} \tag{5.17}$$

Then the corrections (5.9) *converge on* $\rho_0/2$ *to real* $G_\infty = \mathrm{id} + g_\infty$ *and* λ_∞ *with* $E(G_\infty, \lambda_\infty) = 0$, *with preserved symmetries,* $|g_\infty|_{\rho_0/2,1} \leq \varsigma$ *and*

$$|G_\infty - G_0|_{\rho_0/2,1} \leq A_D \gamma^{-3} e_0, \qquad |\lambda_\infty - \lambda_0| \leq A_p \gamma^{-1} e_0. \tag{5.18}$$

Proof. Put $\rho_j = \rho_0/2 + \rho_0/2^{j+1}$ and $\eta_j = (\rho_j - \rho_{j+1})/4 = \rho_0 2^{-j-4}$; then $\eta_j \leq 1$ and $\eta_j < \rho_j/4$. We show inductively $e_j \leq e_0 2^{-10j}$, together with the class conditions along every correction segment, $|\lambda_j - \lambda_{\mathrm{b}}| \leq 1$ and $|\Lambda(G_j, \nu)| \geq b_*$. By (5.14) the ratio of the bound for e_{j+1} to $e_0 2^{-10(j+1)}$ is at most $C_B(16/\rho_0)^6 \gamma^{-6} e_0 2^{10-4j} \leq 1$ by the first inequality in (5.16). The total C^1 correction on each later strip is at most $C_D(16/\rho_0)^4 \gamma^{-3} e_0 \sum_j 2^{-6j} = A_D \gamma^{-3} e_0 \leq A_D c_* \leq \varsigma/2$; the total parameter change is at most $A_p \gamma^{-1} e_0 \leq A_p c_* \leq \frac{1}{2}$; and the total scalar change, by (5.15), is at most $C_L A_D \gamma^{-4} e_0 \leq C_L A_D c_* \leq b_*$. Thus all conditions persist, and the estimates used for the next step are justified. The corrections converge absolutely in the C^1 Wiener norm on $\rho_0/2$ and in the parameter; continuity of E on that space and $e_j \to 0$ give $E = 0$; summing proves (5.18). □

The constant c_* in (5.16) depends only on the class (5.1); it is independent of m, γ and the iterate. The lemma is applied along a sequence for which $e_0/\gamma^6 \to 0$.

6. Proof of the exact sequence theorem

6.1. **The nonvanishing scalar.** For the prepared G_m of (4.7), the expansion (4.9) in a fixed Wiener norm, together with its derivative, gives

$$\sin G_m(y) = \big(1 - \tfrac{z_m}{12}\big)\sin y - \tfrac{z_m}{12}\sin 3y + O(z_m^2).$$

Since

$$Q_\lambda(s) = -\frac{(\sin G_m)'(s+\nu_m/2)}{2\pi},$$
$$D_\nu \sin(ns) = -2\sin(n\nu/2)\cos\big(n(s+\nu/2)\big),$$

the first-order terms give the sine coefficients of Y_λ

$$Y_1 = \frac{1 - z_m/12}{4\pi\sin(\nu_m/2)}, \qquad Y_3 = -\frac{z_m}{16\pi\sin(3\nu_m/2)}. \tag{6.1}$$

As $m \to \infty$, $\sin(3\nu_m/2) = -\frac{\sqrt{3}}{16} z_m + O(z_m^2)$, whence

$$Y_1 \to -\frac{1}{2\pi\sqrt{3}}, \qquad Y_3 \to \frac{1}{\pi\sqrt{3}}. \tag{6.2}$$

The entire remaining tail of Q_λ is $O(z_m^2)$ in the initial Wiener norm. On any fixed smaller strip its cohomological inverse and the subsequent division by sin cost at most $C\gamma_m^{-1}$, so the whole contribution of the tail is $O(z_m^2/\gamma_m) = O(z_m)$; this uses $\gamma_m \geq c'_* z_m$ and estimates all Fourier modes, including all higher near-resonances. Further, $B = P/\sin = 4\sqrt{3} + O(z_m)$ with inverse converging in the Wiener norm on fixed smaller strips, and $\mathrm{mean}(\sin s/\sin s) = \mathrm{mean}(\sin 3s/\sin s) = 1$. Combining,

$$\Lambda(G_m, \nu_m) = \frac{1}{4\sqrt{3}}\Big(-\frac{1}{2\pi\sqrt{3}} + \frac{1}{\pi\sqrt{3}}\Big) + O(z_m) = \frac{1}{24\pi} + O(z_m), \tag{6.3}$$

so eventually $|\Lambda| \geq 1/(48\pi) = 2b_*$, which is the condition required in (5.17). The frequency-three coefficient contributes to the leading value: it comes from the first-order term $-\frac{z_m}{6}\sin 2y$ of G_m, divided by the small denominator $\sin(3\nu_m/2)$ of order z_m.

6.2. **Application of the Newton lemma.** Take $G_0 = G_m$, $\nu = \nu_m$, $\gamma = \gamma_m$, $\lambda_0 = \lambda_{z_m^*}$. By (4.9), (4.10), (6.3) and $z_m \to 0$ all smallness and scalar assumptions of (5.17) hold eventually; by (3.5) and Section 4, $e_0 = O(z_m^{13})$, while γ_m is comparable to z_m. Hence $e_0/\gamma_m^6 = O(z_m^7) \to 0$ and Lemma 5.4 applies for all m above a common m_0. Write the exact pair as (G_m^∞, λ_m). On the strip $\rho_0/2$,

$$G_m^\infty - G_m = O(z_m^{10}) \text{ in } C^1 \text{ Wiener norm}, \qquad \lambda_m - \lambda_{z_m^*} = O(z_m^{12}). \tag{6.4}$$

Shrinking to a smaller fixed strip gives the same order for every fixed number of derivatives. The maps k reconstructed via (5.3) differ by $O(z_m^{10})$ from k_m, their derivatives are bounded away from zero, and their inverses on the real circle and the compositions differ by $O(z_m^{10})$ in every fixed C^j norm (use one additional fixed analytic strip for derivatives). Consequently the

reconstructed meridian lift h_m differs from $h_{z_m^*}$ by $O(z_m^{10})$ in real C^j; in particular

$$h_m(t) = t - \alpha + \sqrt{3}\, z_m\big[\tfrac{1}{48} + \tfrac{\cos 2t}{12}\big] + O(z_m^2). \tag{6.5}$$

The $\cos 2t$ coefficient is $\sqrt{3}z_m/12 + O(z_m^2)$, so $\psi_m = t - h_m$ is nonconstant for all large m and tends to α in every fixed C^j norm. The chosen branch has $0 < \psi_m < \pi/2$, so sec is injective on its range, and the radial function is nonconstant as well.

6.3. **Global geometry.** The analytic inverse function theorem on a fixed smaller strip applied to k in (5.3) shows that h_m and ψ_m are real analytic, with ψ_m even and π-periodic, with uniform holomorphic bounds on a fixed strip and convergence to α. Thus the radial function $r_m(t) = \sec \psi_m(t)$ is even, π-periodic, analytic and close to 2. By Lemma 2.1 it is real analytic on all of S^3, and $r_m \to 2$ in every fixed $C^j(S^3)$ norm. Let $K_m = \{su : u \in S^3, 0 \le s \le r_m(u)\}$. By Lemma 2.2, K_m is strictly convex with analytic boundary and B^4 lies in its interior. Origin symmetry implies that a Euclidean ball with this radial function would be centered at the origin, and nonconstancy in (6.5) excludes it.

6.4. **Conclusion.** By Lemma 5.1, $E(G_m^\infty, \lambda_m) = 0$ implies $W(h_m) = \lambda_m \cos$ on $\mathbb{R}$. Proposition 3.1, applied to the strictly convex analytic body K_m, gives that all distance-one section volumes equal λ_m. Positivity of λ_m and $\lambda_m \to \lambda_{\mathrm{b}}$ follow from (6.4) and the definition of λ_z. This proves Theorem 1.1. □

7. Positive measure of the volume set in dimension four

In this section the rotation number varies in a whole interval, so the single-frequency scheme of Section 5 is carried out for families indexed by a compact set of rotation numbers, with weighted Lipschitz norms. We keep the notation of Part I: ε is the width of the rotation interval, θ the unit-circle rotation, $\nu = 2\pi\theta$, ρ an angular complex strip width, z the parameter of the finite body, and λ the actual section volume. The rotation θ of the endpoint map is different from the angle that [7] computes from λ.

7.1. **A compact frequency set of positive measure.** Let $\theta_0 = -1/3$ and $I_\varepsilon^+ = [\theta_0 + \varepsilon, \theta_0 + 2\varepsilon]$ or $I_\varepsilon^- = [\theta_0 - 2\varepsilon, \theta_0 - \varepsilon]$, with ε small enough that the chosen interval lies in $(-1, 0)$. Put $\gamma = \varepsilon/16$ and

$$\Theta_\varepsilon = I_\varepsilon \cap \{\theta : \ \|n\theta\| \ge \gamma/n^2 \text{ for all } n \ge 1\}.$$

This set is compact and contains only irrationals. For each n the excluded subset of one unit interval consists of n circle intervals of length $2\gamma/n^3$, so has measure at most $2\gamma/n^2$; countable subadditivity gives excluded measure at most $(\pi^2/3)\gamma < \varepsilon/4$. Hence

$$\mathrm{Leb}(\Theta_\varepsilon) \ge \tfrac{3\varepsilon}{4}, \qquad |1 - e^{in\nu}| \ge \frac{4\gamma}{n^2} \quad (n \ne 0,\ \nu = 2\pi\theta \in 2\pi\Theta_\varepsilon). \tag{7.1}$$

The measure of the set of actual volumes is obtained from this in Section 7.8.

7.2. **A Lipschitz family of approximate conjugacies.** Use the finite order-12 initializer h_z and $\lambda_z = \lambda_{\mathrm{b}} + \sqrt{3}\pi z$ of Section 3.2. The geometric residual $R_z = W(h_z) - \lambda_z \cos$ is holomorphic on a fixed strip with $R_z = O(z^{13})$, $\partial_z R_z = O(z^{12})$, and (4.4) gives k_z and $\bar{\nu}(z)$. For $\theta \in I_\varepsilon$ let $z_0(\theta)$ be the real inverse of $\bar{\nu}(z) = 2\pi\theta$. It is analytic on an interval containing both intervals $I_\varepsilon^{\pm}$, $z_0 = O(\varepsilon)$, $\mathrm{Lip}(z_0) = O(1)$ and

$$z_0'(\theta) = \frac{48\pi}{\sqrt{3}} + O(\varepsilon). \tag{7.2}$$

All constants below are uniform in ε after fixing the sign; one may take the maximum over the two signs. The initial circle error $e(\theta) = \mathcal{S}_{z_0} \circ k_{z_0} - k_{z_0} \circ T_{2\pi\theta}$ has supremum norm $O(\varepsilon^3)$ and Lipschitz constant $O(\varepsilon^2)$ on fixed smaller strips, as it is an analytic function of z vanishing through order two and z_0 has bounded derivative.

We apply [4, Theorem 4.1] and its proof to $F_z(x) = \mathcal{S}_z(2\pi x)/(2\pi)$, $H_\theta(x) = k_{z_0(\theta)}(2\pi x)/(2\pi)$ and the parameter $z_0(\theta)$, with the fixed widths of (4.5). Here the strip loss δ of [4] is fixed and unrelated to ε. Condition G1 asks for family derivatives through order three, supplied by the finite holomorphic family and the inverse-domain buffers. Condition G2 holds with $\tau = 2$ and $\gamma = \varepsilon/16$ by (7.1). Condition H1 has uniformly bounded first derivative, reciprocal and second derivative with fixed slack. Condition H2 has base mean parameter derivative $\sqrt{3}/(48\pi) > 0$ and the same slack as in Section 4. Condition H3 holds with fixed bounds strictly above $\mathrm{Lip}(H - \mathrm{id})$, $\mathrm{Lip}(\partial_x H)$ and $\mathrm{Lip}(z_0)$, all $O(1)$. The parameter stays in the middle half of the fixed interval and the image margins are unchanged. The weighted initial error $\max\{\sup|e|, \gamma\delta^{\tau+1}\,\mathrm{Lip}(e)\}$ is $O(\varepsilon^3)$. The constants of the proof of [4, Theorem 4.1] are bounded uniformly as $\gamma \to 0$: after their explicit inverse powers, the quantities in [4, (4.9)–(4.13), (4.20), (4.31), (4.34), (4.37)] carry nonnegative powers of γ, and the fixed H3 slacks keep [4, (4.39)] bounded. Both iteration ratios of that proof are $O(\varepsilon^3/\gamma^2) = O(\varepsilon)$, so the sums occurring there, including the negative-index sums in [4, (4.33) and (4.35)], are bounded by fixed constants for small ε. The parameter budget is treated as in Remark 4.2.

Thus there are a Lipschitz family of exact normalized conjugacies k_θ and parameters $z^*(\theta)$ with, on fixed angular strips,

$$\begin{aligned} &\sup|z^* - z_0| = O(\varepsilon^3), \qquad \mathrm{Lip}(z^* - z_0) = O(\varepsilon^2), \\ &\sup\|k - k_{z_0}\| = O(\varepsilon^2), \qquad \mathrm{Lip}(k - k_{z_0}) = O(\varepsilon). \end{aligned} \tag{7.3}$$

The same orders hold for any fixed number of spatial derivatives after a further fixed strip restriction. For the parameter Lipschitz bound one uses [4, (4.40)], whose correction is of the order of the weighted error divided by γ, i.e. $O(\varepsilon^2)$; for the conjugacy one uses [4, (4.33), (4.35)], whose correction is of the order of the weighted error divided by γ^2, i.e. $O(\varepsilon)$. Normalization, uniqueness of irrational conjugacies, reversibility and π-equivariance give

the same odd, π-equivariant geometry as in Proposition 4.3 for every θ. Put

$$G_\theta(y) = h_{z^*(\theta)}\big(k_\theta(y-\nu/2)\big), \qquad \lambda_{\mathrm{pre}}(\theta) = \lambda_{\mathrm{b}} + \sqrt{3}\pi z^*(\theta).$$

Then G_θ is odd and π-equivariant, reconstructs h_{z^*} exactly, and on a common complex strip

$$G_\theta = \mathrm{id} - \tfrac{z_0}{6}\sin 2y + \mathcal{E}_\theta, \qquad \sup|\mathcal{E}| = O(\varepsilon^2), \quad \mathrm{Lip}(\mathcal{E}) = O(\varepsilon). \tag{7.4}$$

All fixed-derivative versions hold on smaller strips. Suprema and Lipschitz seminorms transfer to Wiener norms by the same fixed Fourier coefficient buffer as in Section 4, applied also to difference quotients. No differentiability of the exact family in θ is needed.

7.3. **Weighted algebra and a shift-stable class.** For a family f_θ of periodic functions on a compact set Θ_ε write

$$\mathrm{Lip}_\rho(f) = \sup_{\theta\neq\theta'} \frac{|f_\theta - f_{\theta'}|_\rho}{|\theta-\theta'|}, \qquad N_\rho(f) = \sup_\theta |f_\theta|_\rho + \varepsilon\,\mathrm{Lip}_\rho(f).$$

Thus Lip_ρ is the Lipschitz seminorm of $\theta \mapsto f_\theta$ with values in the Wiener space, not the Lipschitz seminorm of the scalar function $\theta \mapsto |f_\theta|_\rho$. In the same way $\mathrm{Lip}_\Theta(g; |\cdot|_\rho + |\cdot'|_\rho)$ is the supremum of $(|g_\theta - g_{\theta'}|_\rho + |g'_\theta - g'_{\theta'}|_\rho)/|\theta-\theta'|$. The norm N is submultiplicative, since the two product Lipschitz terms are covered by the product of the two N norms, constant scalar families use the same definition, and Cauchy's estimate gives $N_{\rho-\eta}(f') \le \eta^{-1} N_\rho(f)$. For $g = G - \mathrm{id}$ use the stronger mixed control

$$S_\rho(g) = \sup_\theta\big(|g|_\rho + |g'|_\rho + |g''|_\rho\big) + \varepsilon\ \mathrm{Lip}_\Theta\big(g; |\cdot|_\rho + |\cdot'|_\rho\big), \tag{7.5}$$

that is, supremum C^2 plus weighted Lipschitz C^1. For a fixed l with $|l| \le 1$, $\varepsilon\,\mathrm{Lip}[g'_\theta(\cdot + l\nu(\theta))] \le \varepsilon\,\mathrm{Lip}(g') + 2\pi|l|\varepsilon\sup|g''|$, and the analogous bound for g itself needs only g'; hence translated g' factors and their differences are controlled by S, with constants independent of $\varepsilon \le 1$. The affine part of G contributes only the bounded scalar slope $2\pi l$, separated from periodic functions. For a correction u,

$$N(u(\cdot + l\nu)) \le N(u) + 2\pi|l|\,\varepsilon\sup|u'| \le C\,S(u). \tag{7.6}$$

Take the core width $\rho_0 = 1/128$ and a small fixed $\varsigma_1 > 0$ so that $S(g) \le \varsigma_1$ implies the class (5.1). Require also $\varepsilon \le \varsigma_1/(4\pi)$, $N(\lambda - \lambda_{\mathrm{b}}) \le 1$ and $|\nu + 2\pi/3| \le \varsigma$. The small initial family satisfies these with arbitrarily large relative slack as $\varepsilon \to 0$.

For $k = \frac{1}{2}(G(\cdot - \frac{\nu}{2}) + G(\cdot + \frac{\nu}{2}))$ and $q = \frac{1}{2}(G(\cdot - \frac{\nu}{2}) - G(\cdot + \frac{\nu}{2}))$ define P exactly as in (5.4), so that $P = \sin(s)\,B$. Pointwise in θ, Lemma 5.2 gives $|B - 4\sqrt{3}| < \frac{1}{10}$ and B^{-1} bounded. In the weighted norm the same expressions have fixed bounds under $S(g) \le \varsigma_1$: the removable division of $k - \mathrm{id}$ by sin uses the weighted C^1 bound just established, and all other products and analytic functions use submultiplicativity. Thus $N(B), N(B^{-1}) \le C$

uniformly in ε, θ and the iterate. The reciprocal Lipschitz bound follows from $B_\theta^{-1} - B_{\theta'}^{-1} = B_\theta^{-1}(B_{\theta'} - B_\theta)B_{\theta'}^{-1}$. Similarly $N(B_{\mathrm{new}} - B_{\mathrm{old}}) \le CS(G_{\mathrm{new}} - G_{\mathrm{old}})$, with constants depending only on the fixed strip and class.

7.4. **Weighted difference inverses.** Consider the zero-mean Fourier inverse of either $D_\nu f = f - f(\cdot + \nu)$ or the centered difference $f(\cdot + \frac{\nu}{2}) - f(\cdot - \frac{\nu}{2})$ (absorb the half-shift into the first multiplier). In each case the multiplier $\mathfrak{m}_n(\theta)$ satisfies

$$|\mathfrak{m}_n| \le C\gamma^{-1}n^2, \qquad \frac{|\mathfrak{m}_n(\theta) - \mathfrak{m}_n(\theta')|}{|\theta - \theta'|} \le C\gamma^{-2}n^5 \qquad (\theta, \theta' \in \Theta_\varepsilon). \tag{7.7}$$

For the first inverse subtract the two endpoint fractions: the numerator exponential difference is $\le 2\pi n|\theta - \theta'|$ and both endpoint denominators obey (7.1); the half-shift adds only a term $\le C\gamma^{-1}n^3$; the centered sine multiplier has the same endpoint-product estimate. Only the two endpoints $\theta, \theta' \in \Theta_\varepsilon$ are used. Since $\varepsilon/\gamma = 16$, for $0 < \eta \le 1$ the Fourier sum and $\sup_t t^j e^{-\eta t} \le C_j \eta^{-j}$ give

$$N_{\rho-\eta}(\text{inverse of } f) \le C\gamma^{-1}\eta^{-5}N_\rho(f). \tag{7.8}$$

For an odd π-antiperiodic family, by the coefficient identity for $\sin(ns)/\sin s$,

$$N_{\rho-\eta}(f/\sin) \le \eta^{-1}N_\rho(f). \tag{7.9}$$

All frequencies, including arbitrarily high near-resonances, are covered.

7.5. **Weighted scalar and one quasi-Newton step.** The equation and the derivative identity are those of Section 5, separately for every θ, and preserve the odd, π-equivariant shape. Let $\mathcal{C}_\nu$ denote the first inverse composed with its half-shift. Then $Y_0 = \mathcal{C}_\nu(G'E/(2\pi))$, $Y_\lambda = \mathcal{C}_\nu(-G'\cos G/(2\pi))$ and $\Lambda = \mathrm{mean}(Y_\lambda/P)$. Use fixed inner strips below $\rho_0/2$ to define and estimate scalar means; these gaps do not accumulate. Assume initially $\inf|\Lambda| \ge 2b_*$ with $b_* = 1/(96\pi)$ and $N(\Lambda) \le L_\Lambda$ for a fixed L_Λ; we maintain $\inf|\Lambda| \ge b_*$ and $N(\Lambda) \le L_\Lambda + b_*$, so $N(1/\Lambda) \le 1/b_* + (L_\Lambda + b_*)/b_*^2$ is fixed. The correction is (5.9). All zero means and reflection symmetries hold exactly as before, being algebraic identities at each θ.

Let $e = N_\rho(E)$ with $\rho_0/2 \le \rho \le \rho_0$. By (7.8) and (7.9) with fixed gaps, $N(\delta\lambda) \le C_p\gamma^{-1}e$. For the shape correction pay three successive losses η (first inverse $\gamma^{-1}\eta^{-5}$, sine division η^{-1}, second inverse $\gamma^{-1}\eta^{-5}$), with one extra γ^{-1} in the $\delta\lambda\, Y_\lambda$ term; hence $N_{\rho-3\eta}(\delta G) \le C\gamma^{-3}\eta^{-11}e$. Two further losses η supply the supremum second derivative and the weighted Lipschitz first derivative:

$$S_{\rho-5\eta}(\delta G) \le C_D\gamma^{-3}\eta^{-13}e. \tag{7.10}$$

The principal derivative cancels E exactly and leaves $E'v$, with weighted bound $\le C\gamma^{-3}\eta^{-12}e^2$. For the Taylor remainder, E is an analytic function of three translated values of G and of λ with coefficient families of bounded weighted norms on the input class; by (7.6), translated correction norms are bounded by $S(\delta G)$ and the scalar correction by $N(\delta\lambda)$, so the quadratic

remainder is $\le C[S(\delta G)+N(\delta\lambda)]^2$. A conservative bound for the new residual is

$$N_{\rho-5\eta}(E_{\mathrm{new}}) \le C_B\gamma^{-6}\eta^{-26}e^2. \tag{7.11}$$

Finally, comparing two families G at the same θ, using fixed inner strips, (7.8), (7.9) and the variation bound for B,

$$N(\Lambda_{\mathrm{new}}-\Lambda_{\mathrm{old}}) \le C_L\gamma^{-1}S_{\rho_*}(\delta G), \qquad \rho_*=\rho_0/4, \tag{7.12}$$

where $Y_\lambda/\sin$ has norm $O(\gamma^{-1})$ on fixed gaps. This gives the weighted scalar stability, including the reciprocal Lipschitz control.

7.6. **A strip induction for the weighted scheme.** Put $\rho_j=\rho_0/2+\rho_0/2^{j+1}$ and $\eta_j=(\rho_0/20)2^{-j}$, so that $\rho_j-\rho_{j+1}=5\eta_j$; enlarge C_p, C_D, C_B, C_L to at least one, and set $A_D=C_D(20/\rho_0)^{13}/(1-2^{-51})$, $A_p=C_p/(1-2^{-64})$. Choose a fixed $c_*>0$ with

$$c_* \le \frac{2^{-64}}{C_B(20/\rho_0)^{26}}, \quad c_* \le \frac{\varsigma_1}{2A_D}, \quad c_* \le \frac{1}{2A_p}, \quad c_* \le \frac{b_*}{C_LA_D}. \tag{7.13}$$

Lemma 7.1. *Assume $S(g_0)\le\varsigma_1/2$, $N(\lambda_0-\lambda_{\mathrm{b}})\le\frac12$, $\inf|\Lambda_0|\ge 2b_*$, $N(\Lambda_0)\le L_\Lambda$ and $e_0\le c_*\gamma^6$. Then the corrections converge in supremum C^2 plus weighted Lipschitz C^1 on $\rho_0/2$, λ converges in its weighted Lipschitz norm, $E=0$ in the limit for every θ, and*

$$N(\lambda_\infty-\lambda_0)\le A_p\gamma^{-1}e_0, \qquad \mathrm{Lip}(\lambda_\infty-\lambda_0)\le A_p\gamma^{-1}e_0/\varepsilon. \tag{7.14}$$

Proof. By (7.11) one shows inductively $e_j\le e_0 2^{-64j}$: the ratio to the desired bound at $j+1$ is at most $C_B(20/\rho_0)^{26}\gamma^{-6}e_0 2^{64-38j}\le 1$. The shape corrections sum to $\le A_D\gamma^{-3}e_0$, the parameter corrections to $\le A_p\gamma^{-1}e_0$, and the scalar drift is $\le C_LA_D\gamma^{-4}e_0$. Since $\gamma\le 1$, (7.13) maintains all three margins and $N(\Lambda)\le L_\Lambda+b_*$ throughout, which proves the assumptions before each step and along every correction segment. The remaining statements follow as in Lemma 5.4. □

7.7. **Uniform initialization with the entire weighted resonant tail.** Equation (7.4) is valid in the weighted norm: the remainder is $O(\varepsilon^2)$, since its supremum is $O(\varepsilon^2)$ and ε times its Lipschitz constant is $O(\varepsilon^2)$; the same holds after fixed spatial derivatives. The scalar computation of Section 6 is repeated in this norm. With $G=\mathrm{id}-(z_0/6)\sin 2y+O_N(\varepsilon^2)$ and $\nu=\bar\nu(z_0)$, the two leading sine coefficients of $\sin G$ sit at frequencies 1 and 3, the third being $-z_0/12+O_N(\varepsilon^2)$. Solving the first difference gives for Y_λ the coefficient $na_n/(4\pi\sin(n\nu/2))$, where a_n is the n-th sine coefficient of $\sin G$. Since $\sin(3\nu/2)=-\frac{\sqrt3}{16}z_0+O(z_0^2)$, the third leading quotient tends to $1/(\sqrt3\pi)$ and the first to $-1/(2\sqrt3\pi)$. The smooth ratios $z_0/\sin(3\nu/2)$ have weighted errors $O(\varepsilon)$ uniformly on each interval, including their sign, and their difference quotients are bounded by analytic cancellation at $z_0=0$. The entire differentiated remainder has weighted norm $O(\varepsilon^2)$, so by (7.8),

(7.9) across fixed gaps its total scalar contribution is $O(\varepsilon^2/\gamma) = O(\varepsilon)$, which controls every higher resonance. Since moreover $B = 4\sqrt{3} + O_N(\varepsilon)$,

$$N\big(\Lambda - \tfrac{1}{24\pi}\big) = O(\varepsilon). \tag{7.15}$$

Hence $\inf|\Lambda| \geq 2b_*$ and $N(\Lambda) \leq L_\Lambda$ with a fixed L_Λ for small ε; the weighted bound (7.15) also controls the Lipschitz seminorm of $1/\Lambda$.

The prepared geometric residual is $E_{\mathrm{pre}}(\theta) = R_{z^*(\theta)} \circ G_\theta$. On fixed strips $R_z = O(z^{13})$, $\partial_z R_z = O(z^{12})$ and the spatial derivative is $O(z^{13})$; by (7.3), $\mathrm{Lip}(z^*) = O(1)$ and $\mathrm{Lip}(G) = O(1)$. Therefore $\sup|E_{\mathrm{pre}}| = O(\varepsilon^{13})$, $\mathrm{Lip}(E_{\mathrm{pre}}) = O(\varepsilon^{12})$, and after the fixed supremum-to-Wiener buffers

$$e_0 = N(E_{\mathrm{pre}}) = O(\varepsilon^{13}). \tag{7.16}$$

Since $\gamma = \varepsilon/16$, $e_0/\gamma^6 = O(\varepsilon^7) \to 0$. Also $S(g_0) = O(\varepsilon)$ and $N(\lambda_{\mathrm{pre}} - \lambda_{\mathrm{b}}) = O(\varepsilon)$, so all initial margins hold for small ε, and Lemma 7.1 applies uniformly to every $\theta \in \Theta_\varepsilon$.

7.8. **Noncollapse of the actual volume map.** The prepared volume is $\lambda_{\mathrm{pre}} = \lambda_{\mathrm{b}} + \sqrt{3}\pi z^*$. By (7.2), $\frac{d}{d\theta}[\lambda_{\mathrm{b}} + \sqrt{3}\pi z_0(\theta)] = 48\pi^2 + O(\varepsilon)$ on the whole interval I_ε, so its lower Lipschitz constant on Θ_ε is also $48\pi^2 + O(\varepsilon)$. The Lipschitz contribution of $\sqrt{3}\pi(z^* - z_0)$ is $O(\varepsilon^2)$ by (7.3), and the weighted Newton estimate (7.14), (7.16) gives $\mathrm{Lip}(\lambda_\infty - \lambda_{\mathrm{pre}}) = O(\varepsilon^{11})$. Hence for small ε and distinct $\theta, \theta' \in \Theta_\varepsilon$,

$$|\lambda_\infty(\theta) - \lambda_\infty(\theta')| \geq 24\pi^2|\theta - \theta'|. \tag{7.17}$$

The map λ_∞ is Lipschitz and strictly increasing on this compact set, since both perturbations have smaller Lipschitz constant than the increasing base map. Its range V_ε is compact, and applying the Lipschitz bound to its inverse gives

$$\mathrm{Leb}(V_\varepsilon) \geq 24\pi^2\,\mathrm{Leb}(\Theta_\varepsilon) \geq 18\pi^2\varepsilon > 0. \tag{7.18}$$

The leading expansion $\lambda_\infty(\theta) - \lambda_{\mathrm{b}} = 48\pi^2(\theta - \theta_0) + O(\varepsilon^2)$ also shows that V_ε lies on the chosen side of λ_{b} at distances comparable to ε, namely $\pm(\lambda - \lambda_{\mathrm{b}}) \in [40\pi^2\varepsilon, 100\pi^2\varepsilon]$ for small ε.

7.9. **Proof of the positive-measure theorem in dimension four.** The fixed-strip shape correction is $O(\varepsilon^{10})$, while the prepared meridian is the finite $h_{z^*} = t - \pi/3 + \sqrt{3}z_0R_1 + O(\varepsilon^2)$. The $\cos 2t$ coefficient is $\sqrt{3}z_0/12 + O(\varepsilon^2)$, uniformly nonzero because $|z_0|$ is bounded below by a fixed multiple of ε on either interval; so no output is a ball. The reconstruction, Lemma 2.1, Lemma 2.2 and the integrated section-volume identity of Proposition 3.1 apply pointwise with uniform smallness: the bodies are analytic, globally strictly convex, origin-symmetric bodies of revolution, strictly contain B^4 and satisfy every distance-one section equality with volume $\lambda_\infty(\theta)$. Fixed complex widths give uniform convergence to the ball of radius two in every fixed C^j norm of the radial function. Together with (7.18) this proves Theorem 1.2, with $V_\varepsilon^{\pm}$ the set V_ε for the corresponding sign. □

Part II. Dimensions at least five

Throughout Part II, $d \geq 5$ is a fixed integer, and

$$\lambda = \tfrac{d-2}{2}, \qquad \beta = \lambda - 1 = \tfrac{d-4}{2} > 0, \qquad R_0 = \sqrt{d},$$
$$\theta_0 = \arccos \tfrac{1}{\sqrt{d}}, \qquad \omega_0 = 2\theta_0, \qquad I_h = [\omega_0 + h, \omega_0 + 2h].$$

In this part λ is the Gegenbauer parameter; section volumes are denoted by A or V. For a frequency $\omega = 2\theta$ we write $R = \sec\theta$ for the radius of the round reference ball; the physical mean radius R_* of a body is distinguished from R in Section 12. All constants may depend on d, and no uniformity in d is claimed. We use the zonal polynomials $Y_n = C_n^\lambda / C_n^\lambda(1)$ in the variable $x = \cos t$, where C_n^λ are the Gegenbauer polynomials, normalized by the generating function $(1 - 2xs + s^2)^{-\lambda} = \sum_n C_n^\lambda(x) s^n$ [3, (18.12.4)]. Only even degrees n occur, because all bodies are origin symmetric. We write Π_n for the orthogonal projection onto the multiples of Y_n in $L^2(S^{d-1})$ (physical projections), and Π_0 for the spherical mean.

8. Section operators

8.1. **The section volume as an analytic functional.** Let r be the radial function of a body of revolution close to the ball of radius $R > 1$, written as a function of the axial cosine $s = \langle u, e_1 \rangle$ of $u \in S^{d-1}$. The hyperplane with unit normal $\nu_\phi = (\cos\phi, \sin\phi, 0, \dots, 0)$ at distance one is parametrized by $x = \nu_\phi + q\,\varpi$ with ϖ a unit vector orthogonal to ν_ϕ. Write $z \in [-1, 1]$ for the cosine of the angle between ϖ and the normalized projection of $-e_1$ to $\nu_\phi^\perp$. The point x has $|x| = \sqrt{1+q^2}$ and axial cosine $s = (\cos\phi - q \sin\phi\, z)/\sqrt{1+q^2}$. The boundary of the section in direction (ϕ, z) is the positive root $q = q(\phi, z)$ of

$$\sqrt{1+q^2} = r(s), \qquad s = \frac{\cos\phi - q\sin\phi\, z}{\sqrt{1+q^2}}. \tag{8.1}$$

Lemma 8.1 (Fixed-ray formulas)**.** *Fix $d \geq 4$, a compact interval $\mathcal{J} \subset (1, \infty)$ and a Bernstein ellipse $\mathfrak{E} \supset [-1, 1]$. There are $\delta_2 > 0$ and a complex neighborhood U of $\mathbb{R} \times [-1, 1]$, both depending only on d, $\mathcal{J}$ and $\mathfrak{E}$, with the following property. Let $R \in \mathcal{J}$ and let r be holomorphic on $\mathfrak{E}$, real on $[-1, 1]$, with $\sup_{\mathfrak{E}} |r - R| \leq \delta_2$. Then (8.1) has, for $(\phi, z) \in U$, a unique solution $q(\phi, z)$ with $|q - \sqrt{R^2 - 1}| < \frac{1}{2}\sqrt{R^2 - 1}$; it is holomorphic in (ϕ, z), depends holomorphically on r, and for real (ϕ, z) it is the length of the section ray. The section volume $A_r(\phi) = \mathrm{vol}_{d-1}(K_r \cap \{x \cdot \nu_\phi = 1\})$ and the radial*

derivative $T_r f(\phi) = \frac{d}{d\varepsilon} A_{r+\varepsilon f}(\phi)\big|_{\varepsilon=0}$ *are*

$$A_r(\phi) = \frac{|S^{d-3}|}{d-1} \int_{-1}^{1} q^{d-1}(1-z^2)^{\beta}\, dz,$$

$$T_r f(\phi) = |S^{d-3}| \int_{-1}^{1} \frac{q^{d-2} f(s)}{q/\sqrt{1+q^2} - r'(s)\, \partial_q s} (1-z^2)^{\beta}\, dz, \tag{8.2}$$

$$\partial_q s = \frac{-\sin\phi\, z - q\cos\phi}{(1+q^2)^{3/2}}.$$

Both are holomorphic in ϕ *on the projection of* U*, including* $\phi \in \pi\mathbb{Z}$*. For* $0 < \phi < \pi$ *the equivalent coarea formula in the axial variable* s *is*

$$T_r f(\phi) = \frac{|S^{d-3}|}{(\sin\phi)^{2\lambda-1}} \int r(s)^2\, \mathfrak{D}(\phi, s)^{\beta} f(s)\, ds, \tag{8.3}$$

$$\mathfrak{D} = r(s)^2 \sin^2\phi\,(1-s^2) - (1 - r(s) s \cos\phi)^2,$$

the integral being over the interval where $\mathfrak{D} \geq 0$*.*

Proof. For real (ϕ, z), Cauchy–Schwarz gives $|\cos\phi - q\sin\phi z| \leq \sqrt{1+q^2}$, so $s \in [-1, 1]$; for (ϕ, z) in a small complex neighborhood of the real set, s stays in $\mathfrak{E}$. At the ball, $q_0 = \sqrt{R^2 - 1}$ solves (8.1) for all (ϕ, z), and the derivative of $\sqrt{1+q^2} - r(s)$ in q is $q_0/R > 0$. The holomorphic implicit function theorem, with the uniform bounds on $\mathcal{J}$ and $\mathfrak{E}$, gives δ_2, U and the solution; for real data, convexity of K_r (Lemma 2.2) identifies the root with the section ray. Polar coordinates in the $(d-1)$-dimensional section give $q^{d-1}/(d-1)$ for each direction; integrating over the latitude of the unit sphere S^{d-2} of directions with respect to the axial direction gives the factor $|S^{d-3}|(1-z^2)^{\beta} dz$ with $\beta = (d-4)/2 \geq 0$. Implicit differentiation of (8.1) with $r \to r + \varepsilon f$ gives the denominator in (8.2), which equals q_0/R at the ball and stays nonzero. Since $(1-z^2)^{\beta} dz$ is a finite measure and the integrands are holomorphic, A_r and $T_r f$ are holomorphic on the projection of U. For (8.3) we write the variation of the volume as an integral over the boundary of the section: the normal speed of the boundary point of the section with axial cosine s under $r \to r + \varepsilon f$ is $f(s)$ times the ratio of the area elements, and integrating out the transverse latitude, with Jacobian $1/(r\sin\phi\sqrt{1-s^2})$, leaves exactly $r^2 \mathfrak{D}^{\beta}$. □

Lemma 8.2 (Slice formula)**.** *In the situation of Lemma 8.1, write the meridian curve as* $t \mapsto r(t)(\cos t, \sin t)$*, put* $X(t) = r(t)\cos t$ *and* $\mathfrak{D}(\phi, t) = r(t)^2 \sin^2\phi \sin^2 t - (1 - r(t)\cos t \cos\phi)^2$*. For* $0 < \phi < \pi$*,*

$$A_r(\phi) = \kappa_{d-2}\, (\sin\phi)^{-2\lambda-1} \int \mathfrak{D}(\phi, t)^{\lambda}\, |X'(t)|\, dt, \tag{8.4}$$

the integral being over the t*-interval between the two points where the line* $\{x_1 \cos\phi + x_2 \sin\phi = 1\}$ *meets the meridian curve.*

Proof. Let $x = (x_1, x_2, y) \in \mathbb{R} \times \mathbb{R} \times \mathbb{R}^{d-2}$. On the hyperplane $x_1 \cos\phi + x_2 \sin\phi = 1$ we have $x_2 = (1 - x_1 \cos\phi)/\sin\phi$, and the slice at fixed x_1 of

K_r is $\{y:\ x_2^2+|y|^2\le \varrho(x_1)^2\}$, where $\varrho(x_1)=r(t)\sin t$ is the radius of K_r at the axial coordinate $x_1=r(t)\cos t$. Hence the slice is a $(d-2)$-ball of radius $\sqrt{\mathfrak{D}}/\sin\phi$. The arclength in the hyperplane along the x_1-direction is $|dx_1|/\sin\phi$, and near the ball $t\mapsto X(t)$ is monotone. Integrating the slice volumes gives (8.4), with $(d-2)/2=\lambda$. □

By Lemma 8.1, A_r is holomorphic through $\phi\in\pi\mathbb{Z}$, so the apparent singularity of (8.4) at $\sin\phi=0$ is removable; quantitatively, if the numerator of (8.4) is bounded on a strip, the quotient is bounded on a narrower strip with a loss $C_k\sigma^{-k}$ for a width loss σ and $k=2\lambda+1$, by the maximum principle on circles of radius σ around the zeros of sin.

8.2. **The round operator.** Let $L_R=T_R$ be the derivative at the ball of radius $R>1$.

Lemma 8.3 (Round multipliers)**.** *If $f=\sum_n f_nY_n$ is an even zonal function expanded in zonal polynomials, then*

$$\begin{aligned} L_Rf(\phi)&=\sum_n L_n(R)f_nY_n(\cos\phi),\\ L_n(R)&=|S^{d-2}|\,R(R^2-1)^{\frac{d-3}{2}}\,Y_n(1/R). \end{aligned}\tag{8.5}$$

Proof. The boundary of $B_R\cap H_\phi$ is the $(d-2)$-sphere of radius $\varrho_0=\sqrt{R^2-1}$ about the foot point ν_ϕ. A radial change $\varepsilon f(u)$ changes q by $R\varepsilon f/\varrho_0$ because $|x|^2=1+q^2$. Hence $L_Rf(\phi)=|S^{d-2}|\varrho_0^{d-2}(R/\varrho_0)\,\mathrm{avg}\{f(u):u\in S^{d-1},\langle u,\nu_\phi\rangle=1/R\}$. By the Funk–Hecke formula the average of a zonal harmonic $Y_n(\langle u,e_1\rangle)$ over this latitude sphere equals $Y_n(1/R)Y_n(\langle\nu_\phi,e_1\rangle)$. □

Lemma 8.4 (Radial Taylor formula)**.** *For $r=R+f$ with f analytic and small and every $N\ge1$,*

$$A_{R+f}=A_R+\sum_{j=1}^{N}\frac{1}{j!}\,\partial_R^{j-1}L_R\big(f^j\big)+O\big(\|f\|^{N+1}\big),\tag{8.6}$$

where ∂_R acts only on the R-dependence of the operator L_R and f^j is held fixed; the remainder is bounded in every fixed analytic norm.

Proof. The set $K_r\cap H_\phi$ is the central projection from the origin of $\{u\in S^{d-1}:\langle u,\nu_\phi\rangle\,r(u)\ge1\}$ to H_ϕ, so $A_r(\phi)=\int_{S^{d-1}}\Phi_\phi(r(u),u)\,du$ with $\Phi_\phi(\varrho,u)=\mathbf{1}_{\varrho\langle u,\nu_\phi\rangle\ge1}\langle u,\nu_\phi\rangle^{-d}$, a distribution in ϱ. The j-th Taylor term of $\varrho\mapsto\Phi(\varrho+f(u),u)$ at $\varrho=R$ is $f(u)^j\partial_\varrho^j\Phi/j!$, and $\int f^j\partial_\varrho^j\Phi\,du=\partial_R^{j-1}\big[\int f^j\,\partial_\varrho\Phi\,du\big]$ with f^j independent of R; the bracket is $L_R(f^j)$. The analytic dependence of A_r on r in Lemma 8.1 gives the remainder bound. □

9. Resonances, arithmetic and finite initializers

9.1. **Low-degree identities.** Write c_{nm} for the coefficients in $Y_2 Y_n = \sum_m c_{nm} Y_m$. Put $L_n = L_n(R_0)$ and $L_n' = \partial_R L_n(R_0)$.

Lemma 9.1 (Exact low coefficients)**.** (a) *The first zonal polynomials are*

$$Y_2(x) = \frac{dx^2 - 1}{d-1}, \qquad Y_4(x) = \frac{(d+2)(d+4)x^4 - 6(d+2)x^2 + 3}{(d-1)(d+1)},$$

and

$$Y_2^2 = \frac{2}{(d-1)(d+2)} + \frac{4(d-2)}{(d-1)(d+4)} Y_2 + \frac{d^2(d+1)}{(d-1)(d+2)(d+4)} Y_4. \tag{9.1}$$

(b) *At $R_0 = \sqrt{d}$ one has $Y_2(1/R_0) = 0$, $L_0 = |S^{d-2}| \sqrt{d}\,(d-1)^{(d-3)/2} \neq 0$, $L_2 = 0$ and $L_2' = -2|S^{d-2}|(d-1)^{(d-5)/2} \neq 0$.*

(c) *The values at $1/\sqrt{d}$ of the next two zonal polynomials are*

$$Y_4(1/\sqrt{d}) = -\frac{2(d+4)}{d^2(d+1)} \neq 0, \qquad Y_6(1/\sqrt{d}) = \frac{16(d-2)(d+6)}{d^3(d+1)(d+3)} \neq 0.$$

(d) *The diagonal coefficient of multiplication by Y_2 is*

$$c_{nn} = \frac{\lambda\, n(n+2\lambda)}{(2\lambda+1)\big((n+\lambda)^2 - 1\big)} = \frac{\lambda}{2\lambda+1} - \frac{\lambda(\lambda^2-1)}{(2\lambda+1)\big((n+\lambda)^2-1\big)}, \tag{9.2}$$

strictly increasing in $n \geq 0$ since $\lambda > 1$.

Proof. (a) is a direct computation with the explicit Gegenbauer polynomials. (b) follows from (8.5): $Y_2(1/R) = (d/R^2 - 1)/(d-1)$ has derivative $-2/(\sqrt{d}(d-1))$ at R_0. (c): by the generating function at $x = 1/\sqrt{d}$ put $g_n = d^{n/2} C_n^\lambda(1/\sqrt{d})$, so $\sum g_n s^n = (1 - 2s + ds^2)^{-\lambda}$, and read off g_4, g_6. (d): apply the three-term recurrence $(n+1)C_{n+1} = 2(n+\lambda)xC_n - (n+2\lambda-1)C_{n-1}$ twice; the coefficient of C_n in $x^2 C_n$ is $\frac{(n+1)(n+2\lambda)}{4(n+\lambda)(n+\lambda+1)} + \frac{n(n+2\lambda-1)}{4(n+\lambda)(n+\lambda-1)}$, and substitution into Y_2 gives (9.2). The derivative in n is $2\lambda(n+\lambda)(\lambda^2-1)/[(2\lambda+1)((n+\lambda)^2-1)^2] > 0$. □

Definition 9.2. Put $r_1 = -\frac{1}{2}c_{22} = -\dfrac{2(d-2)}{(d-1)(d+4)}$, $w_1 = r_1 + Y_2$, and

$$a_{4,d} = -\frac{d^{3/2}(d+1)(d^3 - 2d^2 - 9d + 12)}{2(d-1)^2(d+2)(d+4)^2}, \qquad h_2 = a_{4,d}\, Y_4.$$

The value r_1 is the branch direction of the degree-two equation: by (9.1) the quadratic Taylor term of (8.6) along w_1 has degree-two part $\frac{1}{2}L_2'(2r_1 + c_{22})$. Similarly the degree-four equation gives $a_{4,d} = -c_{24}L_4'/(2L_4)$, with c_{24} the Y_4 coefficient in (9.1), and logarithmic differentiation of L_4 at R_0 gives $L_4'/L_4 = (d^3 - 2d^2 - 9d + 12)/[\sqrt{d}(d-1)(d+4)]$. For instance $a_{4,5} = -\frac{5\sqrt{5}}{72}$ and $a_{4,6} = -\frac{1071\sqrt{6}}{10000}$.

9.2. **Resonant degrees.** A degree $n \geq 4$ is *resonant* if $C_n^\lambda(1/\sqrt{d}) = 0$, equivalently $L_n(R_0) = 0$.

Lemma 9.3 (Structure of the round kernel)**.** (a) *Every zero of C_n^λ in $(-1,1)$ is simple; hence $L_n' \neq 0$ at every resonant degree.*

(b) *Two resonant even degrees do not differ by 2; the degrees 4 and 6 are not resonant.*

(c) *For every resonant n the detuning $\delta_n = L_n'(r_1 + c_{nn})$ is nonzero, since $c_{nn} > c_{22}$ gives $r_1 + c_{nn} > c_{22}/2 > 0$. Distinct resonances are not coupled to first order in the direction w_1, nor are they coupled to degree two.*

Proof. (a) An interior zero with zero derivative of a solution of $(1-x^2)y'' - (2\lambda+1)xy' + n(n+2\lambda)y = 0$ forces $y \equiv 0$ by uniqueness for this regular second-order equation. (b) If $C_n(x_0) = C_{n+2}(x_0) = 0$ with $x_0 = 1/\sqrt{d} > 0$ then the recurrence at the intervening degree gives $C_{n+1}(x_0) = 0$, and repeating backward gives $C_0(x_0) = 0$, a contradiction. Degrees 4 and 6 are nonresonant by Lemma 9.1(c). (c) By (9.1) and the recurrence, multiplication by Y_2 changes the degree by at most two, so by (b) it does not couple two distinct resonant degrees, and it does not couple degree two to a resonant degree since resonant degrees are at least eight. □

For odd d the full zero set $\{n : C_n^\lambda(1/\sqrt{d}) = 0\}$ is not classified, and it is not asserted to be finite. For even d it is finite, by Proposition 9.5 below.

9.3. **The Darboux representation in even dimensions.** For even d we have $\lambda \in \mathbb{N}$, and we put $k = \lambda - 1 \geq 1$ and $k_\lambda = 2^k k!$. Let $\mathsf{D}_j = j\cot t - \partial_t$.

Lemma 9.4 (Darboux representation)**.** *Let $d = 2\lambda + 2 \geq 6$ be even and $m \geq \lambda$ an integer.*

(a) $C_{m-\lambda}^\lambda(\cos t) = \dfrac{\mathsf{D}_k \cdots \mathsf{D}_1 \sin(mt)}{k_\lambda \sin^\lambda t}$.

(b) $\mathsf{D}_k \cdots \mathsf{D}_1 \sin(mt) = a_k(m, \cot t)\sin(mt) + b_k(m, \cot t)\cos(mt)$, *where $a_0 = 1$, $b_0 = 0$ and, with $\mathsf{q} = \cot t$,*

(9.3)
$$a_j = j\mathsf{q}a_{j-1} + (1+\mathsf{q}^2)\partial_{\mathsf{q}}a_{j-1} + mb_{j-1}, \quad b_j = j\mathsf{q}b_{j-1} + (1+\mathsf{q}^2)\partial_{\mathsf{q}}b_{j-1} - ma_{j-1}.$$
The polynomial a_k is even in m and has parity k in q; b_k is odd in m and has parity $k+1$ in q.

(c) *With $S_k = k(k+1)/2$: if $k = 2\ell - 1$, then $b_k = (-1)^\ell m^k + O(m^{k-2})$ and $a_k = (-1)^{\ell-1} S_k \mathsf{q}\, m^{k-1} + O(m^{k-3})$; if $k = 2\ell$, then $a_k = (-1)^\ell m^k + O(m^{k-2})$ and $b_k = (-1)^\ell S_k \mathsf{q}\, m^{k-1} + O(m^{k-3})$, with coefficients polynomial in q.*

(d) *For $t \in (0,\pi)$ the numbers $a_k(m, \cot t)$ and $b_k(m, \cot t)$ do not vanish simultaneously.*

Proof. (a) The differentiation formula [3, (18.9.19)] reads $\frac{d}{dx}C_N^\mu = 2\mu C_{N-1}^{\mu+1}$. Hence $C_{m-\lambda}^\lambda = (k_\lambda)^{-1}(d/dx)^k C_{m-1}^1$, and $C_{m-1}^1(\cos t) = \sin(mt)/\sin t$. With

$d/dx = -(\sin t)^{-1}\partial_t$, induction on j gives $[-(\sin t)^{-1}\partial_t]^j[\sin(mt)/\sin t] = \mathsf{D}_j\cdots\mathsf{D}_1\sin(mt)/\sin^{j+1}t$, since $-(\sin t)^{-1}\partial_t[g\sin^{-(j+1)}t] = \mathsf{D}_{j+1}g/\sin^{j+2}t$. (b) Apply D_j to $a\sin(mt)+b\cos(mt)$ with a,b functions of q and use $\partial_t\mathsf{q} = -(1+\mathsf{q}^2)$. The parities follow by induction from (9.3). (c) Induction on k using (9.3): the top term of $b_{2\ell}$ or $a_{2\ell+1}$ receives $(k+1)\mathsf{q}$ times the top term of the same function and $\pm m$ times the top term of the other, and $S_k + k + 1 = S_{k+1}$. (d) Put $\mathsf{L}_j = -\partial_t^2 + j(j-1)\csc^2 t$. One checks $\mathsf{D}_j^\dagger\mathsf{D}_j = \mathsf{L}_j - j^2$ and $\mathsf{D}_j\mathsf{D}_j^\dagger = \mathsf{L}_{j+1} - j^2$ with $\mathsf{D}_j^\dagger = j\cot t + \partial_t$, hence $\mathsf{D}_j\mathsf{L}_j = \mathsf{L}_{j+1}\mathsf{D}_j$. Thus $\mathsf{D}_j\cdots\mathsf{D}_1$ maps the solutions $\sin(mt), \cos(mt)$ of $\mathsf{L}_1 y = m^2 y$ to solutions of $\mathsf{L}_{k+1}y = m^2 y$. On solutions of $\mathsf{L}_j y = m^2 y$ the map $(y, y') \mapsto (\mathsf{D}_j y, (\mathsf{D}_j y)')$ has the matrix $\left(\begin{smallmatrix} j\cot t & -1 \\ m^2 - j^2\csc^2 t & j\cot t \end{smallmatrix}\right)$ of determinant $m^2 - j^2$, so the Wronskian of the two transformed solutions is $-m\prod_{j=1}^{k}(m^2-j^2) \neq 0$ for $m \geq k+1$. The transformed cosine is $a_k\cos(mt) - b_k\sin(mt)$. If a_k and b_k vanished at t, both transformed solutions would vanish there, contradicting the nonvanishing Wronskian. □

Proposition 9.5 (Finite resonance in even dimensions)**.** *Let $d = 2\lambda+2 \geq 6$ be even. Then $C_n^\lambda(1/\sqrt{d})$ vanishes for only finitely many n, and the set of zero degrees is effectively computable.*

Proof. Put $\sigma = \sqrt{d-1}$, so $\cot\theta_0 = 1/\sigma$. By Lemma 9.4(b) the parities of a_k, b_k give $\sigma^k a_k(m, 1/\sigma) = \tilde{a}(m)$ and $\sigma^k b_k(m, 1/\sigma) = \sigma\tilde{b}(m)$ with $\tilde{a}, \tilde{b} \in \mathbb{Z}[m]$. Hence, for $m = n + \lambda$, the zero equation is $\tilde{a}(m)\sin m\theta_0 + \sigma\tilde{b}(m)\cos m\theta_0 = 0$, and by Lemma 9.4(d), $N_d(m) := \tilde{a}(m)^2 + (d-1)\tilde{b}(m)^2 > 0$ for $m \geq \lambda$. With $s = i\sigma$ we have $e^{i\theta_0} = (1+s)/\sqrt{d}$, and the zero equation becomes $(\tilde{a} + \tilde{b}s)(1+s)^m = (\tilde{a} - \tilde{b}s)(1-s)^m$. If an odd prime p divides d, Hensel lifting gives r_p modulo p^m with $r_p^2 \equiv 1 - d$ and $r_p \equiv -1$ modulo p (the derivative $2r_p$ is a unit); applying the ring homomorphism $\mathbb{Z}[s] \to \mathbb{Z}/p^m\mathbb{Z}$, $s \mapsto r_p$, the left side vanishes and $(1 - r_p)^m$ on the right is a unit, hence p^m divides $\tilde{a} - \tilde{b}r_p$ and therefore $N_d(m)$. The remaining even dimensions are $d = 2^j$, $j \geq 3$: with $u = (1+s)/2$ one has $u^2 - u + d/4 = 0$ and $u\bar{u} = d/4 \geq 2$; expanding $(u + \bar{u})^{2m-1} = 1$ shows that u^m and $\bar{u}^m$ generate the unit ideal of $\mathbb{Z}[u]$, so u^m divides $\tilde{a} - \tilde{b}s$ in $\mathbb{Z}[u]$, and taking norms $(d/4)^m$ divides $N_d(m)$. Since N_d is a fixed positive polynomial, these exponential divisibilities fail beyond an effectively computable bound. □

9.4. **Circle arithmetic.** Let $z_0 = e^{2i\theta_0} = (2 - d + 2i\sqrt{d-1})/d$. Then $z_0 + z_0^{-1} = 4/d - 2$. For $d \geq 5$ this is rational and not an integer; since the trace of a root of unity is an algebraic integer, z_0 is not a root of unity and θ_0/π is irrational. The number z_0 is a root of $dz^2 + (2d-4)z + d$, whose discriminant $16(1-d)$ is negative; so this polynomial is irreducible over $\mathbb{Q}$, and dividing it by the greatest common divisor $\gcd(d, 4)$ of its coefficients gives the minimal polynomial of z_0 with relatively prime integer coefficients, all bounded in absolute value by $2d$.

Lemma 9.6 (Logarithmic forms [1, p. 20])**.** *Let $\alpha_1, \dots, \alpha_n$ be algebraic numbers different from 0 and 1, let D be the degree over $\mathbb{Q}$ of the field $\mathbb{Q}(\alpha_1, \dots, \alpha_n)$, and let* Log *denote the principal branch of the logarithm. Let $b_1, \dots, b_n$ be integers and $B \geq \max_j |b_j|$, $B \geq e$. For each j let $A_j \geq e$ be an upper bound for the absolute values of the coefficients of the minimal polynomial of α_j over $\mathbb{Z}$ with relatively prime coefficients. If $\Xi = b_1 \operatorname{Log} \alpha_1 + \cdots + b_n \operatorname{Log} \alpha_n \neq 0$, then*

$$\log|\Xi| > -(16nD)^{2(n+2)} \log A_1 \cdots \log A_n \, \log B.$$

Lemma 9.7 (Rotation arithmetic at ω_0)**.** *There are $c_d > 0$ and $\tau_0 = 64^8 \log(2d)$ such that*

$$|\sin k\theta_0| \geq c_d \, k^{-\tau_0} \qquad (k \geq 1). \tag{9.4}$$

Fix $\tau > \tau_0 + 1$ and $0 < \gamma \leq c_d/2$, and set

$$\mathcal{D}_h = \{\omega \in I_h : \ |\sin(k\omega/2)| \geq \gamma k^{-\tau} \text{ for all integers } k \geq 1\}.$$

Then $\mathcal{D}_h$ is compact and $\operatorname{Leb}(I_h \setminus \mathcal{D}_h) = o_d(h)$.

Proof. Apply Lemma 9.6 with $n = 2$, $D = 2$, $\alpha_1 = z_0$, $\alpha_2 = -1$, $A_1 = 2d$, $A_2 = e$, $B = \max(e, 2k)$ to $\Xi = k \operatorname{Log} z_0 - 2j \operatorname{Log}(-1) = 2i(k\theta_0 - j\pi)$, with j the integer nearest to $k\theta_0/\pi$; here $\operatorname{Log} z_0 = 2i\theta_0$ because $0 < 2\theta_0 < \pi$, and $\operatorname{Log}(-1) = i\pi$. The hypotheses hold: $\Xi \neq 0$ as θ_0/π is irrational, $|2j| \leq k + 1 \leq 2k$, both numbers are different from 0 and 1, $\mathbb{Q}(z_0, -1)$ has degree two, and the coefficient bounds were noted above. Hence $|\Xi| \geq (\max(e, 2k))^{-64^8 \log(2d)}$, and $|\sin k\theta_0| \geq 2 \operatorname{dist}(k\theta_0, \pi\mathbb{Z})/\pi = |\Xi|/\pi$.

For $\omega \in I_h$ and $k \leq K := \lfloor (c_d/(2h))^{1/(\tau_0+1)} \rfloor$ we have $|\sin(k\omega/2) - \sin k\theta_0| \leq kh \leq \frac{c_d}{2} k^{-\tau_0}$, hence $|\sin(k\omega/2)| \geq \frac{c_d}{2} k^{-\tau_0} \geq \gamma k^{-\tau}$. For $k > K$ the set $\{\omega \in I_h : |\sin(k\omega/2)| < \gamma k^{-\tau}\}$ consists of at most $kh/(2\pi) + 2$ intervals of length at most $C\gamma k^{-\tau-1}$. Summing over $k > K$ gives at most $C\gamma[hK^{1-\tau} + K^{-\tau}] = o_d(h)$, because $\tau > 1$ and $\tau > \tau_0 + 1$. Compactness is clear. □

Lemma 9.8 (Even base arithmetic)**.** *Let $d \geq 6$ be even. There are $c, C > 0$ such that $|C_n^\lambda(1/\sqrt{d})| \geq c \exp[-C \log^2(n + 2)]$ for every $n \geq 0$ with $C_n^\lambda(1/\sqrt{d}) \neq 0$.*

Proof. Let $m = n + \lambda$ and use the notation of the proof of Proposition 9.5. Writing $a_k = a_k(m, 1/\sigma)$, $b_k = b_k(m, 1/\sigma)$ and $x = m\theta_0$,

$$\sigma^k(a_k \sin x + b_k \cos x) = \frac{(\tilde{a} + \tilde{b}s)e^{-ix}}{2i}\big(e^{2ix} - \xi_m\big), \qquad \xi_m = \frac{\tilde{a} - \tilde{b}s}{\tilde{a} + \tilde{b}s},$$

and $|\tilde{a} + \tilde{b}s| = N_d(m)^{1/2} \geq 1$. Since s is purely imaginary, $|\xi_m| = 1$. By Lemma 9.4(a) it suffices to bound $|z_0^m - \xi_m|$ from below when it is nonzero. If $\tilde{b}(m) = 0$ then $\xi_m = 1$, and if $\tilde{a}(m) = 0$ then $\xi_m = -1$; in these cases $|z_0^m \mp 1|$ is bounded below by the two-logarithm argument of Lemma 9.7, which gives a polynomial bound. Otherwise $\xi_m \neq \pm 1$, so $\xi_m \notin \mathbb{Q}$, and ξ_m is a root of $N_d(m)z^2 - 2(\tilde{a}^2 - (d-1)\tilde{b}^2)z + N_d(m)$; its minimal polynomial with

relatively prime integer coefficients divides this quadratic, so its coefficients are bounded by $2N_d(m) \le C_d m^{2k}$. Choose $\ell \in \mathbb{Z}$ with $|\ell| \le m+1$ such that $\Xi = m \operatorname{Log} z_0 - \operatorname{Log} \xi_m - 2\ell \operatorname{Log}(-1)$ lies in $i(-\pi, \pi]$. Then $|z_0^m - \xi_m| = |e^{\Xi} - 1| \ge \frac{2}{\pi}|\Xi|$, and $\Xi \neq 0$ whenever the Gegenbauer value is nonzero. Lemma 9.6 with $n = 3$, $D = 2$ (all three numbers lie in $\mathbb{Q}(z_0)$), $A_1 = 2d$, $A_2 = \max(e, C_d m^{2k})$, $A_3 = e$ and $B = 2m + 2$ gives $|\Xi| \ge \exp[-C \log^2(m + 2)]$. The factors σ^{-k}, $\frac{1}{2} N_d(m)^{1/2} \ge \frac{1}{2}$ and $(k_\lambda \sin^\lambda \theta_0)^{-1}$ only change the constants. □

9.5. **Integrality of the base values for odd d.** Let d be odd, $m_* = (d-3)/2$ so that $\lambda = m_* + \frac{1}{2}$, and define rational numbers b_n by $(1 - 2s + ds^2)^{-\lambda} = \sum_n b_n s^n$, so that $b_n = d^{n/2} C_n^\lambda(1/\sqrt{d})$. The recurrence is $b_0 = 1$, $b_1 = 2\lambda$, $(n+1)b_{n+1} = 2(n+\lambda)b_n - d(n + 2\lambda - 1)b_{n-1}$.

Lemma 9.9. *With $c = (2m_*)!/m_*!$, $c\,4^n b_n \in \mathbb{Z}$ for all n. Consequently $b_n \neq 0$ implies $|b_n| \ge 1/(c\,4^n)$, and the normalized round value $(1+n)^{\lambda+1} C_n^\lambda(\cos\theta_0)/C_n^\lambda(1)$ has absolute value at least $e^{-C_d n}$ whenever it is nonzero.*

Proof. For every integer $j \ge 0$, $c\,4^j (m_* + \frac{1}{2})_j / j! = \binom{2j+2m_*}{j+m_*}(j+m_*)!/j! \in \mathbb{Z}$. Expand $(1-(2s-ds^2))^{-\lambda} = \sum_j \frac{(\lambda)_j}{j!}(2s - ds^2)^j$; the coefficient of s^n is a finite sum over $j \le n$, and multiplying each term by $c\,4^n$ clears its denominator. Finally $C_n^\lambda(1) = \binom{n+d-3}{n}$ is a positive polynomial in n of degree $d-3$. □

This lemma does not determine which b_n vanish, nor whether their zero set is finite.

9.6. **Finite formal solutions.** For the next proposition we write the physical radial function as a function of the axial angle t, and we use the scalar quantities

$$R_* = \Pi_0 r, \qquad \varepsilon = \text{the coefficient of } Y_2 \text{ in } r,$$
$$a_2 = \operatorname{mean}_t Y_2(\cos t) = \frac{d-2}{2(d-1)},$$

where mean_t is the angular mean in t (not the spherical mean).

Proposition 9.10 (Finite initializers)**.** *Fix $\rho_* \in (0, 1]$. For every $N \ge 2$ there is $h_N > 0$ such that for $0 < h < h_N$ there is a family $v_\omega^{(N)}$, $\omega \in I_h$, polynomial in $\omega - \omega_0$ with coefficients that are real odd π-periodic trigonometric polynomials, such that the body $K(v_\omega^{(N)}, \omega)$ of Definition 2.3 with radial function $r = r_\omega$ has the following properties, uniformly in $\omega \in I_h$:*

(i) *$(A_r - \Pi_0 A_r)/\varepsilon = O(h^N)$ and $h\,\partial_\omega[(A_r - \Pi_0 A_r)/\varepsilon] = O(h^N)$ in every norm $\|\cdot\|_{\rho_*, q}$ of the output angle;*

(ii) *$c_{\mathrm{t}} h \le \varepsilon \le C_{\mathrm{t}} h$, $R_* - R_0 - r_1 \varepsilon = O(h^2)$, and $r = R_* + \varepsilon Y_2 + \varepsilon^2 h_2 + \varepsilon u_*$ with $\Pi_0 u_* = \Pi_2 u_* = 0$ and $u_* = O(h^2)$, together with the same bounds for $h\,\partial_\omega$;*

(iii) *the frequency slope is*

$$\omega_1 = \frac{2(r_1 + a_2)}{\sqrt{d(d-1)}} = \frac{\sqrt{d}\,(d-2)}{(d-1)^{3/2}(d+4)} > 0,$$

and the mean section volume $V(\omega) = \Pi_0 A_r$ *satisfies*

$$V'(\omega) = \frac{L_0\, r_1}{\omega_1} + O(h) = -2|S^{d-2}|(d-1)^{\frac{d-2}{2}} + O(h). \tag{9.5}$$

Proof. Step 1: the radial expansion. We seek $r_\varepsilon = R_0 + \varepsilon w_1 + \sum_{j\geq 2} \varepsilon^j v_j$ with $v_1 := w_1$ and the gauge $\Pi_2 v_j = 0$ for $j \geq 2$, so that ε is the physical degree-two coefficient. Write

$$v_j = c_j + \sum_{n\in\mathcal{Z}} \mathsf{a}_{j,n} Y_n + \sum_{n\in\mathcal{N}} \mathsf{b}_{j,n} Y_n,$$

where $\mathcal{Z}$ is the set of resonant even degrees and $\mathcal{N}$ the set of nonresonant even degrees $n \geq 4$; we put $c_1 = r_1$ and $\mathsf{a}_{1,n} = \mathsf{b}_{1,n} = 0$. By Lemma 8.4, with $L = L_{R_0}$ and $L' = \partial_R L_R|_{R_0}$, the coefficient of ε^j in A_{r_ε} is

$$Lv_j + L'(w_1 v_{j-1}) + \mathcal{F}_j \quad (j \geq 3), \qquad Lv_2 + \tfrac{1}{2}L'(w_1^2) \quad (j = 2),$$

where $\mathcal{F}_j$ is a finite polynomial expression in $v_1, \dots, v_{j-2}$: in the i-th Taylor term the coefficient of ε^j is a sum of products $v_{l_1} \cdots v_{l_i}$ with $l_1 + \cdots + l_i = j$, and only the terms with $i = 1$ or with $i = 2$ and $\{l_1, l_2\} = \{1, j-1\}$ involve v_j or v_{j-1}. We require that all components of degree $n \geq 2$ of these coefficients vanish for $j \leq N$; the degree-zero component is the volume and is free.

At order $j = 2$ the degree-two component is $\frac{1}{2}L_2'(2r_1 + c_{22})$, which vanishes by the choice of r_1; the forcing $\frac{1}{2}L'(w_1^2)$ has degrees $0, 2, 4$ only, so the degree-four equation $L_4 \mathsf{b}_{2,4} + \frac{1}{2}L_4' c_{24} = 0$ gives $\mathsf{b}_{2,4} = a_{4,d}$, all other $\mathsf{b}_{2,n}$ vanish, and the resonant equations at order two read $0 = 0$ since resonant degrees are at least eight.

At order $j \geq 3$ the equations are solved in the following order.

(1) *Mean of* v_{j-1}. The degree-two component is $L_2'\big(c_{j-1} + c_{42}\mathsf{b}_{j-1,4}\big) + \Pi_2 \mathcal{F}_j$, because $L_2 = 0$, $\Pi_2 v_j = 0$ and $Y_2 Y_n$ has a Y_2 component only for $n \in \{0, 2, 4\}$. Since $L_2' \neq 0$ this determines c_{j-1}.

(2) *Resonant part of* v_{j-1}. For $n \in \mathcal{Z}$ the degree-n component is $\delta_n \mathsf{a}_{j-1,n} + L_n'(c_{n-2,n}\mathsf{b}_{j-1,n-2} + c_{n+2,n}\mathsf{b}_{j-1,n+2}) + \Pi_n \mathcal{F}_j$, because $L_n = 0$, the mean contributes only to degrees $0, 2$, and no other resonant degree is within distance two (Lemma 9.3). The first nonzero coefficient of $\mathsf{a}_{j-1,n}$ is the detuning $\delta_n \neq 0$, so this determines $\mathsf{a}_{j-1,n}$.

(3) *Nonresonant part of* v_j. For $n \in \mathcal{N}$ the degree-n component is $L_n \mathsf{b}_{j,n} + \Pi_n[L'(w_1 v_{j-1}) + \mathcal{F}_j]$, where v_{j-1} is now completely known; since $L_n \neq 0$ this determines $\mathsf{b}_{j,n}$.

Inductively every v_l with $l \leq j - 1$ is a finite combination of zonal polynomials, and L_R is diagonal on them, so every forcing is a finite combination and only finitely many coefficients determined at stage j are nonzero, even if $\mathcal{Z}$ is infinite. The procedure determines all coefficients uniquely except

the mean c_N and the resonant part $\mathsf{a}_{N,\cdot}$ of v_N, which enter first at order $N+1$; we set them equal to zero. Then $r_\varepsilon^{(N)} = R_0 + \sum_{j=1}^N \varepsilon^j v_j$ satisfies $A_{r_\varepsilon^{(N)}} - \Pi_0 A_{r_\varepsilon^{(N)}} = O(\varepsilon^{N+1})$ in every analytic norm, by Lemma 8.4 with remainder of order $N+1$. Its physical degree-two coefficient is ε, its mean is $R_0 + r_1\varepsilon + O(\varepsilon^2)$, and $v_2 = c_2 + h_2$ since $\mathsf{a}_{2,n} = 0$ for $n \in \mathcal{Z}$ (the forcing at order three in a degree ≥ 8 vanishes, its degrees being at most six); hence the decomposition in (ii) holds with $\varepsilon u_* = \sum_{j\geq 3} \varepsilon^j (v_j - c_j)$.

Step 2: the centered representation. We seek $p_\varepsilon = u + \sum_{j=1}^N \varepsilon^j p_j$, p_j odd π-periodic trigonometric polynomials, and $\theta_\varepsilon = \theta_0 + \sum_{j=1}^N \theta_j \varepsilon^j$ such that the radial function of Definition 2.3 built from $(p_\varepsilon, \theta_\varepsilon)$ agrees with $r_\varepsilon^{(N)}$ up to $O(\varepsilon^{N+1})$, i.e. $\sec\alpha(y) - r_\varepsilon^{(N)}(\chi(y)) = O(\varepsilon^{N+1})$. At $\varepsilon = 0$ we have $\chi = \mathrm{id}$ and $\alpha = \theta_0$. Expanding in ε, the terms of order j that contain p_j or θ_j are $R_0 \tan\theta_0 \big[\theta_j + \frac{1}{2}(p_j(y+\theta_0) - p_j(y-\theta_0))\big]$, all other terms being determined by $p_1, \dots, p_{j-1}, \theta_1, \dots, \theta_{j-1}$ and $r^{(N)}$. If $p_j = \sum_\ell p_{j,\ell} \sin(\ell u)$, then $\frac{1}{2}(p_j(y+\theta_0) - p_j(y-\theta_0)) = \sum_\ell p_{j,\ell} \sin(\ell\theta_0)\cos(\ell y)$. Hence the angular mean of the order-j equation determines θ_j, and its $\cos(\ell y)$ coefficient determines $p_{j,\ell}$ after division by $R_0 \tan\theta_0 \sin(\ell\theta_0) \neq 0$ ($\ell \geq 2$ even), which is nonzero because θ_0/π is irrational. Each forcing is a trigonometric polynomial, so each p_j is one. At order one the forcing is $w_1(\cos y)$, so $\theta_1 = \mathrm{mean}_y\, w_1/(R_0 \tan\theta_0) = (r_1 + a_2)/\sqrt{d(d-1)}$, i.e. $\omega_1 = 2\theta_1$. Since $p_\varepsilon, \theta_\varepsilon$ are polynomials in ε and the centered construction is analytic in (p, θ), the difference of the two radial functions is an analytic function of ε whose first N Taylor coefficients vanish, hence $O(\varepsilon^{N+1})$ on a fixed strip.

Step 3: the frequency as parameter. Since $\theta_1 = \omega_1/2 > 0$, the polynomial $\varepsilon \mapsto 2\theta_\varepsilon - \omega_0$ has a truncated inverse $\mathfrak{e}(\omega) = \sum_{i=1}^N e_i(\omega - \omega_0)^i$ with $e_1 = 1/\omega_1$ and $2\theta_{\mathfrak{e}(\omega)} = \omega + O((\omega - \omega_0)^{N+1})$. Let $v_\omega^{(N)}$ be the truncation at degree N in $\omega - \omega_0$ of $p_{\mathfrak{e}(\omega)} - u$. Then $p = u + v_\omega^{(N)}$ at $\theta = \omega/2$ differs from $(p_{\mathfrak{e}(\omega)}, \theta_{\mathfrak{e}(\omega)})$ by $O(h^{N+1})$, so its radial function differs from $r_{\mathfrak{e}(\omega)}^{(N)}$ by $O(h^{N+1})$, and $A_r - \Pi_0 A_r = O(h^{N+1})$ by Lemma 8.1. Dividing by $\varepsilon \asymp h$ gives (i) for the values. All quantities are polynomial, hence holomorphic, in complex ω in a disk of radius $3h$ about ω_0; Cauchy's estimate on the disk of radius h about each point of I_h gives the bounds for $h\partial_\omega$. The physical degree-two coefficient of r is $\mathfrak{e}(\omega) + O(h^{N+1}) \asymp h$, which gives (ii) after replacing ε by the exact coefficient. Finally $\Pi_0 A_{r_\varepsilon^{(N)}} = A_{R_0} + \varepsilon L_0 r_1 + O(\varepsilon^2)$, since the degree-zero part of the first-order term is $L_0 c_1$, and $d\varepsilon/d\omega = 1/\omega_1 + O(h)$; with $L_0 = |S^{d-2}|\sqrt{d}(d-1)^{(d-3)/2}$ and $r_1/\omega_1 = -2\sqrt{d-1}/\sqrt{d}$ this gives (9.5). The only choices made are the gauge $\Pi_2 v_j = 0$, the terminal values $c_N = 0$, $\mathsf{a}_{N,\cdot} = 0$, and the two truncations at degree N; the threshold h_N depends on N through the size of the finitely many coefficients, while the width ρ_* is fixed in advance. $\square$

10. Normalized linearizations

The linearization of the section map at a profile near the ball is a nonlocal operator, and in the native coordinates of Definition 2.3 it has small divisors coming from the rotation ω. In this section we construct, for both parities of d, exact global normalizations which exhibit it as a diagonal operator plus a localized $O(h)$ perturbation.

10.1. **Matrix classes.** For $\rho > 0$ and a matrix $B = (B_{mn})$ put

$$N_\rho(B) = \max\Big\{\sup_m \sum_n |B_{mn}| e^{\rho|m-n|},\ \sup_n \sum_m |B_{mn}| e^{\rho|m-n|}\Big\}.$$

N_ρ is submultiplicative. Let $\ell^2_\rho = \{u : \sum e^{2\rho n}|u_n|^2 < \infty\}$. By Schur's test a matrix with $N_\rho(B) < \infty$ acts boundedly on ℓ^2_r for $0 \le r \le \rho$, with norm at most $N_\rho(B)$. Let $\Lambda_0 = \mathrm{diag}(1+n)$.

Definition 10.1 (Normalized families)**.** Fix $M \ge 1$ and $0 < \rho_\infty < \rho_{\max}$. A *normalized family* on I_h is a family of real matrices $K(\omega) = D(\omega) + A(\omega)$ acting on the modes $n \ge M$ of a fixed parity (step two), where $D = \mathrm{diag}(d_n(\omega))$, $\theta = \omega/2$, is of one of the following types.

(O) $d_n = (1+n)^{\lambda+1} C_n^\lambda(\cos\theta)/C_n^\lambda(1)$ (odd d; n is the degree of an orthonormal Gegenbauer polynomial).

(B) $d_m = -m\cot(m\theta) + q_m$, where (q_m) is a fixed real sequence, independent of ω, with $q_m = q_\infty + O(m^{-2})$ (even d with λ even; m labels the mode $\sin(m\,\cdot)$ of a primitive).

(E) $d_m = -\cot(m\theta)/m + q_m$, where (q_m) is a fixed real sequence, independent of ω, with $q_m = -b_E + O(m^{-2})$ and $b_E > 0$ (even d with λ odd).

The perturbation A is real and, for every $\rho \in [\rho_\infty, \rho_{\max}]$,

$$\begin{aligned} N_\rho(A(\omega)) \le Ch, \qquad \partial_\omega A = A_0 + \Lambda_0 A_1, \\ N_\rho(A_0) \le C, \qquad N_\rho(A_1) \le Ch, \end{aligned} \tag{10.1}$$

where $\omega \mapsto A(\omega)$ is C^1 on I_h as a map into the bounded operators $\ell^2_\rho \to \ell^2_{\rho-\sigma}$ for every $\sigma > 0$. In types (B) and (E), $A_1 = 0$.

In type (O) the frequency derivative is of the weaker form $A_0 + \Lambda_0 A_1$, which costs one power of the mode; it is the form provided by the odd normalization below.

10.2. **Odd dimensions.** Let $d \ge 5$ be odd. Fix θ in a compact subinterval of $(0, \pi/2)$, $\omega = 2\theta$ and $R = \sec\theta$. Let $p = u + v$ be as in Definition 2.3, with $|v|_{\rho,1}$ small on a fixed strip, and let G be the odd analytic function on a Bernstein ellipse with $\cos p(u) = G(\cos u)$; it exists because $\cos p(u)$ is even and π-antiperiodic, and $G(\pm 1) = \pm 1$. The quotients $\sin p(u)/\sin u$ and $\sin\chi(y)/\sin y$ are analytic, nonzero and close to 1, including at the meridians $u, y \in \pi\mathbb{Z}$, and fractional powers of them are taken on the branches near 1.

Let $\tilde T$ be the section derivative T_r of Lemma 8.1 acting on the input $\tilde f(y) = f(\cos\chi(y))$, with output at the normal angle $\phi = p(u)$. Put

$$\begin{aligned} a_0 &= R^{2\lambda}(\sin\theta)^\beta, \qquad a_{\rm in}(y) = \hat r^{2\lambda}(\sin\alpha)^\beta \chi'\Big[\frac{\sin\chi}{\sin y}\Big]^\lambda, \\ a_{\rm out}(u) &= \frac{[\sin p/\sin u]^\lambda}{(p')^\beta}, \qquad \mathcal{L} = a_0\, a_{\rm out}\, \tilde T\, a_{\rm in}^{-1}. \end{aligned} \tag{10.2}$$

These are regular invertible analytic multipliers, equal to a_0, a_0 and 1 at the ball. Below $X = \cos u$, $Y = \cos y$, $\xi_\mp = \cos(y \mp \theta)$, and $G[a,b] = (G(a) - G(b))/(a-b) = \int_0^1 G'(b + \tau(a-b))\, d\tau$ is the divided difference.

Proposition 10.2 (Exact global normalization, odd d)**.** (a) *At the output $\phi = p(u)$ and the axial cosine $s = \cos\chi(y)$ the discriminant of* (8.3) *is* $\mathfrak{D} = \hat r^2[G(\xi_-) - G(X)][G(X) - G(\xi_+)]$.

(b) *Let* $\mathfrak{D}_0 = R^2(1 - X^2 - Y^2) + 2RXY - 1 = R^2(\xi_- - X)(X - \xi_+)$ *be the round discriminant, and*

$$\Psi = \frac{G[\xi_-, X]\, G[X, \xi_+]}{G[\xi_-, \xi_+]\, G'(X)}.$$

Then, for F a function of Y,

$$\mathcal{L}F(X) = |S^{d-3}| R^2 (1 - X^2)^{-(\lambda - \frac12)} \int_{\mathfrak{D}_0 \ge 0} \mathfrak{D}_0^\beta\, \Psi^\beta\, F(Y)\, dY. \tag{10.3}$$

For $\Psi \equiv 1$ this is the round derivative $T_{d,R}$.

(c) *There is an analytic function $\mathfrak{E} = \mathfrak{E}_{d,\theta}(X,Y)$, depending analytically on G and vanishing for $G = \mathrm{id}$, such that $\Psi^\beta - 1 = (X - \xi_-)(X - \xi_+)\,\mathfrak{E}$. Consequently*

$$\mathcal{L}F = T_{d,R}F - \frac{|S^{d-3}|}{(1 - X^2)^{\lambda - 1/2}} \int_{\mathfrak{D}_0 \ge 0} \mathfrak{D}_0^\lambda\, \mathfrak{E}(X,Y) F(Y)\, dY, \tag{10.4}$$

and for a separated amplitude $\mathfrak{E} = b_1(X)c_1(Y)$ the correction equals $-\dfrac{|S^{d-3}|}{R^2|S^{d-1}|} M_{b_1} M_{1-X^2} T_{d+2,R} M_{c_1}$, where M_g denotes multiplication by g.

Proof. (a) With $\phi = p(u)$, $s = \cos\chi$ and $\hat r = \sec\alpha$, multiplying $\mathfrak{D} = \hat r^2 \sin^2\phi \sin^2\chi - (1 - \hat r\cos\chi\cos\phi)^2$ by $\cos^2\alpha$ and writing $\sin\phi\sin\chi$ and $\cos\phi\cos\chi$ through $\cos(\phi \mp \chi)$ gives $\mathfrak{D} = \sec^2\alpha\,[\cos(\phi - \chi) - \cos\alpha][\cos\alpha - \cos(\phi + \chi)]$. The product-to-sum formulas show that this equals $\sec^2\alpha\,[\cos(\chi - \alpha) - \cos\phi][\cos\phi - \cos(\chi + \alpha)]$. Finally $\cos(\chi - \alpha)(y) = \cos p(y - \theta) = G(\xi_-)$, $\cos(\chi + \alpha) = G(\xi_+)$ and $\cos\phi = G(X)$.

(b) In the coarea formula (8.3) substitute $s = \cos\chi(y)$, $ds = -\sin\chi\,\chi'\,dy$, $r(s) = \hat r(y)$, and use (a), writing $\Pi = [G(\xi_-) - G(X)][G(X) - G(\xi_+)]$. Since

$G(\xi_-) - G(\xi_+) = 2\sin\chi\sin\alpha$,

$$(\sin\alpha)^{-\beta}\Pi^\beta = 2^\beta(\sin\chi)^\beta\Big[\frac{\Pi}{G(\xi_-) - G(\xi_+)}\Big]^\beta.$$

With the weights (10.2), all powers of $\hat r$ cancel because $2 - 2\lambda + 2\beta = 0$, the factors χ' cancel, and the powers of $\sin p$ and $\sin\chi$ cancel because $1-\lambda+\beta = 0$. What remains is

$$\mathcal{L}F(u) = a_0|S^{d-3}|2^\beta \int \mathcal{K}^\beta\Big(\frac{\sin y}{\sin u}\Big)^\lambda F\,dy,$$
$$\mathcal{K} = \frac{\Pi}{[G(\xi_-) - G(\xi_+)]\sin p(u)\,p'(u)}.$$

Since $\sin p\,p' = \sin u\,G'(X)$, the ratio of $\mathcal{K}$ to its value $\mathcal{K}_0$ for $G = \mathrm{id}$ is exactly Ψ. For $G = \mathrm{id}$ (the ball) the weights are constant and $\mathcal{L} = T_{d,R}$; substituting $dY = -\sin y\,dy$ and $\Pi = \mathfrak{D}_0/R^2$ shows that the kernel with $\mathcal{K}_0$ is the round coarea kernel $|S^{d-3}|R^2(1-X^2)^{-(\lambda-1/2)}\mathfrak{D}_0^\beta$. This gives (10.3). The identity $\mathfrak{D}_0 = R^2(\xi_- - X)(X - \xi_+)$ follows from $\xi_- + \xi_+ = 2Y\cos\theta$, $\xi_-\xi_+ = Y^2 - \sin^2\theta$, $R\cos\theta = 1$ and $R^2\sin^2\theta = R^2 - 1$. The integration domains agree because G is increasing near the identity.

(c) Ψ is analytic near 1 and symmetric in (ξ_-, ξ_+); hence it is an analytic function $\Phi(X, \varsigma_1, \varsigma_2)$ of X and of $\varsigma_1 = \xi_- + \xi_+$, $\varsigma_2 = \xi_-\xi_+$, including at $\xi_- = \xi_+$. Since $G[a,a] = G'(a)$, $\Psi = 1$ when $X = \xi_-$ or $X = \xi_+$, i.e. $\Phi - 1$ vanishes on the hypersurface $\varsigma_2 = \varsigma_1 X - X^2$. Division in the variable ς_2 gives $\Phi - 1 = (\varsigma_2 - \varsigma_1 X + X^2)\,\mathfrak{E}_1$ with $\mathfrak{E}_1 = \int_0^1 \partial_{\varsigma_2}\Phi(X, \varsigma_1, \varsigma_1 X - X^2 + \tau(\varsigma_2 - \varsigma_1 X + X^2))\,d\tau$ analytic, and $\varsigma_2 - \varsigma_1 X + X^2 = (X - \xi_-)(X - \xi_+)$. Put $\mathfrak{E} = \mathfrak{E}_1\int_0^1 \beta[1+\tau(\Psi-1)]^{\beta-1}d\tau$; as $\varsigma_1 = 2Y\cos\theta$ and $\varsigma_2 = Y^2 - \sin^2\theta$, it is an analytic function of (X, Y), including at the meridians where $\xi_- = \xi_+$. For $G = \mathrm{id}$ we have $\Psi \equiv 1$ and $\mathfrak{E} = 0$. Since $\Psi^\beta - 1 = -\mathfrak{D}_0\mathfrak{E}/R^2$, (10.3) becomes (10.4). For separated amplitudes use the coarea expression of $T_{d+2,R}$, whose denominator power is $\lambda + \frac12$ and whose numerator is $|S^{d-1}|R^2\mathfrak{D}_0^\lambda$.

The identities are first proved on a nondegenerate output interval. The left side of (10.4) is globally analytic by (8.2) and the regular weights, the right side by the fixed-ray formula for $T_{d+2,R}$ and the convergent expansion of $\mathfrak{E}$ below, and analytic continuation on the connected common domain gives the identity through every meridian and turning output. It is an identity of operators. □

Lemma 10.3 (Physical section map)**.** *Let* $\mathcal{X}(Y) = \hat r(y)\cos\chi(y)$ *and* $S_G(X) = (1 - G(X)^2)/(1 - X^2)$*, which is analytic, positive and close to* 1*. Then*

$$\mathcal{V}_p(u) := A_r(p(u)) = \frac{\kappa_{d-2}}{|S^{d-1}|R^d}\,S_G(X)^{-\lambda-\frac12}\,T_{d+2,R}\big[\Omega(X,\cdot)\big](X),$$
$$\Omega(X,Y) = \hat r(Y)^{2\lambda}\mathcal{X}'(Y)\big[G[\xi_-, X]G[X, \xi_+]\big]^\lambda. \tag{10.5}$$

For complex u on the working strip the fixed-ray argument $Y = \cos\theta\cos u - \sin\theta\sin u\, z$, $z \in [-1,1]$, of $T_{d+2,R}$ stays in the same Bernstein ellipse, so (10.5) *has a common analytic domain.*

Proof. By Lemma 8.2 in the variable y, $A_r(\phi) = \kappa_{d-2}(\sin\phi)^{-2\lambda-1}\int \mathfrak{D}^\lambda\, d\mathcal{X}$. Insert $\mathfrak{D} = \hat{r}^2 R^{-2}\mathfrak{D}_0\, G[\xi_-, X]G[X,\xi_+]$ from Proposition 10.2(a), $(\sin p)^2 = S_G(X)(1-X^2)$, and the coarea expression $T_{d+2,R}\Omega(X) = |S^{d-1}|R^2(1-X^2)^{-\lambda-1/2}\int \mathfrak{D}_0^\lambda \Omega\, dY$; the power of R is $2\lambda+2 = d$. At the ball $\Omega = R^{2\lambda+1}$ and $T_{d+2,R}[1] = |S^d|R(R^2-1)^{(d-1)/2}$, and (10.5) reduces to $\kappa_{d-1}(R^2-1)^{(d-1)/2}$, as it should. The point Y is a real linear combination $a\cos u + b\sin u$ with $a^2+b^2 = \cos^2\theta + z^2\sin^2\theta \le 1$, i.e. a real contraction of a real translate of $\cos u$; Bernstein ellipses are convex and invariant under these operations. □

Proposition 10.4 (Matrix class (O))**.** *Let $L^2_\lambda = L^2((-1,1),(1-X^2)^{\lambda-1/2}dX)$ with its orthonormal Gegenbauer basis. Fix $s_d = \lceil\lambda\rceil + 5$ and $r_d = \lceil\lambda\rceil + 8$. Suppose the coefficients of $\mathfrak{E} = \sum_{k,l} e_{kl}T_k(X)T_l(Y)$ satisfy $\sum_{k,l}|e_{kl}|(1+k+l)^{s_d}e^{\rho(k+l)} \le C_d h$; by two contour shifts and r_d integrations by parts in each angular variable this follows from bounds on the mixed angular derivatives of order r_d of $\mathfrak{E}(\cos u, \cos y)$ on the closed polystrip of width ρ, because $r_d > s_d + 2$. Then*

$$\begin{aligned} \frac{\Lambda_0^{\lambda+1}\mathcal{L}}{\kappa_d(R)} &= D + A_r, \qquad \kappa_d(R) = |S^{d-2}|R(R^2-1)^{\frac{d-3}{2}}, \\ d_n &= \frac{(1+n)^{\lambda+1}C_n^\lambda(\cos\theta)}{C_n^\lambda(1)}, \qquad N_\rho(A_r) \le C_d h, \end{aligned} \tag{10.6}$$

and if the coefficient fields have first ω-jets of size $O_d(1)$, then $\partial_\omega A_r = A_0 + \Lambda_0 A_1$ with $N_\rho(A_0) \le C_d$, $N_\rho(A_1) \le C_d h$. Reality and the even sector are preserved. At fixed ω, D and the round factor of order $d+2$ are fixed, so A_r is Lipschitz in the coefficient fields at the same radius.

Proof. The inclusion $\mathsf{I} : L^2_\lambda \to L^2_{\lambda+1}$ is a contraction with adjoint $\mathsf{I}^* g = (1-X^2)g$, and the connection formula $C_n^\lambda = \frac{\lambda}{n+\lambda}(C_n^{\lambda+1} - C_{n-2}^{\lambda+1})$ [3, (18.9.7)] makes I two-diagonal in the orthonormal bases; hence $M_{1-X^2}T_{d+2,R} = \mathsf{I}^*\,\mathrm{diag}(\ell_{d+2,n})\mathsf{I}$ with $\ell_{d+2,n} = |S^d|R(R^2-1)^{(d-1)/2}C_n^{\lambda+1}(1/R)/C_n^{\lambda+1}(1)$. For fixed d and θ in a compact subinterval of $(0,\pi/2)$, the fixed-parameter Darboux expansion of Jacobi polynomials [3, (18.15.4_5)], specialized to Gegenbauer polynomials as in [3, §18.15(ii)], gives $|C_N^\mu(\cos\theta)| \le C_\mu N^{\mu-1}$; with $C_N^\mu(1) = \binom{N+2\mu-1}{N}$ this gives $|\ell_{d+2,n}| \le C_d(1+n)^{-\lambda-1}$. Differentiating the exact formula first, by $\frac{d}{dx}C_N^\mu = 2\mu C_{N-1}^{\mu+1}$ [3, (18.9.19)], and then applying the same bound gives $|\partial_\omega \ell_{d+2,n}| \le C_d(1+n)^{-\lambda}$. Thus $\Lambda_0^{\lambda+1}$ times the correction in (10.4) is a band matrix composed with multiplications, and its frequency derivative costs one power of the mode. Multiplication by T_k has bandwidth k, operator norm ≤ 1 on L^2_λ, and hence

$N_\rho(M_{T_k}) \le (2k+1)e^{\rho k}$; conjugation by Λ_0^q costs at most $(1+k)^{|q|}$ more. Summing over (k,l) gives (10.6). The round part $T_{d,R}$ is diagonal by Lemma 8.3, with $L_n(R)/\kappa_d(R) = C_n^\lambda(\cos\theta)/C_n^\lambda(1)$. The frequency derivative falls on the coefficient fields (giving A_0), on $\ell_{d+2,n}$ (giving $\Lambda_0 A_1$ with $A_1 = O(h)$) and on the scalar $\kappa_d(R)$ (giving A_0). □

10.3. **Even dimensions: primitive, regularizer and endpoint formula.** Let $d = 2\lambda + 2 \ge 6$ be even, $k = \lambda - 1$ and $k_\lambda = 2^k k!$, as in Lemma 9.4. For a smooth function H on $(-1,1)$ the identity

$$\prod_{j=1}^{k} (\partial_t^2 + j^2)\big[\sin t\, H(\cos t)\big] = \partial_c^k\big[(1-c^2)^{k+\frac12} H^{(k)}(c)\big]\Big|_{c=\cos t} \tag{10.7}$$

holds. Indeed, for $k = 0$ both sides equal $\sin t\, H(\cos t)$; if it holds for k, put $\mathsf{u} = (1-c^2)^{k+1/2} H^{(k)}$, so that $(1-c^2)^{k+3/2} H^{(k+1)} = (1-c^2)\mathsf{u}' + (2k+1)c\,\mathsf{u}$, and the Leibniz rule gives $\partial_c^{k+1}[(1-c^2)\mathsf{u}' + (2k+1)c\mathsf{u}] = [(1-c^2)\partial_c^2 - c\,\partial_c + (k+1)^2]\partial_c^k \mathsf{u}$, which is $(\partial_t^2 + (k+1)^2)\partial_c^k \mathsf{u}$ since $\partial_t^2 = (1-c^2)\partial_c^2 - c\,\partial_c$.

Input primitive. For an even radial variation f (a function of the axial cosine s) and the profile r, let H be a k-fold primitive, $H^{(k)} = r^{2\lambda} f$, and put $p_f(t) = k_\lambda \sin t\, H(\cos t)$. The polynomial ambiguity of H (degree $< k$) changes p_f by a combination of $\sin(jt)$, $j \le k$, of the parity of λ; we fix it by requiring that these $\lfloor (\lambda-1)/2 \rfloor$ sine coefficients of p_f vanish. The *native primitive* is $v = p_f \circ \chi$. At the ball, by Lemma 9.4(a), $f = R^{-2\lambda} C_{m-\lambda}^\lambda$ corresponds to $p_f = \sin(mt)$.

Output regularizer. For an even function g of the normal angle ϕ let $S_\lambda g = k_\lambda \sin\phi\, H_g(\cos\phi)$ with $H_g^{(k)} = g$, and $P_\lambda = \partial_\phi \prod_{j=1}^k (\partial_\phi^2 + j^2)$. By (10.7),

$$P_\lambda S_\lambda g(\phi) = -k_\lambda \sin\phi\; \partial_c^\lambda\big[(1-c^2)^{\lambda-\frac12} g(c)\big]_{c=\cos\phi}, \tag{10.8}$$

which does not depend on the polynomial ambiguity of H_g. Since $P_\lambda \sin(mt) = (-1)^k m \prod_{j=1}^k (m^2 - j^2)\cos(mt)$, the regularized output has no cosine modes $\cos(jt)$ with $j \le k$ of the parity of λ.

Native operator. Put $\Phi = \chi + \alpha$, so that $\Phi(y) = p(y+\theta)$ and $(\chi - \alpha)(y + \omega) = \Phi(y)$. Let T be the translation $(Tv)(y) = v(y+\omega)$, $\Delta = I - T$, P_0 the angular mean (zero on antiperiodic functions), $\Delta^\dagger$ the zero-mean inverse of Δ on periodic and the inverse on antiperiodic functions (where it exists), and $\mathfrak{A} = (I+T)\Delta^\dagger$. Then $\mathfrak{A}$ vanishes on constants and $\mathcal{B} := \mathfrak{A}\partial_y$ acts on $\sin(my)$ and $\cos(my)$ as multiplication by $-m\cot(m\theta)$. The native derivative is

$$\mathcal{L}v(y) = \Phi'(y)\,\big(P_\lambda S_\lambda T_r f\big)(\Phi(y)), \qquad v = p_f \circ \chi.$$

With $\mathfrak{q}(c,s) = 1 - s^2 - r(s)^{-2} + 2(s/r(s))c - c^2$ we have $\mathfrak{D} = r^2\mathfrak{q}$ in (8.3), and we put $\mathcal{Q} = \partial_c^\lambda \mathfrak{q}^k$, a polynomial in c of degree at most $\lambda - 2$.

Lemma 10.5 (Endpoint formula)**.** *Let $I(c) = \int f\, r^2 \mathfrak{D}^k\, ds$ be the coarea numerator, the integral being over $[\mathsf{a}(c), \mathsf{b}(c)] = \{\mathfrak{D} \ge 0\}$. Then*

$P_\lambda S_\lambda(T_r f)(\phi) = -k_\lambda |S^{2\lambda-1}| \sin\phi\, I^{(\lambda)}(\cos\phi)$ *and*

$$I^{(\lambda)} = \big[k!\, f\, r^2 (\partial_c \mathfrak{D})^k\, \mathsf{s}'(c)\big]_{\mathsf{a}}^{\mathsf{b}} + \int_{\mathsf{a}}^{\mathsf{b}} H^{(k)}(s)\, \mathcal{Q}(c,s)\, ds, \tag{10.9}$$

where $\mathsf{s}'(c)$ *denotes the derivative of the respective endpoint. Integrating the last term* k *times by parts gives finitely many local endpoint derivatives of* H *of order* $\le k-1$ *and finitely many terms* $J_i(y)\int_y^{y+\omega} g_i(\tilde y) v(\tilde y)\, d\tilde y$ *with* $J_i = \Phi' \sin\Phi\, (\cos\Phi)^i$, $0 \le i \le \lambda - 2$, *and real analytic native coefficients* g_i. *Consequently*

$$\mathcal{L} v = E_+ v - T(E_- v) + \sum_i J_i \int_y^{y+\omega} g_i v, \tag{10.10}$$

where $E_\pm$ *are differential operators of order* k *in the native variable, with coefficients analytic away from the meridians.*

Proof. The first identity is (10.8) combined with (8.3), since $(\sin\phi)^{2\lambda-1} = (1-c^2)^{\lambda-1/2}$. Since $\mathfrak{D}^k$ vanishes to order k at both endpoints, $I^{(j)} = \int f r^2 \partial_c^j \mathfrak{D}^k\, ds$ for $j \le k$; one more derivative produces the boundary term, where $\partial_c^k \mathfrak{D}^k = k!(\partial_c \mathfrak{D})^k$, and the interior term $\int f r^2 \partial_c^\lambda \mathfrak{D}^k\, ds = \int H^{(k)} \mathcal{Q}\, ds$ since $f r^{2+2k} = H^{(k)}$. Integrating by parts k times,

$$\int H^{(k)} \mathcal{Q}\, ds = \sum_{i=0}^{k-1} (-1)^i \big[H^{(k-1-i)} \partial_s^i \mathcal{Q}\big]_{\mathsf{a}}^{\mathsf{b}} + (-1)^k \int H\, \partial_s^k \mathcal{Q}\, ds.$$

In native coordinates $H(\cos\chi) = v/(k_\lambda \sin\chi)$ and $H\, ds = -v\, \chi'\, dy/k_\lambda$, so no division by $\sin\chi$ remains in the residual integral. The upper endpoint b (larger s) is the profile point with $\chi + \alpha = \phi$, i.e. $y = x$ when $\phi = \Phi(x)$; the lower endpoint is the point with $\chi - \alpha = \phi$, i.e. $y = x + \omega$. Writing $\partial_s^k \mathcal{Q} = \sum_{i \le \lambda-2} c^i \mathfrak{g}_i(s)$ and multiplying by the output factor $-k_\lambda |S^{2\lambda-1}| \Phi' \sin\Phi$ gives (10.10). □

Lemma 10.6 (Removability at the meridians)**.** *The coefficients of the differential operators* $E_\pm$ *are real analytic; in particular they have no poles at the meridians.*

Proof. It suffices to work near $t = 0$. The profile is even and analytic, hence analytic in $s = \cos t$ near $s = 1$; put $\mathsf{y} = 1 - s$. Let $\mathsf{b}(c)$ be the endpoint crossing the meridian and $\delta(c) = 1 - \mathsf{b}(c)$. The root is simple: $\partial_s \mathfrak{D} = -2(r_{\mathrm{tip}}^2 - 1) \ne 0$ at the meridian, so $\mathfrak{D} = (s - \mathsf{b}(c))\, \mathsf{d}_1(s,c)$ with d_1 jointly analytic and nonzero. The endpoint angle t_{b} is analytic in c with nonzero derivative, since $c = \cos(t_{\mathsf{b}} + \alpha(t_{\mathsf{b}}))$ has nonzero t-derivative at the crossing, and $\delta = t_{\mathsf{b}}^2$ times an analytic unit.

Test the local endpoint expression on $p(t) = t^j$, $0 \le j \le k$. For odd j, $H = p/(k_\lambda \sin t)$ is analytic in s and all moving-endpoint integrals are analytic. For even j put $\beta_j = (j-1)/2 \ge -\frac{1}{2}$: the local expansion of H has powers $\mathsf{y}^{\beta_j + i}$, $i \ge 0$; after k axial derivatives they become $\mathsf{y}^{\beta_j - k + i}$.

Expand the analytic unit d_1^k in nonnegative powers y^ι and $(\delta-\mathsf{y})^k = \sum_l \binom{k}{l}\delta^{k-l}(-\mathsf{y})^l$. A primitive term evaluated at the moving endpoint has the power $\delta^{k-l}\delta^{\beta_j-k+i+\iota+l+1} = \delta^{\beta_j+i+\iota+1}$, with the nonzero half-integer integration denominator $\beta_j-k+i+\iota+l+1$; so no logarithms occur, and the endpoint exponent starts at $\beta_j+1=(j+1)/2>0$. Since $\delta = t_{\mathsf{b}}^2$ times a unit, these are integer powers $t_{\mathsf{b}}^{j+1+2i+2\iota}$ of positive degree. Hence the whole finite-part primitive is analytic before the λ derivatives in c are taken. The residual integral is analytic for every test since $H\,ds = -p\,dt/k_\lambda$. An operator of order k with meromorphic coefficients whose values on $1, t, \dots, t^k$ are analytic has analytic coefficients (recover the coefficients triangularly from $E(1)$, $E(t)$, ...). The finite-part primitive is used only for these test functions; on physical inputs all integrals are ordinary convergent integrals. □

Proposition 10.7 (Cohomological pull-through)**.** *Write $E_+ - TE_- = \Delta\mathsf{M} + (I+T)\mathsf{N}$ with $\mathsf{M} = (E_+ + E_-)/2$ and $\mathsf{N} = (E_+ - E_-)/2$. For an analytic coefficient J put $M_J = (J+T^{-1}J)/2$ and $Y_J = (J-T^{-1}J)/2$, as multiplication operators. Then, on the relevant parity sector,*

$$\Delta^\dagger J\Delta = (I-P_0)M_J + \mathfrak{A}Y_J, \qquad \Delta^\dagger J_i = \tfrac{1}{2}(\mathfrak{A}+I)J_i,$$
$$\int_y^{y+\omega} u = -\Delta\partial_y^\dagger(u-P_0u) + \omega P_0 u, \tag{10.11}$$

where $\partial_y^\dagger$ is the zero-mean primitive. Consequently the normalized derivative is

$$K_{\mathrm{raw}} := \Delta^\dagger\mathcal{L} = V + \mathfrak{A}C = V + \mathcal{B}W, \qquad W = \partial_y^\dagger(I-P_0)C, \tag{10.12}$$

where V, C, W are finite sums of differential operators, multiplications, zero-mean primitives, translations and finite-rank mean operators with real analytic coefficients depending on the native profile; they contain no cotangent multiplier.

Proof. Direct multiplication gives $\Delta M_J + (I+T)Y_J = J\Delta$ and $P_0J\Delta = 2P_0Y_J$. Since $\Delta\mathfrak{A} = I+T-2P_0$, applying Δ to the right side of the first identity gives $(I-P_0)J\Delta$, and both sides have zero mean. The interval integral splits as stated because $\Delta\partial_y^\dagger(u-P_0u)$ is minus the integral of $u-P_0u$ over $[y, y+\omega]$. Each $J_i = -(\cos^{i+1}\Phi)'/(i+1)$ is a derivative, so $P_0J_i = 0$ and $\Delta^\dagger J_i = \frac{1}{2}(\mathfrak{A}+I)J_i$. Combining these with the residual terms $J_i\int_y^{y+\omega} g_iv$, the term $\omega P_0(g_iv)\cdot\frac{1}{2}(\mathfrak{A}+I)J_i$ equals $\mathfrak{A}[\frac{\omega}{2}P_0(g_iv)J_i] + \frac{\omega}{2}P_0(g_iv)J_i$ because $P_0(g_iv)$ is a scalar; it contributes the same rank-one operator to V and to C. With $\Delta^\dagger\Delta = I-P_0$ and $\Delta^\dagger(I+T) = \mathfrak{A}$ for the local terms we obtain $K_{\mathrm{raw}} = V + \mathfrak{A}C$. Finally $\mathfrak{A}$ kills constants and commutes with ∂_y, so $\mathfrak{A}C = \mathfrak{A}\partial_y\partial_y^\dagger(I-P_0)C = \mathcal{B}W$. □

10.4. **Even dimensions: orders and ball symbols.** For an operator of the type in Proposition 10.7 the *order* is the highest order of differentiation that occurs, primitives counting -1.

Lemma 10.8 (Orders and ball symbols)**.** (a) *For $\sigma = \pm$ the coefficient of ∂_y^k in E_σ is $\mathfrak{c}_\sigma = -k_\lambda |S^{2\lambda-1}|(-\sigma)^k \sin(\alpha)^k (\chi')^{1-k}$, evaluated at the native point of the endpoint. Consequently, if λ is even (k odd), C has order k with nonvanishing leading coefficient, W has order $\lambda - 2$ and V has order at most $k-1$; if λ is odd (k even), V has order k, C has order at most $k-1$ and W has order at most $\lambda - 3$.*

(b) *At the ball of radius $R = \sec\theta$ the operators $E_\pm$, M, N, V, C, W have constant coefficients, the residual terms vanish, and after division by the nonzero scalar $\mathfrak{n}(\theta)$ below, V and W act on $\sin(my)$, $m \geq \lambda$ of the parity of λ, as multiplication by*

$$V_0(m) = a_k(m, \cot\theta), \qquad W_0(m) = -b_k(m, \cot\theta)/m. \tag{10.13}$$

(c) *Consequently W_0 is a polynomial in m of exact degree $\lambda - 2$ (λ even), resp. $\lambda - 3$ (λ odd), and $q_m^\circ(\theta) := V_0(m)/W_0(m)$ satisfies*

$$q_m^\circ(\theta) = \begin{cases} S_k \cot\theta + O(m^{-2}), & \lambda \text{ even}, \\ -\dfrac{\tan\theta}{S_k}\, m^2 + O(1), & \lambda \text{ odd}, \end{cases} \tag{10.14}$$

uniformly for θ in compact subsets of $(0, \pi/2)$, together with the θ-derivative.

Proof. (a) At the endpoint σ we have $\phi = \chi + \sigma\alpha$, $s/r = \cos\chi\cos\alpha$, hence $\partial_c \mathfrak{D} = r^2 \partial_c \mathfrak{q} = 2r^2(\cos\chi\cos\alpha - \cos(\chi + \sigma\alpha)) = 2\sigma r^2 \sin\chi \sin\alpha$. The endpoint derivative is $\mathsf{s}'(c) = \sin\chi\,\chi'/(\sin\Phi\,\Phi')$; for the lower endpoint this uses $(\chi - \alpha)'(y + \omega) = \Phi'(y)$. The highest term of $H^{(k)}$ is $(-1)^k \partial_y^k v/(k_\lambda (\chi')^k \sin^{k+1}\chi)$, because $d/ds = -(\chi' \sin\chi)^{-1}\partial_y$. Hence the boundary term of (10.9) has highest part

$$k!\, \frac{(-1)^k \partial_y^k v}{k_\lambda (\chi')^k \sin^{k+1}\chi}\, r^{2-2\lambda}\, (2\sigma r^2 \sin\chi \sin\alpha)^k\, \frac{\sin\chi\,\chi'}{\sin\Phi\,\Phi'} = \frac{(-\sigma)^k \sin^k\alpha\, (\chi')^{1-k}}{\sin\Phi\,\Phi'}\, \partial_y^k v,$$

using $r^{2-2\lambda+2k} = 1$ and $k!\, 2^k = k_\lambda$. Multiplying by the output factor $-k_\lambda |S^{2\lambda-1}| \sin\Phi\,\Phi'$ gives $\mathfrak{c}_\sigma$. All other endpoint terms involve $H^{(j)}$ with $j \leq k-1$, and the residual terms have order -1. As functions of the native variable, $\mathfrak{c}_+ = (-1)^k \mathfrak{c}_-$; therefore the top coefficient of M vanishes for odd k and that of N for even k. By Proposition 10.7, V contains M and terms of order -1, C contains N and terms of order -1, and $W = \partial_y^\dagger (I - P_0) C$. This gives the orders. That W has exactly the stated order near the ball follows from (b) and (c).

(b) At the ball $\chi = \mathrm{id}$, $\alpha = \theta$, $\Phi(y) = y + \theta$ and $r \equiv R$. Then $\mathfrak{q} = \sin^2\theta(1 - c^2) - (s - c\cos\theta)^2$ has total degree two in (c, s), so $\mathcal{Q}$ has degree at most $k-1$ in s and $\partial_s^k \mathcal{Q} = 0$: the residual terms vanish. By Lemma 8.3, Lemma 9.4(a), $P_\lambda \sin(mt) = (-1)^k m \prod_j (m^2 - j^2) \cos(mt)$ and $C_{m-\lambda}^\lambda(1) = m \prod_{j=1}^k (m^2 -$

$j^2)/(2\lambda-1)!$, the regularized round operator maps the primitive $p=\sin(m\,\cdot)$ to

$$\kappa(\theta)\,[\mathsf{D}_k\cdots\mathsf{D}_1\sin(m\,\cdot)](\theta)\,\cos(m\phi),$$
$$\kappa(\theta)=\frac{(-1)^k(2\lambda-1)!\,|S^{d-2}|R^{1-2\lambda}(R^2-1)^{\frac{d-3}{2}}}{k_\lambda\sin^\lambda\theta}\neq 0.$$

Since D_j acts on the variable θ and $\sin(m\theta)\cos(m\phi)=\frac12[\sin m(\phi+\theta)-\sin m(\phi-\theta)]$, this equals $\frac{\kappa(\theta)}{2}(\mathsf{D}_k\cdots\mathsf{D}_1)_\theta[p(\phi+\theta)-p(\phi-\theta)]=\frac{\kappa(\theta)}{2}\sum_{j=0}^k\mathfrak{d}_j(\theta)\big[(-1)^jp^{(j)}(\phi+\theta)-p^{(j)}(\phi-\theta)\big]$, where $\mathsf{D}_k\cdots\mathsf{D}_1=\sum_j\mathfrak{d}_j(\theta)(-\partial_\theta)^j$ with $\mathfrak{d}_k=1$. On the other hand, by Lemma 10.5 the same operator is $\sum_j[\alpha_j(\phi)p^{(j)}(\phi-\theta)+\beta_j(\phi)p^{(j)}(\phi+\theta)]$ with coefficients analytic in ϕ. The difference of the two expressions vanishes on $p=\sin(m\,\cdot)$ for all m in the progression $m_0+2\mathbb{N}$; writing $\partial^j\sin(mt)=\mathrm{Im}((im)^je^{imt})$, it is a combination $\sum_\zeta P_\zeta(m)\zeta^m$ with polynomial coefficients P_ζ and $\zeta\in\{e^{\pm i(\phi-\theta)},e^{\pm i(\phi+\theta)}\}$. For ϕ outside a discrete set the four numbers ζ^2 are distinct, and the sequences $\ell\mapsto\ell^i\mu^\ell$ with distinct μ are linearly independent; hence all P_ζ vanish, i.e. $\alpha_j(\phi)=-\frac{\kappa}{2}\mathfrak{d}_j(\theta)$ and $\beta_j(\phi)=\frac{\kappa}{2}(-1)^j\mathfrak{d}_j(\theta)$ for all ϕ. (For $j=k$ this agrees with (a).) Thus $E_\pm$ have constant coefficients at the ball, and so do M, N; then $V=(I-P_0)\mathsf{M}$, $C=\mathsf{N}$ and W are Fourier multipliers. Put $\mathfrak{n}(\theta)=-\kappa(\theta)/2$. On $\sin(my)$, $\Delta\sin(my)=-2\sin(m\theta)\cos(m(y+\theta))$, so the full normalized operator acts as multiplication by $\mathfrak{n}(\theta)[a_k(m)+b_k(m)\cot(m\theta)]$, using Lemma 9.4(b). Writing $\mathsf{M}\sin(my)=\mu_1(m)\sin(my)+\mu_2(m)\cos(my)$ and $\mathsf{N}\sin(my)=\nu_1(m)\sin(my)+\nu_2(m)\cos(my)$ with polynomials μ_i,ν_i, and using that $\mathcal{B}$ is multiplication by $-m\cot(m\theta)$ on sines and cosines, we obtain

$$\begin{aligned}[\mu_1(m)-\cot(m\theta)\nu_2(m)]\sin(my)+[\mu_2(m)+\cot(m\theta)\nu_1(m)]\cos(my)\\ =\mathfrak{n}(\theta)[a_k(m)+b_k(m)\cot(m\theta)]\sin(my).\end{aligned}$$

If θ/π is irrational, $\cot(m\theta)$ takes values arbitrarily close to every real number along $m\in m_0+2\mathbb{N}$, which is impossible for a nonzero rational function of m; hence $\mu_2=\nu_1=0$, $\mu_1=\mathfrak{n}a_k$ and $\nu_2=-\mathfrak{n}b_k$. This gives (10.13) for θ/π irrational, and for all θ by continuity, since all symbols are polynomial in m with coefficients analytic in θ.

(c) follows from Lemma 9.4(c): for $k=2\ell-1$, $W_0=-(-1)^\ell m^{k-1}+O(m^{k-3})$ and $V_0=(-1)^{\ell-1}S_k\cot\theta\,m^{k-1}+O(m^{k-3})$; for $k=2\ell$, $W_0=-(-1)^\ell S_k\cot\theta\,m^{k-2}+O(m^{k-4})$ and $V_0=(-1)^\ell m^k+O(m^{k-2})$. The quotients are rational in m with coefficients polynomial in $\cot\theta$, which gives the uniformity in θ. □

From now on we replace $\mathcal{L}$ by $\mathfrak{n}(\theta)^{-1}\mathcal{L}$ and K_{raw} by $\mathfrak{n}(\theta)^{-1}K_{\mathrm{raw}}$; this does not change the equations, and the ball symbols are (10.13).

10.5. **Even dimensions: localization and the matrix classes (B) and (E).** We work in the sine basis $\sin(my)$ of the native primitive, $m \geq 1$ of the parity of λ, and in the Wiener norms $\|a\|_{\rho,q}$ of coefficient functions.

Lemma 10.9 (Same-radius localization)**.** *Let a be an analytic π-periodic or π-antiperiodic coefficient with $\|a\|_{\rho,q} < \infty$, let $j \geq 0$, and let $\mathfrak{w}_{\mathrm{in}}(n)$, $\mathfrak{w}_{\mathrm{out}}(m)$ be positive weights with $\mathfrak{w}_{\mathrm{out}}(m)\, n^j\, \mathfrak{w}_{\mathrm{in}}(n) \leq C_0(1+|m-n|)^q$ for all $m, n \geq 1$. Let B be the matrix of $v \mapsto \mathfrak{w}_{\mathrm{out}} \cdot \Pi_{\sin}[a\, \partial_y^j(\mathfrak{w}_{\mathrm{in}} v)]$ in the sine basis, where $\Pi_{\sin}$ is the projection onto the sine modes and the weights act diagonally on the modes. Then $N_\rho(B) \leq 4C_0\|a\|_{\rho,q}$. The same bound holds with ∂_y^j replaced by $\partial_y^\dagger$ (with $j=-1$), by a translation $T^{\pm 1}$, or with a replaced by a finite-rank operator $v \mapsto P_0(gv)\, a$.*

Proof. We have $a(y)\sin(ny) = \sum_k \hat{a}_k (e^{i(k+n)y} - e^{i(k-n)y})/(2i)$, whose $\sin(my)$ coefficient involves only $\hat{a}_k$ with $|k| \in \{|m-n|, m+n\}$: the first case gives Toeplitz entries, the second Hankel entries. In both cases $e^{\rho|m-n|} \leq e^{\rho|k|}$ and, since $|m-n| \leq |k|$, the weight factor is at most $C_0(1+|k|)^q$. For fixed m and each k at most two n occur, so each row sum is at most $4C_0 \sum_k (1+|k|)^q e^{\rho|k|}|\hat{a}_k|$, and the same holds for column sums. The derivative contributes the factor n^j already included; primitives contribute n^{-1}, translations unimodular factors, and the finite-rank case is a product of two bounded analytic vectors. □

The estimate uses the coefficient function at the same radius ρ, with q additional powers of the mode, i.e. finitely many derivatives of the coefficient; no width is lost. It is distinct from the composition estimates of Section 13, which lose width.

Proposition 10.10 (Matrix classes (B) and (E))**.** *Choose a finite cutoff $m_\sharp$ above every zero of the polynomial $W_0(m)$ (uniformly for θ near θ_0) and above all $m = n + \lambda$ with n resonant, and let F be a fixed diagonal operator supported on the modes $m < m_\sharp$ such that $W_0 + F$ is invertible on the whole sine sector. Put $\widetilde{W} = W + F$. Suppose the native coefficient fields of V, C are within $O(h)$ of their values at the ball of radius $\sec\theta$ in the norm $\|\cdot\|_{\rho,q_W}$ for a fixed q_W and all $\rho \in [\rho_\infty, \rho_{\max}]$, with first ω-jets of size $O(1)$. Then:*

(i) *$\widetilde{W}(W_0+F)^{-1} = I + O(h)$ in N_ρ; hence $\widetilde{W}$ has a Neumann inverse of order minus the order of W_0, without rotation small divisors.*

(ii) *$K_{\mathrm{raw}}\widetilde{W}^{-1} = \mathcal{B} + Q$ with $Q = V\widetilde{W}^{-1} - \mathcal{B}F\widetilde{W}^{-1}$; the last term has output in the finite set of modes $m < m_\sharp$, on which $\mathcal{B}$ is analytic for ω near ω_0.*

(iii) *If λ is even, the compression of $\mathcal{B}+Q$ to the modes $m \geq M \geq m_\sharp$ is $\mathrm{diag}(-m\cot(m\theta) + q_m) + A$ with the frozen sequence $q_m = q_m^\circ(\theta_0)$ of* (10.14) *and a real matrix A satisfying* (10.1) *with $A_1 = 0$. This is type* (B)*.*

(iv) *If λ is odd, with $\Lambda = \operatorname{diag}(m)$, the compression of $\Lambda^{-1}(\mathcal{B}+Q)\Lambda^{-1}$ to $m \ge M$ is $\operatorname{diag}(-\cot(m\theta)/m + q_m) + A$ with the frozen sequence $q_m = q_m^\circ(\theta_0)/m^2$ and A as in* (iii)*. This is type* (E) *with $b_E = \tan\theta_0/S_k$.*

The matrices are real and in general nonsymmetric.

Proof. (i) By Lemma 10.8, $W - W_0$ is a finite sum of terms $(a-a^0)\partial_y^j$ with $j \le \operatorname{ord} W_0$, primitives, translations and finite-rank terms, all with coefficient differences $O(h)$; composed with $(W_0+F)^{-1}$, whose symbol is $O(m^{-\operatorname{ord} W_0})$, Lemma 10.9 with $\mathfrak{w}_{\mathrm{in}} = |W_0+F|^{-1}$ gives $N_\rho = O(h)$; the mismatch between input and output powers is bounded by a power of $1+|k|$ and absorbed by q_W. (ii) is (10.12). (iii) By Lemma 10.8 and (10.13), at the ball Q is diagonal on $m \ge m_\sharp$ with entries $q_m^\circ(\theta)$. Away from the ball, $Q - Q_{\mathrm{ball}} = (V-V_0)\widetilde{W}^{-1} + V_0(\widetilde{W}^{-1} - (W_0+F)^{-1}) - \mathcal{B}F(\widetilde{W}^{-1} - (W_0+F)^{-1})$ has $N_\rho = O(h)$ by (i) and Lemma 10.9, because $\operatorname{ord} V \le \operatorname{ord} W$ for even λ; the last term has finite output support and does not meet the high compression. Put $A = [Q - Q_{\mathrm{ball}}]_{\mathrm{high}} + \operatorname{diag}(q_m^\circ(\theta) - q_m^\circ(\theta_0))$. By (10.14) and its θ-derivative, $|q_m^\circ(\theta) - q_m^\circ(\theta_0)| \le C|\theta-\theta_0| \le Ch$ and $|\partial_\theta q_m^\circ(\theta)| \le C$ uniformly in $m \ge m_\sharp$; the coefficient jets give $N_\rho(\partial_\omega A) = O(1)$. (iv) is identical after the congruence by Λ^{-1}, which turns $-m\cot(m\theta)$ into $-\cot(m\theta)/m$ and $q_m^\circ(\theta)$ into $q_m^\circ(\theta)/m^2 = -\tan\theta/S_k + O(m^{-2})$; here $\operatorname{ord} V = \operatorname{ord} W + 2$, and the two factors Λ^{-1} compensate the extra two powers in Lemma 10.9. □

Remark 10.11 (Global coefficient families). The coefficient families V, C, $\widetilde{W}$ and Q are given by explicit finite formulas in terms of p, χ, α, Φ, translations, primitives and means, for *all* $\omega \in I_h$ and all small v; they are C^1 in ω. The rotation small divisors enter only through the diagonal symbol $-m\cot(m\theta)$ of $\mathcal{B}$, which has poles at isolated frequencies. The physical equation is solved only on the retained set of Section 11, where all divisors are bounded below.

11. The high-mode inverse

In this section we prove an inverse theorem for the normalized families of Definition 10.1. The matrices are real and not assumed to be symmetric; no spectral theorem is used.

11.1. **Hypotheses on the diagonal.** Put $L_h = \log(1/h)$, $\delta = 1/\tau$ with τ from Lemma 9.7, and $N_h = \lfloor L_h/(4C_1)\rfloor$ with C_1 from (S3) below. Let $d_n(\omega)$ be the diagonal of a normalized family on I_h; in types (B) and (E) it is meromorphic with poles at the zeros of $\sin(n\omega/2)$. Fix $\mu > 0$, with $\mu \le b_E/8$ in type (E). We consider the following hypotheses, for all $n \ge M$.

(S1) *Transversality.* (a) At nonpole $\omega \in I_h$ with $|d_n(\omega)| \le 2\mu$: $c_S n^2 \le |\partial_\omega d_n(\omega)| \le C_S n^2$. (b) At nonpole $\omega \in I_h$ with $|d_n(\omega)| \ge \mu$: $|\partial_\omega d_n(\omega)| \le C_S n^2 d_n(\omega)^2$. (c) The open set $\{\omega \in I_h : \ \omega \text{ nonpole}, \ |d_n(\omega)| < 2\mu\}$ has at most $C_S(hn+1)$ connected components, the *active cells*, and on each of them $\partial_\omega d_n$ has a constant sign.

(S2) *Separation.* For $\omega \in \mathcal{D}_h$ and distinct $n, l \geq M$ with $|d_n(\omega)|, |d_l(\omega)| < 2\mu$: $|n - l| \geq c_S \min(n, l)^\delta$.

(S3) *Base values.* (a) If $d_n(\omega_0) \neq 0$ then $|d_n(\omega_0)| \geq c_S e^{-C_1 n}$. (b) If $n \leq N_h$, $\omega \in I_h$, and $|d_n(\omega)| < 2\mu$ or $|d_n(\omega_0)| < \mu$, then the closed segment between ω_0 and ω contains no pole of d_n, and $|\partial_\omega d_n| \leq C_S n^2$ on it.

The sign in (S1)(c) may change from one active cell to the next; only its constancy on each cell is used.

Lemma 11.1 (Verification for type (O))**.** *For odd d, the diagonal* (O) *satisfies* (S1)–(S3) *once M is large, and moreover $|\partial_\omega d_n| \leq Cn^2$ and $|\partial_\omega^2 d_n| \leq Cn^3$ on all of I_h.*

Proof. By the fixed-parameter interior Darboux expansion [3, (18.15.4_5) and §18.15(ii)], applied to C_n^λ and, after differentiating the exact formula by [3, (18.9.19)], to $C_{n-1}^{\lambda+1}$, uniformly for θ near θ_0,

$$d_n = \mathsf{c}(\theta)\, n \cos\varphi_n + O(1), \qquad \partial_\omega d_n = -\tfrac{1}{2}\mathsf{c}(\theta)\, n^2 \sin\varphi_n + O(n), \qquad \varphi_n = (n+\lambda)\theta - \tfrac{\lambda\pi}{2}, \tag{11.1}$$

with $\mathsf{c}(\theta) = 2\Gamma(2\lambda)/(\Gamma(\lambda)(2\sin\theta)^\lambda) > 0$. Differentiating once more with [3, (18.9.19)] gives $|\partial_\omega^2 d_n| \leq Cn^3$. Choose C' with $\mathsf{c}(\theta)C' > 2\mu + C$, where C is the constant in (11.1). Outside the phase windows $\{|\cos\varphi_n| \leq C'/n\}$ we have $|d_n| > 2\mu$; on each window $|\sin\varphi_n| \geq \frac{1}{2}$ for $n \geq M$, so $|\partial_\omega d_n| \geq \frac{1}{8}\mathsf{c}\, n^2$ there and $\partial_\omega d_n$ has the fixed sign of $-\sin\varphi_n$, which alternates from one window to the next. Each window is an interval, the windows are disjoint, and the phase changes by $(n+\lambda)h/2$ across I_h, so at most $(n+\lambda)h/(2\pi) + 2$ windows meet I_h; on each window d_n is strictly monotone, so $\{|d_n| < 2\mu\}$ meets it in one interval. This gives (S1)(a),(c); (S1)(b) follows from $|\partial_\omega d_n| \leq Cn^2 \leq C\mu^{-2}n^2 d_n^2$. For (S2), two active indices have phases within $O(1/n + 1/l)$ of $\frac{\pi}{2} + \pi\mathbb{Z}$, so $\operatorname{dist}((n-l)\theta, \pi\mathbb{Z}) \leq C(1/n + 1/l)$, while on $\mathcal{D}_h$, $|\sin((n-l)\theta)| \geq \gamma|n-l|^{-\tau}$. (S3)(a) is Lemma 9.9, and (S3)(b) follows from the global bound on $\partial_\omega d_n$, since there are no poles. □

Lemma 11.2 (Verification for types (B) and (E))**.** *For even d, the diagonals* (B) *and* (E) *of Proposition 10.10 satisfy* (S1)–(S3) *once M exceeds $m_\sharp$ and is large. In both types $\partial_\omega d_m > 0$ at every nonpole frequency.*

Proof. Type (B). Here $\partial_\omega d_m = \frac{m^2}{2}\csc^2(m\omega/2) > 0$. If $|d_m| \leq 2\mu$ then $|m\cot(m\omega/2)| \leq 2\mu + \sup|q_m|$, so $|\cot(m\omega/2)| \leq C/m$ and $\frac{m^2}{2} \leq \partial_\omega d_m \leq m^2$ for $m \geq M$. If $|d_m| \geq \mu$, then $\csc^2 = 1 + \cot^2 = 1 + (q_m - d_m)^2/m^2 \leq 1 + 2(q_m^2 + d_m^2)/m^2$, whence $\partial_\omega d_m / d_m^2 \leq \frac{m^2}{2}\big[\mu^{-2}(1 + 2\sup q^2/m^2) + 2/m^2\big] \leq Cm^2$. On each cotangent cell $\{2\pi j < m\omega < 2\pi(j+1)\}$ the function d_m increases from $-\infty$ to $+\infty$, so $\{|d_m| < 2\mu\}$ meets it in one interval, and at most $mh/(2\pi) + 2$ cells meet I_h. An active index has $m\omega$ within C/m of an odd multiple of π; for two active indices $(m-n)\omega$ is within $C(1/m + 1/n)$ of $2\pi\mathbb{Z}$, while on $\mathcal{D}_h$, $|\sin((m-n)\omega/2)| \geq \gamma|m-n|^{-\tau}$. This gives (S2).

Type (E). Here $\partial_\omega d_m = \frac{1}{2}\csc^2(m\omega/2) > 0$. If $|d_m| \le 2\mu \le b_E/4$, then $q_m = -b_E + O(m^{-2})$ forces $\frac{1}{2}b_E m \le |\cot(m\omega/2)| \le 2b_E m$ for $m \ge M$, hence $cm^2 \le \partial_\omega d_m \le Cm^2$. If $|d_m| \ge \mu$, then $\cot^2 = m^2(q_m - d_m)^2 \le 2m^2(q_m^2 + d_m^2)$ gives $\partial_\omega d_m / d_m^2 \le Cm^2$. Cells are as in type (B). An active index has $m\omega$ within C/m of $2\pi\mathbb{Z}$, which gives (S2) as before.

(S3) for both types. By Lemma 10.8 and Lemma 9.4,

$$\begin{aligned} d_m(\omega_0) &= -\frac{m\,[a_k \sin m\theta_0 + b_k \cos m\theta_0]}{b_k \sin m\theta_0} \\ &= -\frac{m\,k_\lambda \sin^\lambda \theta_0\, C^\lambda_{m-\lambda}(\cos\theta_0)}{b_k(m)\sin m\theta_0} \qquad \text{(type (B))}, \end{aligned}$$

and the same expression divided by m^2 in type (E), with a_k, b_k evaluated at $(m, \cot\theta_0)$. Since $|b_k(m)| \le Cm^k$ and $|\sin m\theta_0| \le 1$, Lemma 9.8 gives $|d_m(\omega_0)| \ge c\exp[-C\log^2(m+2)] \ge c_S e^{-C_1 m}$ whenever $d_m(\omega_0) \ne 0$; for $m \ge M > m_\sharp$ it is nonzero. For (S3)(b) let $m \le N_h$. In type (B), the condition $|d_m| < 2\mu$ at ω, or $|d_m| < \mu$ at ω_0, gives $|\sin(m\varpi/2)| \ge c_0 > 0$ at that endpoint ϖ; the phase $m\omega/2$ changes by at most $mh \le N_h h \to 0$ along the segment, so $|\sin| \ge c_0/2$ there and $\partial_\omega d_m \le 2m^2/c_0^2$. In type (E) the same condition gives $c/m \le |\sin(m\varpi/2)|$ and $|\cot(m\varpi/2)| \le Cm$ at that endpoint, i.e. a phase distance comparable to $1/m$ from the pole; since $mh \le N_h h = o(1/N_h)$, the segment keeps a distance at least half of that, and $\partial_\omega d_m \le Cm^2$ on it. □

11.2. **Smooth diagonal elimination.** Let ψ_μ be a fixed smooth even function with $0 \le \psi_\mu \le 1$, $\psi_\mu = 0$ on $|t| \le \mu$ and $\psi_\mu = 1$ on $|t| \ge 2\mu$. Put $g_\mu(t) = \psi_\mu(t)/t$ and $\nu_\mu(t) = 1 - \psi_\mu(t)$, and

$$\mathcal{G} = \operatorname{diag} g_\mu(d_n), \qquad \mathcal{P} = \operatorname{diag} \nu_\mu(d_n),$$

with $\mathcal{G}_n = \mathcal{P}_n = 0$ at poles. Near a pole $\psi_\mu(d_n) = 1$, so $\mathcal{G}_n = 1/d_n$ is a quotient of trigonometric functions with nonzero denominator and $\mathcal{P}_n \equiv 0$; thus $\omega \mapsto \mathcal{G}_n(\omega), \mathcal{P}_n(\omega)$ are C^1 on all of I_h. Moreover $\|\mathcal{G}\| \le \mu^{-1}$ and, by (S1)(a),(b),

$$|\partial_\omega \mathcal{G}_n| + |\partial_\omega \mathcal{P}_n| \le Cn^2 : \tag{11.2}$$

on $\mu \le |d_n| \le 2\mu$ we use $|\partial_\omega d_n| \le C_S n^2$, and on $|d_n| \ge 2\mu$ we use $|\partial_\omega d_n|/d_n^2 \le C_S n^2$.

For small h, $N_\rho(\mathcal{G}A) < \frac{1}{2}$. Set

$$\begin{gathered} \mathcal{J} = (I + \mathcal{G}A)^{-1}, \qquad \mathcal{C} = A\mathcal{J} = (I + A\mathcal{G})^{-1}A, \\ (D + A)\mathcal{J} = D + \mathcal{P}\mathcal{C}, \end{gathered} \tag{11.3}$$

where the identity follows from $\mathcal{J} = I - \mathcal{G}\mathcal{C}$: the left side is $D - D\mathcal{G}\mathcal{C} + \mathcal{C}$ and $I - D\mathcal{G} = \mathcal{P}$ (on finite analytic vectors, then on the analytic diagonal domain). Then $N_\rho(\mathcal{J}) \le 2$ and $N_\rho(\mathcal{C}) \le 2Ch$. No circle condition is needed to define (11.3). Define

$$c_n = \mathcal{C}_{nn}, \qquad e_n = d_n + \nu_\mu(d_n)\, c_n. \tag{11.4}$$

Outside the active cells $e_n = d_n$, and $|e_n| \geq 2\mu$ there. Differentiating (11.3),

$$\begin{aligned} \partial_\omega \mathcal{C} &= (I - \mathcal{C}\mathcal{G})\,\partial_\omega A\,(I - \mathcal{G}\mathcal{C}) - \mathcal{C}\,(\partial_\omega \mathcal{G})\,\mathcal{C}, \\ |(\partial_\omega \mathcal{C})_{nn}| &\leq C(1 + hn + h^2 n^2). \end{aligned} \tag{11.5}$$

For the diagonal bound, the first term has entries $O(1+hn)$ since $\partial_\omega A = A_0 + \Lambda_0 A_1$ and an inserted Λ_0 is localized near n; the last term is bounded by $\sum_l (Ch)^2 e^{-2\rho_\infty |n-l|} C(1+l)^2 \leq Ch^2(1+n)^2$ by (11.2).

Lemma 11.3 (Cellwise transversality of the effective diagonal)**.** *There are M and $h_0(M)$ such that for $n \geq M$, $h < h_0$ and every active cell $\mathcal{O}$ of d_n: e_n is C^1 on $\mathcal{O}$, $\partial_\omega e_n$ has the sign of $\partial_\omega d_n$ on $\mathcal{O}$, and $|\partial_\omega e_n| \geq \frac{c_S}{2} n^2$. Consequently, for $0 < \varepsilon' \leq \mu$ and $\varsigma > 0$,*

$$\begin{aligned} \mathrm{Leb}\{\omega \in I_h : |e_n(\omega)| < \varepsilon'\} &\leq \frac{4C_S}{c_S}\,\varepsilon' n^{-2}(hn+1), \\ \sum_{n>L} \mathrm{Leb}\{|e_n| < \eta n^{-\varsigma}\} &\leq C\eta\big[L^{-\varsigma-1} + hL^{-\varsigma}\big]. \end{aligned} \tag{11.6}$$

Proof. On $\mathcal{O}$, $\partial_\omega e_n = \partial_\omega d_n\,(1 + \nu_\mu'(d_n)c_n) + \nu_\mu(d_n)\partial_\omega c_n$. Here $|\nu_\mu'(d_n)c_n| \leq C\mu^{-1}h \leq \frac{1}{4}$, and by (11.5) $|\partial_\omega c_n| \leq C(1+hn+h^2n^2) \leq \frac{c_S}{4}n^2$ once $n \geq M$ and $h \leq h_0$. Since $|\partial_\omega d_n| \geq c_S n^2$ on $\mathcal{O}$, we get $|\partial_\omega e_n - \partial_\omega d_n| \leq \frac{1}{2}|\partial_\omega d_n|$, which gives the sign and the lower bound. Thus e_n is strictly monotone on each active cell, and $\{|e_n| < \varepsilon'\} \cap \mathcal{O}$ is an interval of length at most $4\varepsilon'/(c_S n^2)$; outside the cells $|e_n| \geq 2\mu > \varepsilon'$. The number of cells is at most $C_S(hn+1)$ by (S1)(c). Summing with $\varepsilon' = \eta n^{-\varsigma}$ gives the second bound. □

Lemma 11.4 (Same-frequency stability)**.** *For two coefficient matrices $A_{\mathrm{old}}, A_{\mathrm{new}}$ at the same ω (so $D, \mathcal{G}, \mathcal{P}$ coincide), $\mathcal{C}_{\mathrm{new}} - \mathcal{C}_{\mathrm{old}} = (I + A_{\mathrm{new}}\mathcal{G})^{-1}(A_{\mathrm{new}} - A_{\mathrm{old}})(I + \mathcal{G}A_{\mathrm{old}})^{-1}$, hence*

$$\begin{aligned} N_\rho(\mathcal{C}_{\mathrm{new}} - \mathcal{C}_{\mathrm{old}}) &\leq C N_\rho(A_{\mathrm{new}} - A_{\mathrm{old}}), \\ \sup_n |e_{n,\mathrm{new}} - e_{n,\mathrm{old}}| &\leq C N_\rho(A_{\mathrm{new}} - A_{\mathrm{old}}), \end{aligned} \tag{11.7}$$

uniformly in n.

11.3. **The active equation and three classes of indices.** At a nonpole frequency let $\mathcal{B}_\omega = \{n : \nu_\mu(d_n) \neq 0\} = \{n : |d_n| < 2\mu\}$ be the active set and $\mathcal{I}_\omega$ its complement. By (11.3) the equation $Ku = f$ is equivalent to $(D + \mathcal{P}\mathcal{C})w = f$ with $u = \mathcal{J}w$, and

$$d_n w_n = f_n \ (n \in \mathcal{I}_\omega), \qquad (D_{\mathcal{B}} + \mathcal{P}_{\mathcal{B}}\mathcal{C}_{\mathcal{B}\mathcal{B}})w_{\mathcal{B}} = f_{\mathcal{B}} - \mathcal{P}_{\mathcal{B}}\mathcal{C}_{\mathcal{B}\mathcal{I}} D_{\mathcal{I}}^{-1} f_{\mathcal{I}}. \tag{11.8}$$

Every inactive $|d_n| \geq 2\mu$, so (11.8), followed by inactive reconstruction and $u = \mathcal{J}w$, is equivalent to the original equation. The active matrix has diagonal e_n and off-diagonal part E_{off}, $(E_{\mathrm{off}})_{nl} = \nu_\mu(d_n)\mathcal{C}_{nl}$ for $n \neq l$, with $N_\rho(E_{\mathrm{off}}) \leq Ch$. With $\varpi_n = |e_n|^{-1/2}$, $\varpi = \mathrm{diag}(\varpi_n)$ and $S = \mathrm{diag}(\operatorname{sgn} e_n)$,

$$D_{\mathcal{B}} + \mathcal{P}_{\mathcal{B}}\mathcal{C}_{\mathcal{B}\mathcal{B}} = \varpi^{-1}\big[S + \varpi E_{\mathrm{off}}\varpi\big]\varpi^{-1}, \tag{11.9}$$

valid for real nonsymmetric E_{off} since $S^{-1} = S$; if the middle perturbation has norm < 1 the Neumann series applies.

Put $\eta_h = h\exp[-L_h^{\delta}/(\log L_h)^4]$ and fix $\varsigma > 0$. A frequency ω is *retained* if $\omega \in \mathcal{D}_h$ and $|e_n(\omega)| \geq \eta_h n^{-\varsigma}$ for all $n \geq M$. On $\mathcal{D}_h$ there are no poles. By (11.6) the exclusion costs $O(\eta_h(1+h)) = o(h)$, and the retained set is compact, because each set $\{|e_n| < \eta_h n^{-\varsigma}\}$ is relatively open: it lies in the pole-free open set $\{|d_n| < \mu\}$, on which e_n is continuous. For a retained ω and an active index n:

$$\begin{aligned} &\mathsf{Z}: \ n \leq N_h,\ d_n(\omega_0) = 0, && |e_n| \geq \tfrac{c_S}{2}\, hn^2; \\ &\mathsf{P}: \ n \leq N_h,\ d_n(\omega_0) \neq 0, && |e_n| \geq h^{1/2}; \\ &\mathsf{H}: \ n > N_h, && |e_n| \geq \eta_h n^{-\varsigma}. \end{aligned} \tag{11.10}$$

Indeed, if $d_n(\omega_0) = 0$ and $n \leq N_h$, then by (S3)(b) $|d_n| \leq 2C_S n^2 h < \mu$ on $[\omega_0, \omega_0 + 2h]$ for small h, since $hN_h^2 \to 0$; so this segment lies in one active cell, where by (S1) the sign of $\partial_\omega d_n$ is fixed and $|\partial_\omega d_n| \geq c_S n^2$. Hence $|d_n(\omega)| \geq c_S h n^2$ on I_h, and $|e_n| \geq c_S h n^2 - 2Ch \geq \frac{c_S}{2} hn^2$ after increasing M. If $d_n(\omega_0) \neq 0$, $n \leq N_h$ and n is active, (S3) gives $|d_n(\omega)| \geq c_S e^{-C_1 N_h} - 2C_S h N_h^2 \geq \frac{c_S}{2} h^{1/4}$, so $|e_n| \geq h^{1/2}$ for small h. All bounds concern absolute values.

11.4. **A smooth extension of the cotangent diagonal.** For types (B) and (E) the diagonal has poles between retained frequencies, so its parameter jets on the retained set are compared through a smooth extension. Fix an even $\zeta \in C^\infty(\mathbb{R})$ with $0 \leq \zeta \leq 1$, $\zeta = 0$ on $[-\frac{1}{2}, \frac{1}{2}]$ and $\zeta = 1$ on $|s| \geq 1$, and put $F_\varepsilon(s) = \zeta(s/\varepsilon)/s$.

Lemma 11.5 (Smooth reciprocal cutoff)**.** *For $m \geq 1$ let $\varepsilon_m = \gamma m^{-\tau}$ and define, in type* (B)*,*

$$d_m^\sharp(\omega) = -m\cos(m\omega/2)\, F_{\varepsilon_m}\big(\sin(m\omega/2)\big) + q_m,$$

and in type (E) *the same expression with $-m$ replaced by $-1/m$. Then $d_m^\sharp \in C^\infty(\mathbb{R})$; $d_m^\sharp = d_m$ and $\partial_\omega d_m^\sharp = \partial_\omega d_m$ on the closed set $\{|\sin(m\omega/2)| \geq \varepsilon_m\} \supset \mathcal{D}_h$; and*

$$|d_m^\sharp - q_m| \leq C\gamma^{-1} m^{\tau+1}, \quad |\partial_\omega d_m^\sharp| \leq C\gamma^{-2} m^{2\tau+2}, \quad |\partial_\omega^2 d_m^\sharp| \leq C\gamma^{-3} m^{3\tau+3},$$

with C depending only on ζ. Consequently $D^\sharp = \mathrm{diag}(d_m^\sharp)$ is a C^2 family of operators $\ell^2_\rho \to \ell^2_{\rho-\sigma}$ with $\|\partial_\omega^j (D^\sharp - q)\|_{\rho\to\rho-\sigma} \leq C_j \gamma^{-j-1}(1 + \sigma^{-(j+1)(\tau+1)})$, $j = 0, 1, 2$.

Proof. Since $F_\varepsilon = 0$ near 0, $d_m^\sharp$ is smooth. Where $|\sin(m\omega/2)| > \varepsilon_m$ we have $\zeta = 1$ in a neighborhood, so $d_m^\sharp = d_m$ there; at points with $|\sin(m\omega/2)| = \varepsilon_m$ all derivatives of $\zeta(s/\varepsilon_m)$ vanish, so value and first derivative still agree. On $\mathcal{D}_h$, $|\sin(m\omega/2)| \geq \gamma m^{-\tau}$. The bounds follow from $|F_\varepsilon| \leq 2/\varepsilon$, $|F_\varepsilon'| \leq C/\varepsilon^2$, $|F_\varepsilon''| \leq C/\varepsilon^3$ and the chain rule, each ω-derivative producing a factor $m/2$. Finally $\sup_m m^a e^{-\sigma m} \leq C_a \sigma^{-a}$. □

In type (O) we put $D^\sharp = D$; by Lemma 11.1, $\|\partial_\omega^j D\|_{\rho\to\rho-\sigma} \le C(1+\sigma^{-j-1})$ for $j = 1, 2$. In all types let $K^\sharp = D^\sharp + A$. Then $K^\sharp = K$ and $\partial_\omega K^\sharp = \partial_\omega K$ on $\mathcal{D}_h$.

11.5. **The inverse theorem.**

Theorem 11.6 (High-mode inverse)**.** *Let $K = D+A$ be a normalized family satisfying* (S1)–(S3)*. There exist M_* and, for each $M \ge M_*$, $h_*(M) > 0$ such that for $h < h_*$ and every retained ω:*

(i) (Inverse.) *$K(\omega)$ is invertible on analytic sequences. For $\sigma > 0$ and $\rho, \rho-\sigma \in [\rho_\infty, \rho_{\max}]$,*

$$\log\|K(\omega)^{-1}\|_{\ell^2_\rho\to\ell^2_{\rho-\sigma}} \le C\sigma^{-2\tau}\big[L_h + |\log\sigma| + 1\big]^{2\tau} =: \log B(\sigma, h), \tag{11.11}$$

and for every fixed $A_* > 0$, if $\sigma \ge c(A_*\log L_h + 1)^{-2}$, then $B(\sigma, h) \le Ch^{-2}\sigma^{-\varsigma}$, with C depending on A_*.

(ii) (Fixed-width response.) *Let $0 < \mu_0 \le \rho_\infty/2$ and $s \ge 0$. Put $\zeta_0 = \min\{\mu_0/(24C_1), 1/8\}$ and $\kappa_* = 1 - 2\zeta_0 < 1$. Then*

$$\|K(\omega)^{-1}f\|_{H^s} \le \big[\epsilon_M h^{-1} + C_s h^{-\kappa_*}\big]\|f\|_{\ell^2_{\mu_0}},$$

where $\epsilon_M \to 0$ as $M \to \infty$, for fixed μ_0 and s.

(iii) (Jets.) *On the retained set put $(K^{-1})^{[1]} = -K^{-1}(\partial_\omega K)K^{-1}$. With three equal portions $\sigma/3$ of the loss,*

$$\|(K^{-1})^{[1]}\|_{\rho\to\rho-\sigma} \le \mathfrak{g}_1(\sigma)\, B(\sigma/3, h)^2,$$

$$\mathfrak{g}_1(\sigma) = \begin{cases} C(1+\sigma^{-2}) & \text{type (O)},\\ C\gamma^{-2}(1+\sigma^{-2\tau-2}) & \text{types (B), (E)}. \end{cases} \tag{11.12}$$

For retained a, b, $K(b)^{-1} - K(a)^{-1} = -K(b)^{-1}[K^\sharp(b) - K^\sharp(a)]K(a)^{-1}$, and with five portions $\sigma/5$,

$$\begin{aligned}\|K(b)^{-1} - K(a)^{-1} - (b-a)(K^{-1})^{[1]}(a)\|_{\rho\to\rho-\sigma}&\\ \le |b-a|\,\big[\mathfrak{m}_A(|b-a|) + |b-a|\big]\,\mathfrak{g}_2(\sigma)B(\sigma/5, h)^3,&\end{aligned} \tag{11.13}$$

where $\mathfrak{m}_A$ is the modulus of continuity of $\partial_\omega A$ as a map into the bounded operators $\ell^2_\rho \to \ell^2_{\rho-\sigma/5}$, and $\mathfrak{g}_2(\sigma) = C\big(1 + \mathfrak{g}_1(\sigma/5)^2 + \gamma^{-3}\sigma^{-3\tau-3}\big)$ in types (B), (E), resp. $C(1+\mathfrak{g}_1(\sigma/5)^2+\sigma^{-3})$ in type (O). Thus K^{-1} with the jet $(K^{-1})^{[1]}$ is a Whitney C^1 family on the retained set with values in the bounded operators $\ell^2_\rho \to \ell^2_{\rho-\sigma}$, for every $\sigma > 0$.

11.6. **Proof of the high-mode inverse theorem.** *Middle factor at a fixed interior radius.* Take a radius $r \in (0, \rho_\infty)$ and work in ℓ^2_r, so that the coefficient matrices have the margin $\rho_\infty - r$. The weights satisfy $\varpi_n \le Ch^{-1/2}/n$ on Z, $\varpi_n \le h^{-1/4}$ on P, and $\varpi_n \le \eta_h^{-1/2}n^{\varsigma/2}$ on H. The blocks of $\varpi E_{\mathrm{off}}\varpi$ below N_h are bounded by

$$\mathsf{ZZ}:\ C/M^2, \qquad \mathsf{ZP}:\ Ch^{1/4}/M, \qquad \mathsf{PP}:\ Ch^{1/2}.$$

For a pair (n,l) of active indices with $\max(n,l) > N_h$: if $\min(n,l) \le \frac{1}{2}\max(n,l)$ then $|n-l| \ge N_h/2$; otherwise $\min(n,l) > N_h/2$ and (S2) gives $|n-l| \ge c_S(N_h/2)^\delta$. In both cases $|n-l| \ge cN_h^\delta$, and polynomial factors in n,l are absorbed by a fixed portion of $e^{-(\rho_\infty - r)|n-l|}$, since near-shell pairs have $n \le 2l \le 4n$ and $|n-l| \ge c\min(n,l)^\delta$. Hence, up to fixed polynomial factors in N_h and $(\rho_\infty - r)^{-1}$,

$$\begin{aligned} &\mathsf{ZH}:\ \sqrt{h/\eta_h}\, e^{-c(\rho_\infty - r)N_h^\delta}, \qquad \mathsf{PH}:\ h^{3/4}\eta_h^{-1/2} e^{-c(\rho_\infty - r)N_h^\delta}, \\ &\qquad\qquad \mathsf{HH}:\ (h/\eta_h)\, e^{-c(\rho_\infty - r)N_h^\delta}. \end{aligned} \tag{11.14}$$

The logarithm of the largest of these is at most $L_h^\delta/(\log L_h)^4 - c(\rho_\infty - r)(L_h/(4C_1))^\delta + O(\log L_h) \to -\infty$. Choose M once so that $C/M^2 < \frac{1}{8}$. Then the perturbation in (11.9) has norm $< \frac{1}{2}$ on ℓ^2_r, for every $r \in (0, \rho_\infty - \sigma_0]$ with a fixed $\sigma_0 > 0$, and the middle matrix has a Neumann inverse of norm ≤ 2.

Interior solution and fixed-width response. This gives, through (11.8) and $u = \mathcal{J}w$, a unique solution in ℓ^2_r, with $w_{\mathcal{B}} = \varpi(S + \varpi E_{\mathrm{off}}\varpi)^{-1}\varpi(\cdots)$. Let $f \in \ell^2_{\mu_0}$. Use the middle inverse on $\ell^2_{2\mu_0/3}$; the right factor ϖ maps $\ell^2_{\mu_0}$ to $\ell^2_{2\mu_0/3}$ with norm $\sup_n \varpi_n e^{-\mu_0 n/3}$, and the left factor maps $\ell^2_{2\mu_0/3}$ to $\ell^2_{\mu_0/3}$, which embeds into H^s with norm $C_s\mu_0^{-s}$. With the buffer $e^{-\mu_0 n/3}$ the weights cost at most

$$\mathsf{Z}:\ \epsilon'_M h^{-1/2}, \qquad \mathsf{P}:\ h^{-1/4}, \qquad \mathsf{H}:\ \eta_h^{-1/2} \sup_{n > N_h} n^{\varsigma/2} e^{-\mu_0 n/3} \le C h^{-\frac{1}{2}+\zeta_0},$$

where $\epsilon'_M = C\sup_{n \ge M} n^{-1} e^{-\mu_0 n/3} \to 0$. For the last bound, $e^{-\mu_0 N_h/3} \le Ch^{\mu_0/(12C_1)}$, while $\eta_h^{-1/2} N_h^{\varsigma/2} \le h^{-1/2-\mu_0/(24C_1)}$ for small h. Each of the two external factors is therefore bounded by $\epsilon'_M h^{-1/2} + h^{-1/4} + Ch^{-1/2+\zeta_0}$. Their product is at most

$$(\epsilon'^2_M + C\epsilon'_M)h^{-1} + C\big(h^{-1/2} + h^{-3/4+\zeta_0} + h^{-1+2\zeta_0}\big) \le \epsilon_M h^{-1} + C_s h^{-\kappa_*},$$

since every term containing a Z weight carries a factor ϵ'_M, and $\max(\frac{1}{2}, \frac{3}{4} - \zeta_0, 1 - 2\zeta_0) = 1 - 2\zeta_0$ for $\zeta_0 \le \frac{1}{4}$. The inactive reconstruction and $\mathcal{J}$ are bounded. This proves (ii).

Analytic width loss. Let $\sigma > 0$ and $N_\sigma = \lceil C\sigma^{-2/\delta}[L_h + |\log\sigma| + 1]^{2/\delta}\rceil$. The tail split is made in the already formed system $D + \mathcal{PC}$ of (11.3), with the same $\mathcal{C}$ and the same retained effective diagonals e_n. Let $w = \mathcal{J}^{-1}u$ be the interior solution, keep its components with $n \le N_\sigma$, and solve the remaining active equation for $n > N_\sigma$ with these components moved to the right side; inactive coordinates keep their diagonal reconstruction. For active $n, l > N_\sigma$ the separation is at least cN_σ^δ as above, so the tail middle factor has norm $< \frac{1}{2}$ at the target radius $\rho - \sigma$, the coefficient margin being σ: the factor $(h/\eta_h)e^{-c\sigma N_\sigma^\delta}$ and the polynomial weights are absorbed by the choice of N_σ. Embedding the known components $n \le N_\sigma$ into $\ell^2_{\rho-\sigma}$ costs e^{CN_σ}, and the interior bound costs a power of h^{-1}. Uniqueness on the interior radius

identifies the result with u. This gives (11.11). For $\sigma \geq c(A_* \log L_h + 1)^{-2}$ the middle factor is small directly at the radius $\rho - \sigma$, without the tail split: the exponent in (11.14) with $\rho_\infty - r$ replaced by σ is at most $L_h^\delta/(\log L_h)^4 - c' L_h^\delta/(\log L_h)^2 + O(\log L_h) \to -\infty$. The two external weights then cost at most $h^{-1/2}$ on $\mathsf{Z} \cup \mathsf{P}$ and $\eta_h^{-1/2} \sup_n n^{\varsigma/2} e^{-\sigma n/3} \leq C h^{-1} \sigma^{-\varsigma/2}$ on H, since $\eta_h \geq h^2$ for small h; this gives $B(\sigma, h) \leq C h^{-2} \sigma^{-\varsigma}$.

Jets and Whitney property. On the retained set, K is differentiable with $\partial_\omega K = \partial_\omega D + A_0 + \Lambda_0 A_1$. In type (O), $|\partial_\omega d_n| \leq C(1+n)^2$ gives $\|\partial_\omega K\|_{\rho \to \rho - \sigma} \leq C(1 + \sigma^{-2})$. In types (B) and (E), on $\mathcal{D}_h$, $|\partial_\omega d_m| \leq \frac{m^2}{2} \csc^2(m\omega/2) \leq \frac{1}{2}\gamma^{-2} m^{2\tau+2}$, giving $\mathfrak{g}_1$. The identity $(K^{-1})^{[1]} = -K^{-1}(\partial_\omega K) K^{-1}$ with three portions of the loss gives (11.12). For retained a, b the resolvent identity holds on nested scales, and $K(b) - K(a) = K^\sharp(b) - K^\sharp(a)$ because $D^\sharp = D$ on $\mathcal{D}_h$. Writing $\Delta_{ab} = b - a$,

$$\begin{aligned} &K(b)^{-1} - K(a)^{-1} - \Delta_{ab}(K^{-1})^{[1]}(a) \\ &\qquad = -K(b)^{-1} \mathsf{r}_{ab} K(a)^{-1} - \Delta_{ab}\big[K(b)^{-1} - K(a)^{-1}\big](\partial_\omega K)(a) K(a)^{-1}, \end{aligned}$$

with $\mathsf{r}_{ab} = K^\sharp(b) - K^\sharp(a) - \Delta_{ab}\partial_\omega K^\sharp(a)$. By Lemma 11.5 the diagonal part of r_{ab} is bounded by $\frac{1}{2}\Delta_{ab}^2 \sup \|\partial_\omega^2 D^\sharp\|$, and the part from A by $|\Delta_{ab}|$ times the modulus of continuity of $\partial_\omega A$. The first term is bounded by $B^2 \|\mathsf{r}_{ab}\|$, the second by $\Delta_{ab}^2 B^3 \mathfrak{g}_1^2$, using three, resp. five, portions of the loss. This gives (11.13); the same argument applied to $(K^{-1})^{[1]}$ gives continuity of the jet. No indicator function of the active set is differentiated. □

12. The finite physical center

The normalized high matrix of Section 10 acts on the modes $n \geq M$. The remaining finite system couples the scalar variables of the physical problem with the low modes and, for even d, with the primitive and output constraints. In this section we keep the notation $r_1, w_1, h_2, \omega_1, a_2$ of Section 9.

12.1. **The tangent equation and the frequency functional.**

Lemma 12.1 (Centered tangent equation)**.** *Let $p = u + v$ be as in Definition 2.3 at $\theta = \omega/2$, and let $(\mathsf{k}, \delta\omega)$ be a variation of (p, ω) with k odd and π-periodic. Put $p'_\pm = p'(y \pm \theta)$, $\mathsf{w} = \mathsf{k}/p'$, and let $\eta = \delta r \circ \chi$ be the induced physical radial variation, at fixed physical angle. Then*

$$F_\eta := \frac{2\chi'\,\eta}{\sec\alpha \tan\alpha\; p'_- p'_+} = \mathsf{w}(y+\theta) - \mathsf{w}(y-\theta) + \delta\omega. \tag{12.1}$$

In particular the frequency functional $\mathcal{R}_p(\eta) := \operatorname{mean}_y F_\eta$ *equals $\delta\omega$. Conversely, let $\omega \in \mathcal{D}_h$ and let η be an even, π-periodic, analytic physical radial variation with $\mathcal{R}_p(\eta) = 0$; write $F_\eta = \sum_{\ell \geq 2 \ even} f_\ell \cos(\ell y)$. Then $\mathsf{k} = p'\mathsf{w}$ with*

$$\mathsf{w} = \sum_\ell \frac{f_\ell}{2\sin(\ell\theta)} \sin(\ell y) \tag{12.2}$$

is the unique odd π-periodic variation at fixed ω inducing η, and $\|\mathsf{w}\|_{\rho-\sigma} \le C\gamma^{-1}\sigma^{-\tau}\|F_\eta\|_\rho$. Its first ω-jet on $\mathcal{D}_h$ costs $C\gamma^{-2}\sigma^{-2\tau-1}$, and replacing $1/(2\sin(\ell\theta))$ by the smooth reciprocal cutoff of Lemma 11.5 gives a C^2 extension in ω with the same value and first jet on $\mathcal{D}_h$. At the ball of radius R, $\mathcal{R}(\eta) = 2\,\mathrm{mean}_y\,\eta/(R\sqrt{R^2-1})$.

Proof. With $\mathsf{k}_\pm = \mathsf{k}(y \pm \theta)$ we have $\delta\chi = \frac{1}{2}(\mathsf{k}_+ + \mathsf{k}_-) + \alpha'\delta\theta$ and $\delta\alpha = \frac{1}{2}(\mathsf{k}_+ - \mathsf{k}_-) + \chi'\delta\theta$. Differentiating $r(\chi(y)) = \sec\alpha(y)$ and using $r'(\chi)\chi' = \sec\alpha\tan\alpha\,\alpha'$ gives $\eta = \sec\alpha\tan\alpha\,[\delta\alpha - (\alpha'/\chi')\delta\chi]$. Since $\chi' = \frac{1}{2}(p'_+ + p'_-)$ and $\alpha' = \frac{1}{2}(p'_+ - p'_-)$,

$$\delta\alpha - \frac{\alpha'}{\chi'}\delta\chi = \frac{\mathsf{k}_+p'_- - \mathsf{k}_-p'_+}{2\chi'} + \frac{\chi'^2 - \alpha'^2}{\chi'}\delta\theta = \frac{p'_+p'_-}{2\chi'}\big[\mathsf{w}(y+\theta) - \mathsf{w}(y-\theta) + 2\delta\theta\big],$$

which is (12.1). The mean of $\mathsf{w}(y+\theta) - \mathsf{w}(y-\theta)$ vanishes. For the converse, $\mathsf{w}(y+\theta) - \mathsf{w}(y-\theta) = \sum_\ell 2\mathsf{w}_\ell \sin(\ell\theta)\cos(\ell y)$ for $\mathsf{w} = \sum \mathsf{w}_\ell \sin(\ell y)$; on $\mathcal{D}_h$ all $\sin(\ell\theta)$ are nonzero with $|\sin\ell\theta| \ge \gamma\ell^{-\tau}$, which gives existence, uniqueness and the bound via $\sup_\ell \ell^\tau e^{-\sigma\ell} \le C\sigma^{-\tau}$. The ω-derivative of $1/\sin(\ell\omega/2)$ is bounded by $\ell^{2\tau+1}/(2\gamma^2)$ on $\mathcal{D}_h$. At the ball $\chi' = p'_\pm = 1$ and $\sec\theta\tan\theta = R\sqrt{R^2-1}$. □

At the base radius $R_0 = \sqrt{d}$ we put $q_0 = 2/\sqrt{d(d-1)}$, so that $\mathcal{R} = q_0\,\mathrm{mean}_y$ at the ball, and $\omega_1 = q_0(r_1 + a_2)$ by Proposition 9.10.

12.2. **The physical normalized derivative.** Let the profile lie in the *near-formal tube*

$$\begin{aligned} &c_{\mathrm{t}}h \le \varepsilon \le C_{\mathrm{t}}h, \qquad R_* - R_0 - r_1\varepsilon = O(h^2), \\ &r = R_* + \varepsilon Y_2 + \varepsilon^2 h_2 + \varepsilon u_*, \qquad \Pi_0 u_* = \Pi_2 u_* = 0, \qquad u_* = O(h^2), \end{aligned} \tag{12.3}$$

in fixed strong analytic norms (specified in Section 13) together with the scaled first ω-jets; here $R_* = \Pi_0 r$ and ε is the coefficient of Y_2 in r. Write $\Pi^\perp = I - \Pi_0$ for the projection onto output functions of zero spherical mean, and put

$$\mathcal{F}(r) = \frac{A_r - \Pi_0 A_r}{\varepsilon}, \qquad \mathsf{j} = Y_2 + 2\varepsilon h_2 + u_*.$$

A variation of r preserving the form (12.3) is $\delta r = \delta R_* + \mathsf{j}\,\delta\varepsilon + \varepsilon z$ with $z = \delta u_*$, $\Pi_0 z = \Pi_2 z = 0$. Then

(12.4)

$$\delta\mathcal{F} = \Pi^\perp T_r z + a_R\,\delta R_* + a_\varepsilon\,\delta\varepsilon, \qquad a_R = \frac{\Pi^\perp T_r 1}{\varepsilon}, \qquad a_\varepsilon = \frac{\Pi^\perp T_r \mathsf{j} - \mathcal{F}}{\varepsilon},$$

the term $-\mathcal{F}/\varepsilon$ in the amplitude column coming from the derivative of the factor $1/\varepsilon$. At a fixed frequency the variation must satisfy the frequency compatibility $\mathcal{R}_p(\delta r) = 0$ of Lemma 12.1, i.e.

$$\varepsilon\,\mathcal{R}_p(z) + \mathcal{R}_p(1)\,\delta R_* + \mathcal{R}_p(\mathsf{j})\,\delta\varepsilon = 0; \tag{12.5}$$

the factor ε in front of $\mathcal{R}_p(z)$ is the physical amplitude. The *physical normalized derivative* with target g, $\Pi_0 g = 0$, is the system (12.4) $= g$, (12.5), for the unknowns $(z, \delta R_*, \delta\varepsilon)$.

Lemma 12.2 (Leading scalar structure)**.** *In the tube* (12.3)*, $\mathcal{F} = O(h^2)$, and uniformly*

$$a_R = L_2' Y_2 + O(h), \qquad a_\varepsilon = -r_1 L_2' Y_2 + O(h),$$
$$\mathcal{R}_p(1) = q_0 + O(h), \qquad \mathcal{R}_p(\mathrm{j}) = q_0 a_2 + O(h),$$

in fixed analytic norms. In particular the components of a_R and a_ε of degree ≥ 4 are $O(h)$, and

$$\det \begin{pmatrix} L_2' & -r_1 L_2' \\ q_0 & q_0 a_2 \end{pmatrix} = L_2'\, q_0 (r_1 + a_2) = L_2'\, \omega_1 \neq 0. \tag{12.6}$$

Proof. By Lemma 8.4, with $f = r - R_* = \varepsilon Y_2 + \varepsilon^2 h_2 + \varepsilon u_*$ and $T_{R_*+f} g = L_{R_*} g + L'_{R_*}(fg) + O(\|f\|^2 \|g\|)$,

$$\begin{aligned} \Pi^\perp A_r &= \varepsilon L_2(R_*) Y_2 + \varepsilon^2 L_4 a_{4,d} Y_4 + \tfrac{\varepsilon^2}{2} \Pi^\perp L'(Y_2^2) + O(h^3) \\ &= \varepsilon^2 \big[L_2'(r_1 + \tfrac{c_{22}}{2}) Y_2 + (L_4 a_{4,d} + \tfrac{1}{2} L_4' c_{24}) Y_4 \big] + O(h^3), \end{aligned}$$

using $L_2(R_*) = L_2'(R_* - R_0) + O(h^2)$ and (9.1); both brackets vanish by Definition 9.2. Hence $\mathcal{F} = O(h^2)$. Next $\Pi^\perp T_r 1 = \varepsilon L_2' Y_2 + O(h^2)$, and

$$\begin{aligned} \Pi^\perp T_r \mathrm{j} &= L_2(R_*) Y_2 + \varepsilon\, \Pi^\perp L'(Y_2^2) + 2\varepsilon L_4 h_2 + O(h^2) \\ &= \varepsilon \big[L_2'(r_1 + c_{22}) Y_2 + (L_4' c_{24} + 2 L_4 a_{4,d}) Y_4 \big] + O(h^2), \end{aligned}$$

whose Y_4 coefficient vanishes and whose Y_2 coefficient is $-r_1 L_2' \varepsilon$, since $c_{22} = -2r_1$. With $\mathcal{F}/\varepsilon = O(h)$ this gives a_R and a_ε. Finally $\mathcal{R}_p \to q_0 \,\mathrm{mean}_y$ and $\mathrm{mean}_y \,\mathrm{j} \to \mathrm{mean}_y\, Y_2(\cos y) = a_2$. □

The scalar column a_ε has no degree-four component at leading order because of the equation defining h_2; this is used to show that the high parts of the scalar columns are $O(h)$.

12.3. **Normalized coordinates, constraints and dimension count.** To combine (12.4)–(12.5) with the high matrix we write the system in the normalized native coordinates of Section 10, and we replace the projected output equation by the equivalent unprojected equation

$$T_r z + a_R\, \delta R_* + a_\varepsilon\, \delta\varepsilon - \varkappa = g, \tag{12.7}$$

with an additional scalar unknown $\varkappa$; for $\Pi_0 g = 0$ its spherical mean gives $\varkappa = \Pi_0 T_r z$, and its projection is (12.4). The constraints $\Pi_0 z = \Pi_2 z = 0$ are kept as two linear rows; Π_0 and Π_2 are evaluated natively by the spherical measure $(\sin\chi)^{d-2} \chi'\, dy$.

Odd d. The unknowns are the normalized input $F = a_{\mathrm{in}} \tilde{z}$ (all degrees) and the three scalars $(\delta R_*, \delta\varepsilon, \varkappa)$; the equations are the normalized output of (12.7) (all degrees), the frequency row (12.5) and the two projection rows. Splitting the harmonic coordinates at the degree M, the *finite center*

consists of the degrees $n < M$ and the three scalars, and the rows of output degree $n < M$ together with the three rows; it is square.

Even d. The unknown is the native primitive v of z. By (12.8) below, a primitive determines the physical variation, and two primitives give the same variation exactly when they differ by a combination of the $\lfloor(\lambda-1)/2\rfloor$ modes $\sin(j\chi)$, $j \le k$ of the parity of λ; we add the corresponding primitive rows, which require these sine coefficients of the physical primitive to vanish. They are the native integrals $\int v(y)\sin(j\chi(y))\chi'(y)\,dy$. On the output side, the regularized output has no cosine modes $\cos(l\phi)$, $l \le k$ of the parity of λ (and $l \neq 0$ in the periodic case); in native coordinates these are the functionals $\ell_l(\mathbf{g}) = \int[\Delta\mathbf{g}](y)\cos(l\Phi(y))\,dy$, for which $\ell_l(\sin(m\,\cdot)) = -2\pi\sin(l\theta)\delta_{lm}$ at the ball; only the finitely many indices $1 \le l \le k$ occur, and for each of them $\sin(l\theta_0) \neq 0$ because θ_0/π is irrational (Section 9.4); since $|\theta - \theta_0| = O(h)$ on I_h, continuity gives $|\sin(l\theta)| \ge \frac{1}{2}\min_{1\le l\le k}|\sin(l\theta_0)| > 0$ for all these l, uniformly for small h. Every transformed derivative of a physical input lies in the common kernel of these functionals, since the identity of Proposition 10.7 was derived from the physical derivative. We append $\lfloor(\lambda-1)/2\rfloor$ columns along a fixed low Fourier basis dual to the ℓ_l; applying the ℓ_l to a solved equation with physical source forces the appended multipliers to vanish. The numbers of primitive rows and of appended columns are both $\lfloor(\lambda-1)/2\rfloor$, so the finite center, formed as in the odd case after the input change $\widetilde{W}$ (and the congruence by Λ^{-1} if λ is odd), is again square, and the high matrix is unchanged. The physical radial variation is recovered from the primitive by

$$z \circ \chi = \cos(\alpha)^{2\lambda}\Big[\frac{-1}{\chi'\sin\chi}\partial_y\Big]^k\Big[\frac{v}{k_\lambda\sin\chi}\Big], \tag{12.8}$$

whose divisions by $\sin\chi$ are removable on the parity class of v (finitely many removable divisions and derivatives).

12.4. **The finite matrix.** Fix M above all lower degrees used above, above $m_\sharp$ in the even case, and large enough for Theorem 11.6. Split the finite variables into the extra resonances Z_0 below M (degrees $n < M$ with $C_n^\lambda(1/\sqrt{d}) = 0$, $n \ge 8$) and the primary block P_0 (all other finite variables). On harmonic coordinates use the same exponential weight e^{rn} as for the high system, at a fixed interior radius r; scalar coordinates have fixed weights. The finite matrix has the block form

$$\begin{gathered} F_h = \begin{pmatrix} F_{\mathsf{PP}} & F_{\mathsf{PZ}} \\ F_{\mathsf{ZP}} & F_{\mathsf{ZZ}} \end{pmatrix}, \qquad F_{\mathsf{PP}} = F^0_{\mathsf{PP}} + O_{d,M}(h), \\ F_{\mathsf{PZ}},\, F_{\mathsf{ZP}} = O_{d,M}(h), \qquad F_{\mathsf{ZZ}} = \varepsilon\,\mathrm{diag}(\bar\delta_n) + O_{d,M}(h^2), \end{gathered} \tag{12.9}$$

where $\bar\delta_n$ is the detuning δ_n of Lemma 9.3 multiplied by the fixed normalization factors at the ball.

Lemma 12.3. *The matrix F^0_{PP} is invertible,* (12.9) *holds, and* $\sup_n|\bar\delta_n|^{-1} \le C_d$*, the supremum being over all resonant degrees and C_d independent of M.*

Proof. At the ball limit $h \to 0$ along (12.3), in physical coordinates and the order: projection rows, output degrees $n \geq 4$ nonresonant, output degree two, frequency row, output degree zero, the primary block is block triangular: the projection rows give 1 on the input degrees 0 and 2; a nonresonant input degree n has output only in degree n, with the nonzero entry L_n; the degree-two row and the frequency row restricted to $(\delta R_*, \delta\varepsilon)$ form the matrix (12.6), while the input harmonics enter them only with $L_2 = 0$ or with the factor ε; the degree-zero output row has the entry -1 in the column $\varkappa$, since a_R, a_ε have zero mean. In the even case the primitive rows and appended columns have the nonzero diagonal pairings computed above. The finite changes from physical to normalized native coordinates are at the ball diagonal with nonzero entries on each retained mode (weights $a_{\mathrm{in}}, a_{\mathrm{out}}$, the factors $\Lambda_0^{\lambda+1}/\kappa_d$, the primitive S_λ, the multipliers of P_λ for $m \geq \lambda$, the denominators $\sin(m\theta_0)$ and $W_0 + F$), and no change of coordinates divides by a resonant multiplier. Hence F^0_{PP} is invertible. Both cross blocks vanish at the ball: a resonant input has zero round output and zero projection rows, and the frequency row carries ε; conversely the scalar columns have only degree-two base output, the column $\varkappa$ is constant, and the other low harmonics are diagonal. The first-order part of F_{ZZ} in the direction w_1 is diagonal with entries $\varepsilon\delta_n$ (Lemma 9.3); the tube (12.3) gives the remainder $O(h^2)$. Passing from the ideal to the actually compressed native block changes F_h by uniformly conditioned left and right factors and by $O(h^2)$: a transformed low basis vector has a low part which is an $O(h)$ perturbation of its base image and a high tail of size $O(h)$, and dropping the tail changes each finite row by $O(h^2)$, since all high-to-finite rows vanish at the ball and are $O(h)$. Finally, in the odd case, $\bar\delta_n = (1+n)^{\lambda+1} L_n'(r_1 + c_{nn})/\kappa_d(R_0)$ up to the fixed weights, and at a zero of $d_n(\omega_0)$ the identity $\partial_\theta Y_n(\cos\theta) = -\sin\theta\, Y_n'(\cos\theta)$ and (S1)(a) give $|\bar\delta_n| \geq cn^2(r_1 + c_{nn}) \geq c'n^2$ for $n \geq M_0$; the finitely many lower resonances have $\bar\delta_n \neq 0$. In the even case there are finitely many resonances. □

Proposition 12.4 (Smallest singular value)**.** *For each fixed M and small h, $\|F_h^{-1}\| \leq C_{d,M} h^{-1}$, equivalently $\sigma_{\min}(F_h) \geq c_{d,M} h$. Moreover $\limsup_{h\to 0} h\|F_h^{-1}\| \leq C_d$ with C_d independent of M.*

Proof. Let $S_\varepsilon = \mathrm{diag}(I_{\mathsf{P}}, \varepsilon^{-1} I_{\mathsf{Z}})$. Then

$$S_\varepsilon F_h = \begin{pmatrix} F^0_{\mathsf{PP}} + O(h) & O(h) \\ \varepsilon^{-1} F_{\mathsf{ZP}} & \mathrm{diag}(\bar\delta_n) + O(h) \end{pmatrix}, \qquad \varepsilon^{-1} F_{\mathsf{ZP}} = O_{d,M}(1).$$

The Schur complement of the upper left block is $\Sigma = \mathrm{diag}(\bar\delta_n) + O(h) - \varepsilon^{-1} F_{\mathsf{ZP}} F^{-1}_{\mathsf{PP}} F_{\mathsf{PZ}} = \mathrm{diag}(\bar\delta_n) + O_{d,M}(h)$, which is invertible with $\|\Sigma^{-1}\| \leq 2C_d$ for small h by Lemma 12.3. The block formula

$$(S_\varepsilon F_h)^{-1} = \begin{pmatrix} F^{-1}_{\mathsf{PP}} + F^{-1}_{\mathsf{PP}} F_{\mathsf{PZ}} \Sigma^{-1} \varepsilon^{-1} F_{\mathsf{ZP}} F^{-1}_{\mathsf{PP}} & -F^{-1}_{\mathsf{PP}} F_{\mathsf{PZ}} \Sigma^{-1} \\ -\Sigma^{-1} \varepsilon^{-1} F_{\mathsf{ZP}} F^{-1}_{\mathsf{PP}} & \Sigma^{-1} \end{pmatrix}$$

has norm $\le C_{d,M}$, and $F_h^{-1} = (S_\varepsilon F_h)^{-1} S_\varepsilon$ gives the first claim since $\varepsilon \ge c_{\mathrm{t}} h$. For the second, $hF_h^{-1} = (S_\varepsilon F_h)^{-1} \operatorname{diag}(hI_{\mathsf{P}}, (h/\varepsilon) I_{\mathsf{Z}})$; for fixed M, as $h \to 0$ the first block column tends to 0, and in the second block column the upper entry is $O_{d,M}(h)$ while the lower one is $\Sigma^{-1}(h/\varepsilon)$ with norm $\le 2C_d/c_{\mathrm{t}}$. The diagonal exponential weights do not change diagonal blocks. This is a bound on the operator norm; the conditioning of the coordinate changes was established in Lemma 12.3. □

12.5. **Feedback from the high modes and the full border.** Write the full normalized system as $\left(\begin{smallmatrix} K_{\mathsf{HH}} & K_{\mathsf{HC}} \\ K_{\mathsf{CH}} & F_h \end{smallmatrix}\right)$, with K_{HH} the high matrix of Theorem 11.6 and C the finite center. The columns K_{HC} and rows K_{CH} vanish at the ball: for the amplitude column this uses the equation for h_2 (Lemma 12.2); extra resonant contributions in higher formal terms are annihilated by the round multiplier; the projection rows vanish on high modes at the ball; and the frequency row carries ε. In the weighted coordinates fixed above, the harmonic cross blocks have weighted row and column norms $O(h)$ by (10.1), with constants independent of M; the high parts of the scalar columns are $O(h)$ analytic vectors with source radius above r; the projection rows are analytic dual vectors of size $O(h)$ on high modes; and the frequency row, whose coefficient on a normalized high mode is at most $C\varepsilon(1+n)^{\lambda}$, is exponentially decaying after the weight e^{-rn} and hence $O(h)$.

Proposition 12.5 (Invertibility of the full border)**.** *Let the profile lie in the tube* (12.3) *and let ω be retained. For M fixed large enough and then h small, the full normalized system is invertible. The Schur complement $\Sigma_h = F_h - K_{\mathsf{CH}} K_{\mathsf{HH}}^{-1} K_{\mathsf{HC}}$ satisfies $\|\Sigma_h^{-1}\| \le 2(C_d+1)h^{-1}$ on the fixed interior radius, and the inverse is given by*

$$x_{\mathsf{C}} = \Sigma_h^{-1}\big(f_{\mathsf{C}} - K_{\mathsf{CH}} K_{\mathsf{HH}}^{-1} f_{\mathsf{H}}\big), \qquad x_{\mathsf{H}} = K_{\mathsf{HH}}^{-1}\big(f_{\mathsf{H}} - K_{\mathsf{HC}} x_{\mathsf{C}}\big).$$

For a width loss σ its norm is at most $C_{d,M} h^{-1} \sigma^{-q} [1 + B(\sigma/c, h)]^2$ for fixed q, c.

Proof. By (11.3)–(11.9), $K_{\mathsf{HH}}^{-1} = \mathcal{J}$[inactive diagonal inverse + $\varpi(S + \varpi E_{\mathrm{off}} \varpi)^{-1} \varpi$] up to the bounded inactive correction of (11.8). The inactive part contributes $O(h^2)$ to the feedback $K_{\mathsf{CH}} K_{\mathsf{HH}}^{-1} K_{\mathsf{HC}}$. For the active part, the cross blocks multiplied by one external weight ϖ have norms at most $C_d h^{1/2}/M$ on Z (where $\varpi_n \le C h^{-1/2}/n$ and $n \ge M$), $C_d h^{3/4}$ on P, and $C_{d,M} h\, \eta_h^{-1/2} e^{-c(N_h - M)} = o_{d,M}(h^{1/2})$ on H, using the exponential decay of the coupling between modes below M and above N_h. With the middle inverse of norm ≤ 2,

$$\|K_{\mathsf{CH}} K_{\mathsf{HH}}^{-1} K_{\mathsf{HC}}\| \le C_d h/M^2 + o_{d,M}(h).$$

By Proposition 12.4, $\|F_h^{-1}\| \le (C_d + o_{d,M}(1)) h^{-1}$. Choose M so that $C_d^2/M^2 \le \frac{1}{4}$ and then h small; then $\|F_h^{-1} K_{\mathsf{CH}} K_{\mathsf{HH}}^{-1} K_{\mathsf{HC}}\| \le \frac{1}{2}$ and Σ_h is invertible by a Neumann series with the stated bound. The displayed formulas give the inverse. For a target loss σ both K_{HH}^{-1} factors cost $B(\sigma/c, h)$,

the finite conversions cost fixed powers of σ^{-1}, and Σ_h^{-1} costs h^{-1}; the inverse may contain two high inverse factors. The fixed-width response of Theorem 11.6(ii) gives the alternative bound $C_d(\epsilon_M + h^{1-\kappa_*})$ for the relative feedback when the center is measured with polynomial weights. □

13. The native Newton iteration and proof of the theorem for $d \geq 5$

13.1. **Order of choices.** Fix $d \geq 5$, the base angle θ_0, the exponent τ_0 of Lemma 9.7, $\tau > \tau_0 + 1$, $\gamma \in (0, c_d/2]$ and $\delta = 1/\tau$, and fix analytic radii $0 < \rho_\infty < \rho_0 \leq \rho_{\max}$. Fix the finite list of coefficient fields of Lemma 13.1 below. Choose the high cutoff M large enough for Lemma 11.3, the middle-factor estimate of Theorem 11.6 and Proposition 12.5, and use this compression at every step of the iteration. Then fix the cost exponents of Lemma 13.3, the early-window constant a_*, the exponents $K \geq 8$ and $d_* > (\varsigma + 1)\log 2$, a sufficiently high finite initializer order N in Proposition 9.10, and finally h, which may depend on M, N and all finite coefficients.

13.2. **Spaces and the field ledger.** Let $\mathcal{A}_\rho^q$ be the space of the introduction (real, odd, π-periodic, norm $\|\cdot\|_{\rho,q}$) and $\mathcal{A}_{\rho,\mathrm{e}}^q$ the analogous space of real, even, π-periodic functions. The tracked shape is $v = p - u \in \mathcal{A}_\rho^{q_d}$ (p itself is not periodic); the residual is even and π-periodic. We use three norm levels:

$$\text{residual: } \mathcal{A}_{\rho,\mathrm{e}}^0; \qquad \text{shape and corrections: } \mathcal{A}_\rho^{q_d};$$
$$\text{native tube complement: } \mathcal{A}_{\rho,\mathrm{e}}^{t_d},$$

with $t_d = 8(d+1)$, $m_d = 16(d+1)$, $q_d = 32(d+1)$.

Lemma 13.1 (Field ledger)**.** *There is a finite list of coefficient fields (the weights, the coefficients of the normalized operators of Section 10, the scalar columns and rows of Section 12, the native projections and the volume map), each an analytic expression in v and its angular derivatives of order at most $s_*(d)$, where*

$$s_*(d) = 4r_d + 2\lceil\lambda\rceil + 24 \quad (d \text{ odd}, \ r_d = \lceil\lambda\rceil + 8), \qquad s_*(d) = 4d + 8 \quad (d \text{ even}),$$

such that the following holds: the fields and their first explicit ω-derivatives, and their shape derivatives up to order three, are bounded at the same radius by the norm $\|v\|_{\rho,m_d}$, and $\max\{t_d + 1, s_*(d)\} < m_d$.

Proof. Odd d. By Proposition 10.4 the coefficient norm with weight $s_d = \lceil\lambda\rceil + 5$ of $\mathfrak{E}$ is controlled by the mixed angular derivatives of order r_d in each variable. The construction of $\mathfrak{E}$ (Proposition 10.2(c)) uses G through order three; the mixed derivatives add at most $2r_d$ and one explicit frequency derivative one more, so derivatives of G through order $2r_d + 4$ suffice; an axial derivative of order a costs at most $(1+n)^{2a}$ in the angular coefficient norm (repeated Chebyshev differentiation), so $4r_d + 8$ angular derivatives of v suffice for $\mathfrak{E}$. The weights, the projections, the volume amplitude Ω and S_G, and the scalar columns, including the noncommuting placement of

the multipliers and the factor $\Lambda_0^{\lambda+1}$, use at most $2\lceil\lambda\rceil + 16$ further angular powers. The tube complement (13.3) below is a separate field using at most t_d derivatives, $t_d + 1$ with an explicit frequency derivative. Since $\lceil\lambda\rceil \le d/2$, $s_*(d) = 6\lceil\lambda\rceil + 56 \le 3d + 56 < 16(d+1)$ for $d \ge 5$; for $d = 5$, $s_* = 68 < 96$.

Even d. With $k = \lambda - 1$: the j-th axial derivative of r along the profile uses at most $j + 1$ angular derivatives and j removable divisions by $\sin\chi$, each costing one more derivative (Lemma 5.2); the endpoint coefficients of $E_\pm$ (Lemma 10.5, Lemma 10.8) and the residual coefficients g_i involve $H^{(i)}$, $i \le k$, and $\partial_s^i \mathcal{Q}$, $i \le k$, hence at most $2k + 2$ derivatives, and the removability of Lemma 10.6 at most $2k + 1$ further removable divisions; the pull-through coefficients M_J, Y_J add translations only. This gives order $\le 4k+4 = 4\lambda$ for V, C, W. Lemma 10.9 adds at most $\lambda+3$ powers for the order mismatch and the congruence by Λ^{-1}, and an explicit frequency derivative adds one (translations by $\pm\theta$). The scalar columns, applied through the same normalization to the primitives of 1 and j, and the volume map (8.4) with its removable division by $\sin^{2\lambda+1}$, use fewer than $8\lambda + 16 = 4d + 8$ derivatives. Shape derivatives replace factors by variations and do not increase the maximal order. Finally $4d+8 < 16(d+1)$ and $t_d+1 = 8d+9 < 16(d+1)$. □

Norm conversions. For odd d the high system lives on the orthonormal Gegenbauer coefficients $\ell^2_{\lambda,\rho}$ in the variable Y. The map $\mathcal{A}^0_{\rho,\mathrm{e}} \to \ell^2_{\lambda,\rho}$ is bounded, because T_k has degree k and bounded L^2_λ norm. Conversely an orthonormal Gegenbauer polynomial of degree n has real supremum norm $O_d((1+n)^\lambda)$ and at most $n + 1$ Chebyshev coefficients, each bounded by twice that norm, so Cauchy–Schwarz with a buffer σ gives

$$\|f\|_{\rho-\sigma,q_d} \le C_d\big(1 + \sigma^{-q_d-d-4}\big)\|f\|_{\ell^2_{\lambda,\rho}}. \tag{13.1}$$

For even d the high system lives on the native sine coefficients, $\mathcal{A}^0_\rho \to \ell^2_\rho$ is bounded, and $\|f\|_{\rho-\sigma,q_d} \le C\sigma^{-q_d-1}\|f\|_{\ell^2_\rho}$. These are bounded linear maps; they act on first jets and Whitney remainders, and each use spends a positive buffer.

Same-width Lipschitz bound. At fixed ω and for $\rho \ge \rho_\infty$,

$$\begin{aligned} N_\rho\big(A(v_{\mathrm{new}},\omega) - A(v_{\mathrm{old}},\omega)\big) &\le C_d\|v_{\mathrm{new}} - v_{\mathrm{old}}\|_{\rho,m_d} \\ &\le C_d\|v_{\mathrm{new}} - v_{\mathrm{old}}\|_{\rho,q_d}, \end{aligned} \tag{13.2}$$

by Proposition 10.4, resp. Lemma 10.9 and Proposition 10.10; the frozen diagonal part of A in the even case does not depend on v. This uses the coefficient norms at the same radius and not the frequency derivative of A, which grows with the mode in type (O).

13.3. **Native tube, residual and transport.** For each p use the centered χ, α of Definition 2.3. Define R_* and ε by the physical degree-zero and degree-two projections of r, evaluated natively with the measure

$(\sin\chi)^{d-2}\chi'\,dy$ and the fixed round normalizing constants, and the native complement

$$u_{\mathrm{n}}(y) = \frac{\sec\alpha(y) - R_* - \varepsilon Y_2(\cos\chi(y)) - \varepsilon^2 h_2(\cos\chi(y))}{\varepsilon}. \tag{13.3}$$

Its $\mathcal{A}^{t_d}_{\rho,\mathrm{e}}$ norm is tracked directly, and $\Pi_0 u_* = \Pi_2 u_* = 0$ holds exactly for $u_* = u_{\mathrm{n}} \circ \chi^{-1}$. With one fixed buffer for the identification of the charts, (13.3) gives the tube (12.3) needed in Section 12. The coefficient fields and scalar columns are computed directly from p.

The residual is

$$\mathcal{F}(p,\omega)(u) = \frac{\mathcal{V}_p(u) - \bar{A}_p}{\varepsilon}, \qquad \bar{A}_p = \mathsf{w}_d \int_0^\pi \mathcal{V}_p(u)(\sin p(u))^{d-2} p'(u)\,du, \tag{13.4}$$

where $\mathsf{w}_d^{-1} = \int_0^\pi (\sin u)^{d-2}\,du$ and $\mathcal{V}_p(u) = A_r(p(u))$ is the volume map: (10.5) for odd d, and for even d the slice formula (8.4) in native coordinates,

$$A_r(\Phi(y)) = -\kappa_{2\lambda}(\sin\Phi(y))^{-2\lambda-1} \int_y^{y+\omega} \mathfrak{D}^\lambda\, \mathcal{X}'\, d\tilde{y}, \tag{13.5}$$

with $\mathcal{X} = \sec\alpha\cos\chi$, whose singularity at $\sin\Phi = 0$ is removable (Section 8). The mean $\bar{A}_p$ is the spherical mean of the physical output, pulled back by p. If k is a variation of p at fixed ω and η the induced physical radial variation, the chain rule gives

$$D_p\mathcal{F}[\mathsf{k}] = (\delta_{\mathrm{phys}}\mathcal{F})[\eta] \circ p + \frac{\partial_u \mathcal{F}}{p'}\,\mathsf{k}, \tag{13.6}$$

where $\delta_{\mathrm{phys}}\mathcal{F}$ is the physical normalized differential of (12.4). The second term, the *transport term*, is proportional to the residual.

13.4. **Whitney norms and one global extension.** Let $\mathsf{E} \subset I_h$ be compact and f a Banach-valued family on E with Whitney first jets $f^{[1]}$. Put

$$\|f\|_{\mathrm{Wh}(\mathsf{E})} = \sup_{\mathsf{E}}\|f\| + h\sup_{\mathsf{E}}\|f^{[1]}\| + h\sup_{a\neq b\in\mathsf{E}} \frac{\|f(b) - f(a)\|}{|b-a|}. \tag{13.7}$$

The Whitney remainder property is an additional property, not implied by (13.7). Extend f once: on each bounded gap (a,b) of E, with $\ell = b-a$ and $s = (\omega - a)/\ell$, put

$$\begin{aligned} f^{\mathrm{ext}}(\omega) = (2s^3 - 3s^2 + 1)f(a) + (3s^2 - 2s^3)f(b) \\ + \ell(s^3 - 2s^2 + s)f^{[1]}(a) + \ell(s^3 - s^2)f^{[1]}(b), \end{aligned}$$

and on the outer gaps use the affine extension from the endpoint. The derivative is a fixed polynomial combination of $f^{[1]}(a)$, $f^{[1]}(b)$ and $(f(b) - f(a))/\ell$, so

$$\sup_{I_h}\|f^{\mathrm{ext}}\| + h\sup_{I_h}\|\partial_\omega f^{\mathrm{ext}}\| \le C\|f\|_{\mathrm{Wh}(\mathsf{E})}. \tag{13.8}$$

On shrinking gaps the derivative differs from $f^{[1]}(a)$ by a constant times $\|f(b)-f(a)-\ell f^{[1]}(a)\|/\ell+\|f^{[1]}(b)-f^{[1]}(a)\|$, which tends to zero by the Whitney property; hence f^{ext} is C^1 on I_h in the Banach space, including at accumulation points of E. Reality and odd π-periodicity are preserved by linearity.

13.5. **The correction step.** Let E_j be the retained set at stage j, $\rho_{j+1}=\rho_j-\sigma_j$ with $\sigma_j=c(j+1)^{-2}$ and $\rho_0-\sum_j\sigma_j=\rho_\infty$, and $e_j=\|\mathcal{F}_j\|_{\mathrm{Wh}(\mathsf{E}_j)}$ in $\mathcal{A}^0_{\rho_j,\mathrm{e}}$. Split σ_j into twelve equal blocks. On E_j:

(1) convert the native source $-\mathcal{F}_j$ to normalized coordinates (one block);
(2) solve the full normalized system of Proposition 12.5, including the residual subtraction in the amplitude column and the factor ε in the frequency row, and form its first jet by the resolvent identity (two blocks, subdivided a fixed number of times for the at most two high inverse factors and their jets; in the even case this includes the cutoff losses of Lemma 11.5);
(3) return to physical radial data, by (13.1) and, for even d, by (12.8) (one block); the frequency row gives $\mathcal{R}_p(\eta_j)=0$ for $\eta_j=\varepsilon z_j+\delta R_{*,j}+\mathfrak{j}\,\delta\varepsilon_j$;
(4) obtain the odd π-periodic correction k_j from Lemma 12.1 (one block, including the cutoff of its divisors for the jet);
(5) upgrade k_j and its jet to $\mathcal{A}^{q_d}$ with a Cauchy buffer (one block); repeating this at later stages again produces $\mathcal{A}^{q_d}$;
(6) use the remaining six blocks for the Cauchy derivative in the transport term, the Taylor and jet estimates, and the finitely many norm conversions.

Then extend k_j once by (13.8) (no loss of width), replace p by $p+\mathsf{k}_j^{\mathrm{ext}}$ on all of I_h, and recompute every field, R_*, ε, the physical mean, the normalization, the scalar columns and the projections from this single p. The correction need not solve the equation on the gaps; the centered construction gives an actual body at every $\omega\in I_h$.

Proposition 13.2 (Quadratic recurrence)**.** *There are constants Γ_j, depending on h, σ_j and the inverse bounds, such that*

$$e_{j+1}\le\Gamma_j e_j^2 \quad \text{on } \mathsf{E}_j, \qquad \sup_{I_h}\|\Delta_j\mathsf{f}\|+h\sup_{I_h}\|\partial_\omega\Delta_j\mathsf{f}\|\le\Gamma_j e_j \tag{13.9}$$

for the increment $\Delta_j\mathsf{f}$ of every tracked field f of Lemma 13.1, of the shape in $\mathcal{A}^{q_d}_{\rho_{j+1}}$, and of the tube complement.

Proof. On E_j the physical normalized differential applied to η_j equals $-\mathcal{F}_j$ by construction, so by (13.6) $D_p\mathcal{F}[\mathsf{k}_j]=-\mathcal{F}_j+(\partial_u\mathcal{F}_j/p')\mathsf{k}_j$, and

$$\mathcal{F}(p+\mathsf{k}_j)=\frac{\partial_u\mathcal{F}_j}{p'}\,\mathsf{k}_j+\mathcal{T}_2(p;\mathsf{k}_j),$$

where $\mathcal{T}_2$ is the second-order Taylor remainder. By the Cauchy estimate on one block, $\|\partial_u\mathcal{F}_j\|\le Cb_j^{-1}e_j$ with $b_j=\sigma_j/12$, and $\|\mathsf{k}_j\|\le\Gamma'_j e_j$ with

Γ'_j the product of the inverse and conversion costs of steps (1)–(5); hence the transport term is quadratic in e_j. The exact nonlinear map has a common analytic domain (Lemma 8.1, Lemma 10.3, (13.5)); products, nonzero scalar inverses, analytic powers, translations, fixed β-integrals, removable divided differences and removable divisions by sin give bounded second shape derivatives after positive buffers, with fixed powers of $\varepsilon^{-1} \le Ch^{-1}$; so $\|\mathcal{T}_2\| \le Ch^{-P} b_j^{-P} \|\mathsf{k}_j\|^2$. Three shape derivatives and one explicit frequency derivative suffice for the first-jet and divided-difference parts of (13.7), using products and the resolvent identities; the Whitney remainders follow from (11.13), Lemma 11.5 and Lemma 12.1. The extension agrees with values and jets on E_j, and restriction to a smaller set does not increase the norm. No smallness of the residual on the gaps is used. The increments of the fields are controlled by (13.8) and the same finite calculus. □

Lemma 13.3 (Cost classes)**.** *There are fixed P_0, P such that, with $L = \log(1/h)$, for every $a_* > 0$ and $j_* = \lceil a_* \log L \rceil$,*

$$\begin{aligned} \log \Gamma_j &\le P_0 L + P_0 \log(j+2) + C_{a_*} \quad (j \le j_*), \\ \log \Gamma_j &\le C(L + j + 1)^P \quad (\textit{all } j \ge 0). \end{aligned} \tag{13.10}$$

Proof. The inverse and border costs are those of (11.11), (11.12), (11.13) and Proposition 12.5, evaluated at a fixed fraction of σ_j. For $j \le j_*$ we have $\sigma_j \ge c(a_* \log L + 1)^{-2}$, so the early bound $B \le Ch^{-2}\sigma_j^{-\varsigma}$ of Theorem 11.6(i) applies. For all j, $\log B(\sigma_j/c, h) \le C(j+1)^{4\tau}(L + \log(j+2) + 1)^{2\tau} \le C(L + j + 1)^{6\tau}$. The factors $\mathfrak{g}_1, \mathfrak{g}_2$ of (11.12)–(11.13), which in the even case contain $\gamma^{-4}\sigma_j^{-4\tau-4}$ from the cutoff of the cotangent diagonal, the tangent divisor costs $\gamma^{-2}\sigma_j^{-2\tau-1}$, the conversions (13.1), the normalization order $\lambda + 1$, the indices m_d, q_d, the powers of ε^{-1} and products of at most a fixed number of inverse bounds only enlarge P_0 and P; γ is fixed. □

Proposition 13.4 (Two-regime induction)**.** *Let $K \ge 8$ and $d_* > 0$, and choose a_* with $a_* \log 2 > P - 1$. There are a finite initializer order N and $h_* > 0$ such that for $h < h_*$, for all $j \ge 0$,*

$$\Gamma_j e_j \le h^K \exp[-d_*(j+1)]. \tag{13.11}$$

In particular $e_j \le h^{c_\star 2^j}$ with $c_\star$ as large as required; if some e_j vanishes, the subsequent corrections are taken to be zero.

Proof. Put $a_j = \log \Gamma_j$ and $b_j = -\log e_j$. From (13.9), $b_{j+1} \ge 2b_j - a_j$, whence $b_j \ge 2^j \big[b_0 - \sum_{i<j} 2^{-i-1} a_i\big]$. By (13.10) the part of the sum with $i \le j_*$ is at most $P_0 L + C$, and the part with $i > j_*$ is at most $C 2^{-j_*}(L + j_* + 1)^P \le C L^{P - a_* \log 2}(1 + o(1)) = o(L)$. By Proposition 9.10 the initial residual satisfies $e_0 \le C_N h^N$, including its scaled jet and divided differences, so $b_0 \ge NL - C'_N$. Choose $N \ge P_0 + c_\star + 1$; then $b_j \ge c_\star 2^j L$ for all j and small h. For $j \le j_*$, $a_j - b_j \le P_0 L + P_0 \log(j+2) + C - c_\star 2^j L \le -KL - d_*(j+1)$ if $c_\star > P_0 + K + 1$. For $j > j_*$, the function $j \mapsto (L + j + 1)^P / (2^j L)$ is

decreasing once $L+j+1 > P/\log 2$, and at $j = j_*$ it is $O(L^{P-1-a_*\log 2}) \to 0$; hence $a_j - b_j \le -\frac{1}{2}c_\star 2^j L \le -KL - d_*(j+1)$ for all $j > j_*$ and small h. $\square$

We take $K \ge 8$ and $d_* > (\varsigma+1)\log 2$. The summed corrections are $O(h^K)$ in C^1_h and preserve all native-field, branch and convexity margins and $\varepsilon \asymp h$. The tube (12.3), including its scaled jet, is preserved: the field u_{n} changes by $O(h^{K-1})$, the division by ε costing one power of h. The high scalar columns are initialized directly on the finite native initializer, where their high parts are $O(h)$ on a fixed source strip by Lemma 12.2 (with the cancellation coming from h_2); they belong to the field list, so their summed changes are $O(h^K)$. Hence the finite center and the feedback remain in the range of Proposition 12.4 and Proposition 12.5 at every step.

13.6. **Nested exclusions.** Let $\eta_h = h\exp[-L^\delta/(\log L)^4]$ and fix $\varsigma > 0$. Start with $\mathsf{E}_0 = \mathcal{D}_h \cap \{|e_{n,0}| \ge 2\eta_h n^{-\varsigma}\ \forall n \ge M\}$; by (11.6) and Lemma 9.7, $\mathrm{Leb}(I_h \setminus \mathsf{E}_0) = o_d(h)$. Put $\eta_{h,j} = \eta_h(1+2^{-j})$ and $N_j = N_\star 2^j$ with a fixed $N_\star \ge M$. Given E_j, update p on I_h and define

$$\mathsf{E}_{j+1} = \mathsf{E}_j \cap \{\omega :\ |e_{n,j+1}(\omega)| \ge \eta_{h,j+1} n^{-\varsigma} \text{ for all } n \ge M\},$$

where $e_{n,j+1}$ is formed from the new coefficient family with the same compression and the same smooth elimination (11.3). By Lemma 11.4, (13.2), (13.9) and (13.11),

$$\sup_{n\ge M} |e_{n,j+1}(\omega) - e_{n,j}(\omega)| \le C_d h^K e^{-d_*(j+1)} \qquad (\omega \in \mathsf{E}_j), \tag{13.12}$$

with no factor depending on n (the frequency derivatives are not used here). For $n \le N_{j+1}$ the reserve $(\eta_{h,j} - \eta_{h,j+1})n^{-\varsigma} \ge \eta_h N_\star^{-\varsigma} 2^{-(\varsigma+1)(j+1)}$ exceeds (13.12) for every j, because $d_* > (\varsigma+1)\log 2$ and $h^K/\eta_h \to 0$. So only indices $n > N_{j+1}$ can be newly deleted. The new coefficients are defined by the new p on all of I_h, including the old gaps, and satisfy the hypotheses of Lemma 11.3; therefore the newly deleted part of E_j has measure at most

$$C_d\,\eta_h\big[N_{j+1}^{-\varsigma-1} + hN_{j+1}^{-\varsigma}\big], \tag{13.13}$$

which is summable in j. The sets E_j are compact and nested (the conditions are closed relative to $\mathcal{D}_h$, which contains no pole). With $E_{d,h} = \bigcap_j \mathsf{E}_j$,

$$\mathrm{Leb}(E_{d,h}) \ge h - o_d(h). \tag{13.14}$$

At every fixed mode n, the coefficients converge and $\eta_{h,j} \downarrow \eta_h$, so the limiting gap $\eta_h n^{-\varsigma}$ persists.

13.7. **The exact limit and the volume image.** The corrections $\mathsf{k}_j^{\mathrm{ext}}$ and their first frequency derivatives are summable in $\mathcal{A}^{q_d}_{\rho_\infty}$, the latter being bounded by h^{-1} times (13.11). Hence $v_\omega = p_\infty - u$ is a C^1 map from I_h to $\mathcal{A}^{q_d}_{\rho_\infty}$, small in $\|\cdot\|_{\rho_\infty,1}$. By Lemma 2.4, for every $\omega \in I_h$ the body $K_\omega = K(v_\omega,\omega)$ is an origin-symmetric, strictly convex body of revolution with real-analytic boundary strictly containing B^d, whose radial function is

$O(h)$-close to $\sqrt{d}$ on a fixed complex strip. Its physical degree-two coefficient stays comparable to h, so K_ω is not a ball. On $E_{d,h}$ the residual (13.4) tends to zero in $\mathcal{A}^0_{\rho_\infty,\mathrm{e}}$, and the volume map is continuous on the common domain, so $A_r - \Pi_0 A_r = 0$: all distance-one sections of K_ω have the volume $V(\omega) = \bar{A}_{p_\infty(\omega)}$.

The mean $V(\omega) = \bar{A}_{p_\infty(\omega)}$ is a bounded native integral of the C^1 family, so $V \in C^1(I_h)$. By Proposition 9.10, the initializer has $V'(\omega) = -2|S^{d-2}|(d-1)^{(d-2)/2} + O_d(h)$, and the summed derivative changes are $O_d(h^{K-1})$. For small h, V' is negative and bounded away from zero on I_h, so V is bi-Lipschitz there and $\mathrm{Leb}(V(E_{d,h})) \geq c\,\mathrm{Leb}(E_{d,h}) > 0$ by (13.14).

Proof of Theorem 1.3. Items (i)–(iv) were established above; the measure statement follows from (13.14) and the bi-Lipschitz property of V. □

Remark 13.5 (The case $d = 5$). For odd d the arguments of Part II use only $\lambda \geq \frac{3}{2}$ and inequalities verified for every odd $d \geq 5$. For $d = 5$ the constants are

$$\lambda = \tfrac{3}{2}, \qquad \beta = \tfrac{1}{2}, \qquad R_0 = \sqrt{5}, \qquad r_1 = -\tfrac{1}{6}, \qquad c_{nn} = \frac{3n(n+3)}{2(4n^2+12n+5)},$$

$$a_{4,5} = -\frac{5\sqrt{5}}{72}, \qquad Y_4(1/\sqrt{5}) = -\frac{3}{25}, \qquad Y_6(1/\sqrt{5}) = \frac{11}{125},$$

$$L_2' = -4\pi^2, \qquad \omega_1 = \frac{\sqrt{5}}{24}, \qquad V'(\omega_0) = -32\pi^2;$$

in Lemma 9.9, $m_* = 1$ and $c = 2$; the remainder of Proposition 10.2 is built from $T_{7,R}$, with coarea exponent $\lambda + \frac{1}{2} = 2$ and $\mathfrak{D}_0^{3/2}$; the constant of Lemma 10.3 is $\kappa_3/(|S^4|R^5) = 1/(2\pi R^5)$; and the ledger is $r_5 = 10$, $s_5 = 7$, $t_5 = 48$, $m_5 = 96$, $q_5 = 192$ with $s_*(5) = 68 < 96$.

Use of AI-assisted tools

AI-assisted tools were used in the research and preparation of this manuscript.

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